

39. I²C Bus Interface (IIC)

39.1 Overview

The I²C bus interface (IIC) has 3 channels. The IIC module conforms with and provides a subset of the NXP I²C (Inter-Integrated Circuit) bus interface functions.

Table 39.1 lists the IIC specifications, Figure 39.1 shows a block diagram, and Figure 39.2 shows an example of I/O pin connections to external circuits, with an I²C bus configuration. Table 39.2 lists the I/O pins.

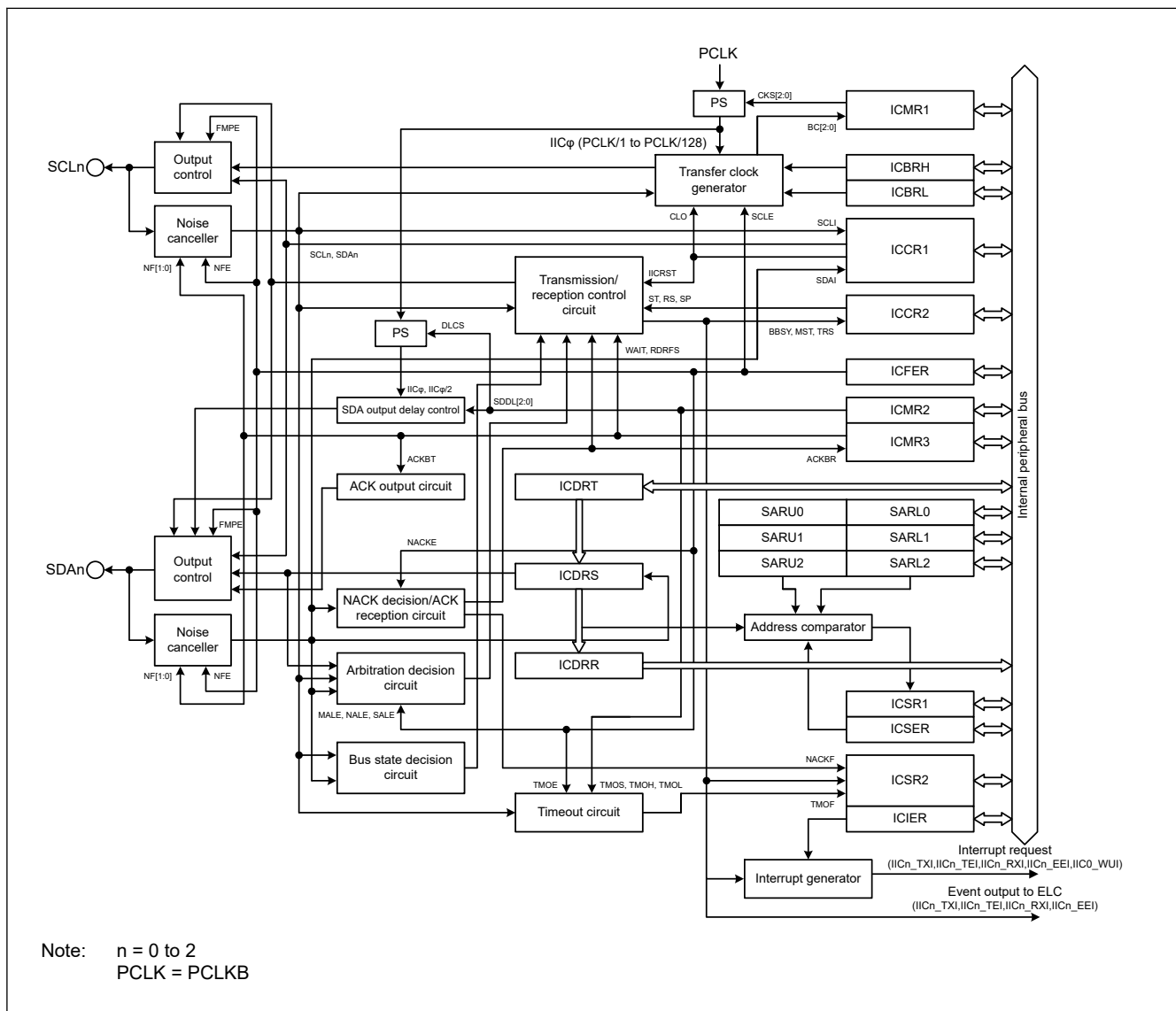
Table 39.1 IIC specifications (1 of 2)

Parameter	Specifications
Communications format	- I²C-bus format or SMBus format - Master or slave mode selectable - Automatic securing of the setup times, hold times, and bus-free times for the transfer rate
Transfer rate	Fast-mode Plus supported, up to 1 Mbps
SCL clock	For master operation, the duty cycle of the SCL clock is selectable in the range from 4% to 96%
Issuing and detecting conditions	- Start, restart, and stop conditions are automatically generated - Start conditions (including restart conditions) and stop conditions are detectable
Slave address	- Configurable for up to three different slave addresses 7- and 10-bit address formats supported, including simultaneous use - General call addresses, device ID addresses, and SMBus host addresses detectable
Acknowledgment	- For transmission, automatic loading of the acknowledge bit Transfer of the next transmit data can be automatically suspended on detection of a not-acknowledge bit. - For reception, automatic transmission of the acknowledge bit If a wait between the 8th and 9th clock cycles is selected, the software can control the value in the acknowledge field in response to the received value.
Wait function	During reception, the following wait periods are available by holding the SCL clock low: - Waiting between the eighth and ninth clock cycles - Waiting between the ninth clock cycle and the 1st clock cycle of the next transfer
SDA output delay function	Output timing of transmitted data, including the acknowledge bit, can be delayed
Arbitration	- For multi-master operation: - SCL clock synchronization is possible when conflict occurs with the SCL signal from another master - When issuing the start condition creates conflict on the bus, loss of arbitration is detected by testing for a mismatch between the internal signal for the SDA line and the level on the SDA line - In master operation, loss of arbitration is detected by testing for non-matching between the signal on the SDA line and the internal signal for the SDA line - Loss of arbitration because the start condition occurs while the bus is busy is detectable, to prevent the issuing of double start conditions - Loss of arbitration is detectable on transfer of a not-acknowledge bit because the internal signal for the SDA line and the level on the SDA line do not match - Loss of arbitration because a mismatch of internal and line levels for data is detectable in slave transmission
Timeout function	Internal detection of long-interval stops of the SCL clock
Noise cancellation	- Digital noise filters for both the SCL and SDA signals - Programmable window for noise cancellation by the filters
Interrupt sources	- Transfer error or event occurrence (arbitration-lost, NACK, timeout, start or restart condition, or stop condition) - Receive data full, including matching with a slave address - Transmit data empty, including matching with a slave address - Transmit end
Module-stop function	Module-stop state can be set
IIC operating modes	- Master transmit - Master receive - Slave transmit - Slave receive

Table 39.1 IIC specifications (2 of 2)

Parameter	Specifications
Event link function (output)	- Transfer error or event occurrence (arbitration-lost, NACK, timeout, start or restart condition, or stop condition) - Receive data full, including matching with a slave address - Transmit data empty, including matching with a slave address - Transmit end
Wakeup function*1	CPU can return from Software Standby mode using a wakeup event
TrustZone Filter	Security and Privilege attribution can be set for each channels

Note 1. This function is only available for IIC channel IIC0. IIC1 and IIC2 are not supported.

**Figure 39.1 IIC block diagram**

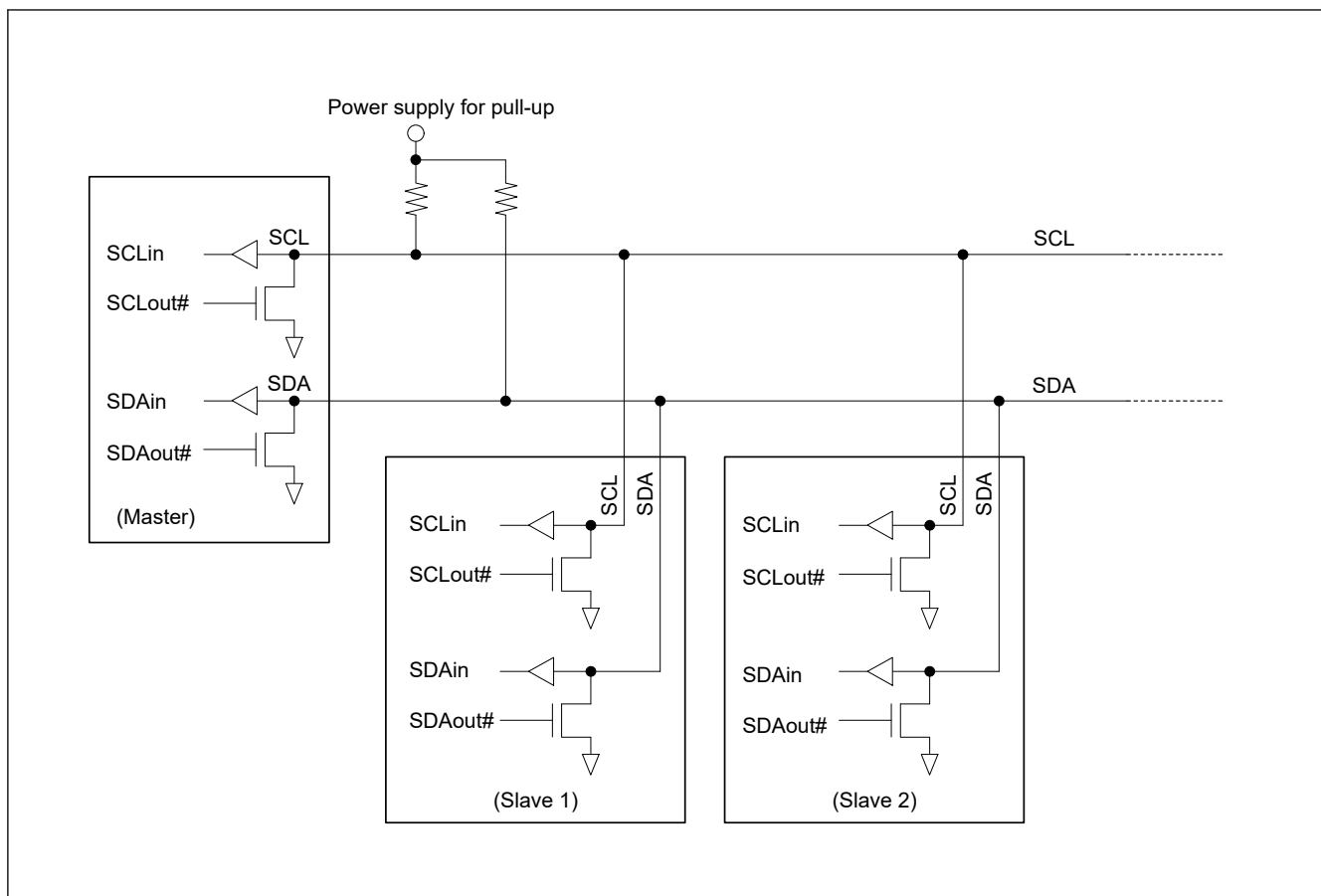


Figure 39.2 I/O pin connection to an external circuit (I²C bus configuration example)

The input level of the signals for IIC is CMOS when I²C bus is selected (ICMR3.SMBS = 0), or TTL when SMBus is selected (ICMR3.SMBS = 1).

Table 39.2 IIC I/O pins

Channel	Pin name	I/O	Function
IICn	SCLn	I/O	IICn serial clock I/O pin
	SDAn	I/O	IICn serial data I/O pin

Note: n = 0 to 2

39.2 Register Descriptions

39.2.1 ICCR1 : I²C Bus Control Register 1

Base address: IICn = 0x4025_E000 + 0x0100 × n (n = 0 to 2)
IICn_NS = 0x5025_E000 + 0x0100 × n (n = 0 to 2)

Offset address: 0x00

Bit position:	7	6	5	4	3	2	1	0
Bit field:	ICE	IICRS T	CLO	SOWP	SCLO	SDAO	SCLI	SDAI

Value after reset: 0 0 0 1 1 1 1 1

Bit	Symbol	Function	R/W
0	SDAI	SDA Line Monitor 0: SDAn line is low 1: SDAn line is high	R

Bit	Symbol	Function	R/W
1	SCLI	SCL Line Monitor 0: SCLn line is low 1: SCLn line is high	R
2	SDAO	SDA Output Control/Monitor 0: Read: IIC drives SDAn pin low Write: IIC drives SDAn pin low 1: Read: IIC releases SDAn pin Write: IIC releases SDAn pin	R/W
3	SCLO	SCL Output Control/Monitor Use an external pull-up resistor to drive the signal high. 0: Read: IIC drives SCLn pin low Write: IIC drives SCLn pin low 1: Read: IIC releases SCLn pin Write: IIC releases SCLn pin	R/W
4	SOWP	SCLO/SDAO Write Protect This bit is read as 1. 0: Write enable SCLO and SDAO bits 1: Write protect SCLO and SDAO bits	W
5	CLO	Extra SCL Clock Cycle Output This bit clears automatically after 1 clock cycle is output. 0: Do not output extra SCL clock cycle (default) 1: Output extra SCL clock cycle	R/W
6	IICRST	I ² C Bus Interface Internal Reset This setting clears the bit counter and the SCLn/SDAn output latch. 0: Release IIC reset or internal reset 1: Initiate IIC reset or internal reset	R/W
7	ICE	I ² C Bus Interface Enable Used in combination with the IICRST bit to select either IIC or internal reset. 0: Disable (SCLn and SDAn pins in inactive state) 1: Enable (SCLn and SDAn pins in active state)	R/W

Note: S-TYPE3, P-TYPE3

SDAO bit (SDA Output Control/Monitor) and SCLO bit (SCL Output Control/Monitor)

The SDAO and SCLO bits directly control the SDAn and SCLn signals output from the IIC. When writing to these bits, also write 0 to the SOWP bit. Setting these bits results in input to the IIC by the input buffer. When slave mode is selected, a start condition might be detected and the bus might be released, depending on the bit settings.

Do not rewrite these bits during a start condition, stop condition, restart condition, transmission, or reception. Operation after rewriting under these conditions is not guaranteed. When reading these bits, the state of signals output from the IIC can be read.

CLO bit (Extra SCL Clock Cycle Output)

The CLO bit allows output of an extra SCL clock cycle for debugging or error processing. Normally, set this bit to 0. Setting the bit to 1 in a normal communication state causes a communication error. For details on this function, see [section 39.12.2. Extra SCL Clock Cycle Output Function](#).

IICRST bit (I²C Bus Interface Internal Reset)

The IICRST bit initiates an internal state reset of the IIC. Setting this bit to 1 initiates an IIC reset or internal reset. Whether an IIC reset or internal reset is initiated is determined by the settings of this bit in combination with the ICE bit. [Table 39.3](#) lists the IIC resets.

The IIC reset initializes all registers except ICCR1.ICE and ICCR1.IICRST bits and internal states of the IIC. In addition to the internal states of the IIC, the internal reset initializes the following:

- Bit counter (ICMR1.BC[2:0] bits)
- I²C Bus Shift Register (ICDRS)
- I²C Bus Status Registers (ICSR1 and ICSR2)
- SDAO and SCLO Output Control/Monitor (ICCR1.SDAO and ICCR1.SCLO bits)

- I²C Bus Control Register 2 (except ICCR2.BBSY bit)

For the reset conditions for each register, see [section 39.15. State of Registers When Issuing Each Condition](#).

An internal reset initiated with the IICRST bit set to 1 during operation (with the ICE bit set to 1) resets the internal states of the IIC without initializing the port settings and the control and setting registers of the IIC. If the IIC hangs in a low-level output state, resetting the internal states cancels the low-level output state and releases the bus with the SCLn pin and SDAn pin at high impedance.

Note: If an internal reset is initiated using the IICRST bit for a bus hang-up that occurs during communication with the master device in slave mode, the slave and master devices might enter different states, because the bit counter information differs. For this reason, do not initiate an internal reset in slave mode. Initiate recovery processing from the master device. If an internal reset is required because the IIC hangs with the SCLn line in a low-level output state in slave mode, initiate an internal reset, then issue a restart condition from the master device, or issue a stop condition and resume communication from the start condition. If communication is restarted by initiating a reset solely in the slave device without issuing a start or restart condition from the master device, synchronization is lost because the master and slave devices operate asynchronously.

Table 39.3 IIC resets

IICRST	ICE	State	Specifications
1	0	IIC reset	Resets all registers except ICCR1.IICRST and ICCR1.ICE bits, and the internal states of the IIC
	1	Internal reset	Reset the following: • ICMR1.BC[2:0] bits • ICSR1, ICSR2, ICDSR registers • SDAO and SCLO Output Control/Monitor (ICCR1.SDAO and ICCR1.SCLO bits) • I²C Bus Control Register 2 (except ICCR2.BBSY bit) • Internal states of the IIC

ICE bit (I²C Bus Interface Enable)

The ICE bit selects the active or inactive state of the SCLn and SDAn pins. It can also be combined with the IICRST bit to initiate two types of resets. See [Table 39.3](#) for the reset descriptions.

Set the ICE bit to 1 when using the IIC. The SCLn and SDAn pins are placed in the active state when the ICE bit is set to 1. Set the ICE bit to 0 when the IIC is not used. The SCLn and SDAn pins are placed in the inactive state when the ICE bit is set to 0. Do not assign the SCLn or SDAn pin to the IIC when setting up the pin function control. Slave address comparison is performed if the pins are assigned to the IIC.

39.2.2 ICCR2 : I²C Bus Control Register 2

Base address: IICn = 0x4025_E000 + 0x0100 × n (n = 0 to 2)
IICn_NS = 0x5025_E000 + 0x0100 × n (n = 0 to 2)

Offset address: 0x01

Bit position:	7	6	5	4	3	2	1	0
Bit field:	BBSY	MST	TRS	—	SP	RS	ST	—
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	—	This bit is read as 0. The write value should be 0.	R/W
1	ST	Start Condition Issuance Request 0: Do not issue a start condition request 1: Issue a start condition request	R/W
2	RS	Restart Condition Issuance Request 0: Do not issue a restart condition request 1: Issue a restart condition request	R/W

Bit	Symbol	Function	R/W
3	SP	Stop Condition Issuance Request 0: Do not issue a stop condition request 1: Issue a stop condition request	R/W
4	—	This bit is read as 0. The write value should be 0.	R/W
5	TRS	Transmit/Receive Mode 0: Receive mode 1: Transmit mode	R/W ¹
6	MST	Master/Slave Mode 0: Slave mode 1: Master mode	R/W ¹
7	BBSY	Bus Busy Detection Flag 0: I ² C bus released (bus free state) 1: I ² C bus occupied (bus busy state)	R

Note: S-TYPE3, P-TYPE3

Note 1. The MST and TRS bits can be written to when the ICMR1.MTWP bit is set to 1.

ST bit (Start Condition Issuance Request)

The ST bit requests transition to master mode and triggers a start condition. When this bit is set to 1, a start condition is issued when the BBSY flag is set to 0 (bus free state). For details on this function, see [section 39.11. Start, Restart, and Stop Condition Issuing Function](#).

[Setting condition]

- When 1 is written to the ST bit.

[Clearing conditions]

- When 0 is written to the ST bit
- When a start condition is issued (a start condition is detected)
- When the AL (arbitration-lost) flag in ICSR2 is set to 1
- When 1 is written to the IICRST bit in ICCR1 to apply an IIC reset or an internal reset.

Note: Only set the ST bit to 1 (start condition request) when the BBSY flag is set to 0 (bus free state). Arbitration might be lost if the ST bit is set to 1 (start condition request) when the BBSY flag is 1 (bus busy state).

RS bit (Restart Condition Issuance Request)

The RS bit requests that a restart condition be issued in master mode. When this bit is set to 1 to request a restart condition, a restart condition is issued when the BBSY flag is set to 1 (bus busy state) and the MST bit is set to 1 (master mode). For details on this function, see [section 39.11. Start, Restart, and Stop Condition Issuing Function](#).

[Setting condition]

- When 1 is written to the RS bit with the BBSY flag in ICCR2 set to 1.

[Clearing conditions]

- When 0 is written to the RS bit
- When a restart condition is issued (a start condition is detected)
- When the AL (arbitration-lost) flag in ICSR2 is set to 1
- When 1 is written to the IICRST bit in ICCR1 to apply an IIC reset or an internal reset.

Note: Do not set the RS bit to 1 while issuing a stop condition.

Note: If 1 (restart condition request) is written to the RS bit in slave mode, the restart condition is not issued, but the RS bit remains set to 1. If the operating mode changes to master mode without the bit being cleared, a restart condition might be issued.

SP bit (Stop Condition Issuance Request)

The SP bit requests that a stop condition be issued in master mode. When this bit is set to 1, a stop condition is issued when the BBSY flag is set to 1 (bus busy state) and the MST bit is set to 1 (master mode). For details on this function, see [section 39.11. Start, Restart, and Stop Condition Issuing Function](#).

[Setting condition]

- When 1 is written to the SP bit with both the BBSY flag and the MST bit in ICCR2 set to 1.

[Clearing conditions]

- When 0 is written to the SP bit
- When a stop condition is issued (a stop condition is detected)
- When the AL (arbitration-lost) flag in ICSR2 is set to 1
- When a start condition and a restart condition are detected
- When 1 is written to the IICRST bit in ICCR1 to apply an IIC reset or an internal reset.

Note: Writing to the SP bit is not possible while the BBSY flag is 0 (bus free state).

Note: Do not set the SP bit to 1 while a restart condition is being issued.

TRS bit (Transmit/Receive Mode)

The TRS bit indicates transmit or receive mode. The IIC is in receive mode when the TRS bit is 0 and in transmit mode when the bit is 1. The combination of this bit and the MST bit indicates the IIC operating mode.

The value of the TRS bit automatically changes to 1 for transmit mode or 0 for receive mode when a start condition is issued or detected and the R/W# bit is set. Although writing to the TRS bit is possible when the MTWP bit in ICMR1 is set to 1, writing to this bit is not required during normal usage.

[Setting conditions]

- When a start condition is issued normally because of a start condition request (when a start condition is detected with the ST bit set to 1)
- When a restart condition is issued normally because of a restart condition request (when a restart condition is detected with the RS bit set to 1)
- When the R/W# bit appended to the slave address is set to 0 in master mode
- When the address received in slave mode matches the address enabled in ICSE, with the R/W# bit set to 1
- When 1 is written to the TRS bit with the MTWP bit in ICMR1 set to 1.

[Clearing conditions]

- When a stop condition is detected
- When the AL (arbitration-lost) flag in ICSR2 is set to 1
- When the R/W# bit appended to the slave address is set to 1 in master mode
- In slave mode, on a match between the received address and the address enabled in ICSE when the value of the received R/W# bit is 0, including when the received address is the general call address
- In slave mode, when a restart condition is detected (a restart condition is detected with ICCR2.BBSY = 1 and ICCR2.MST = 0)
- When 0 is written to the TRS bit with the MTWP bit in ICMR1 set to 1
- When 1 is written to the IICRST bit in ICCR1 to apply an IIC reset or an internal reset.

MST bit (Master/Slave Mode)

The MST bit indicates master or slave mode. The IIC is in slave mode when the MST bit is 0 and is in master mode when the bit is 1. The combination of this bit and the TRS bit indicates the IIC operating mode.

The value of the MST bit automatically changes to 1 for master mode or 0 for slave mode when a start condition is issued or a stop condition is issued or detected. Although writing to the MST bit is possible when the MTWP bit in ICMR1 is set to 1, writing to this bit is not required during normal usage.

[Setting conditions]

- When a start condition is issued normally because of a start condition request (when a start condition is detected with the ST bit set to 1)
- When 1 is written to the MST bit with the MTWP bit in ICMR1 set to 1.

[Clearing conditions]

- When a stop condition is detected
- When the AL (arbitration-lost) flag in ICSR2 is set to 1
- When 0 is written to the MST bit with the MTWP bit in ICMR1 set to 1
- When 1 is written to the IICRST bit in ICCR1 to apply an IIC reset or an internal reset.

BBSY flag (Bus Busy Detection Flag)

The BBSY flag indicates whether the I²C bus is occupied (bus busy state) or released (bus free state). The flag is set to 1 when the SDAn line changes from high to low when the SCLn line is high, assuming that a start condition was issued. The flag is set to 0 if a start condition is not detected for the bus free time (ICBRL setting), assuming that a stop condition was issued.

[Setting condition]

- When a start condition is detected.

[Clearing conditions]

- When a start condition is not detected for the bus free time (ICBRL setting) after detecting a stop condition
- When 1 is written to the IICRST bit in ICCR1 with the ICE bit in ICCR1 set to 0 (IIC reset).

39.2.3 ICMR1 : I²C Bus Mode Register 1

Base address: IICn = 0x4025_E000 + 0x0100 × n (n = 0 to 2)
IICn_NS = 0x5025_E000 + 0x0100 × n (n = 0 to 2)

Offset address: 0x02

Bit position:	7	6	5	4	3	2	1	0
Bit field:	MTWP	CKS[2:0]			BCWP	BC[2:0]		
Value after reset:	0	0	0	0	1	0	0	0

Bit	Symbol	Function	R/W
2:0	BC[2:0]	Bit Counter 0 0 0: 9 bits 0 0 1: 2 bits 0 1 0: 3 bits 0 1 1: 4 bits 1 0 0: 5 bits 1 0 1: 6 bits 1 1 0: 7 bits 1 1 1: 8 bits	R/W ^{*1}
3	BCWP	BC Write Protect This bit is read as 1. 0: Write enable BC[2:0] bits 1: Write protect BC[2:0] bits	W ^{*1}
6:4	CKS[2:0]	Internal Reference Clock Select Select the internal reference clock source (IICφ) for the IIC. IICφ = (PCLKB / 2 ^{CKS[2:0]}) clock	R/W
7	MTWP	MST/TRS Write Protect 0: Write protect MST and TRS bits in ICCR2 1: Write enable MST and TRS bits in ICCR2	R/W

Note: S-TYPE3, P-TYPE3

Note 1. Rewrite the BC[2:0] bits and set the BCWP bit to 0 at the same time.

BC[2:0] bits (Bit Counter)

The BC[2:0] bits function as a counter indicating the number of bits remaining to be transferred on detection of a rising edge on the SCLn line. Although BC[2:0] are read/write bits, it is not required to access these bits under normal conditions.

To write to these bits, specify the number of bits to be transferred plus one, for an additional acknowledge bit, between transferred frames when the SCLn line is at a low level. The value in the BC[2:0] bits returns to 000b at the end of a data transfer, including the acknowledge bit, or when a start or restart condition is detected.

39.2.4 ICMR2 : I²C Bus Mode Register 2

Base address: IICn = 0x4025_E000 + 0x0100 × n (n = 0 to 2)
IICn_NS = 0x5025_E000 + 0x0100 × n (n = 0 to 2)

Offset address: 0x03

Bit position:	7	6	5	4	3	2	1	0
Bit field:	DLCS	SDDL[2:0]			—	TMOH	TMOL	TMOS
Value after reset:	0	0	0	0	0	1	1	0

Bit	Symbol	Function	R/W
0	TMOS	Timeout Detection Time Select 0: Select long mode 1: Select short mode	R/W
1	TMOL	Timeout L Count Control 0: Disable count while SCLn line is low 1: Enable count while SCLn line is low	R/W
2	TMOH	Timeout H Count Control 0: Disable count while SCLn line is high 1: Enable count while SCLn line is high	R/W
3	—	This bit is read as 0. The write value should be 0.	R/W
6:4	SDDL[2:0]	SDA Output Delay Counter 0 0 0: No output delay 0 0 1: 1 IICφ cycle (When ICMR2.DLCS = 0 (IICφ)) 1 or 2 IICφ cycles (When ICMR2.DLCS = 1 (IICφ/2)) 0 1 0: 2 IICφ cycles (When ICMR2.DLCS = 0 (IICφ)) 3 or 4 IICφ cycles (When ICMR2.DLCS = 1 (IICφ/2)) 0 1 1: 3 IICφ cycles (When ICMR2.DLCS = 0 (IICφ)) 5 or 6 IICφ cycles (When ICMR2.DLCS = 1 (IICφ/2)) 1 0 0: 4 IICφ cycles (When ICMR2.DLCS = 0 (IICφ)) 7 or 8 IICφ cycles (When ICMR2.DLCS = 1 (IICφ/2)) 1 0 1: 5 IICφ cycles (When ICMR2.DLCS = 0 (IICφ)) 9 or 10 IICφ cycles (When ICMR2.DLCS = 1 (IICφ/2)) 1 1 0: 6 IICφ cycles (When ICMR2.DLCS = 0 (IICφ)) 11 or 12 IICφ cycles (When ICMR2.DLCS = 1 (IICφ/2)) 1 1 1: 7 IICφ cycles (When ICMR2.DLCS = 0 (IICφ)) 13 or 14 IICφ cycles (When ICMR2.DLCS = 1 (IICφ/2))	R/W
7	DLCS	SDA Output Delay Clock Source Select 0: Select internal reference clock (IICφ) as the clock source for SDA output delay counter 1: Select internal reference clock divided by 2 (IICφ/2) as the clock source for SDA output delay counter ^{*1}	R/W

Note: S-TYPE3, P-TYPE3

Note 1. The setting DLCS = 1 (IICφ/2) is only valid when SCL is low. When SCL is high, the DLCS = 1 setting becomes invalid and the clock source becomes the internal reference clock (IICφ).

TMOS bit (Timeout Detection Time Select)

The TMOS bit selects long or short mode for the timeout detection time when the timeout function is enabled (ICFER.TMOE = 1). When this bit is set to 0, long mode is selected. When it is set to 1, short mode is selected. In long mode, the timeout detection internal counter functions as a 16 bit-counter. In short mode, the counter functions as a

14-bit counter. While the SCLn line is in the state that enables this counter as specified in the TMOH and TMOL bits, the counter counts up in synchronization with the internal reference clock (IIC ϕ) as a count source. For details on this function, see [section 39.12.1. Timeout Function](#).

TMOL bit (Timeout L Count Control)

The TMOL bit enables or disables up-counting on the internal counter of the timeout function while the SCLn line is held low and the timeout function is enabled (ICFER.TMOE = 1).

TMOH bit (Timeout H Count Control)

The TMOH bit enables or disables up-counting on the internal counter of the timeout function while the SCLn line is held high and the timeout function is enabled (ICFER.TMOE = 1).

SDDL[2:0] bits (SDA Output Delay Counter)

The SDDL[2:0] bits can be used to delay the SDA output. This counter works with the clock source selected in the DLCS bit. This setting can be used for all types of SDA output, including transmission of the acknowledge bit.

Set the SDA output delay to meet the I²C bus standard for the data enable time/acknowledge enable time,^{*1} or the SMBus standard, within [data hold time (300 ns or more + the SCL clock low-level period) - the data setup time (250 ns)]. If a value outside the standard is set, communication between devices might malfunction or falsely indicate a start or stop condition, depending on the bus state.

For details on this function, see [section 39.5. SDA Output Delay Function](#).

Note 1. Data enable time/acknowledge enable time.

3,450 ns for up to 100 kbps: Standard mode (Sm)

900 ns for up to 400 kbps: Fast mode (Fm)

450 ns for up to 1 Mbps: Fast-mode Plus (Fm+)

39.2.5 ICMR3 : I²C Bus Mode Register 3

Base address: IICn = 0x4025_E000 + 0x0100 × n (n = 0 to 2)
IICn_NS = 0x5025_E000 + 0x0100 × n (n = 0 to 2)

Offset address: 0x04

Bit position:	7	6	5	4	3	2	1	0
Bit field:	SMBS	WAIT	RDRF S	ACKW P	ACKB T	ACKB R	NF[1:0]	

Value after reset: 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
1:0	NF[1:0]	Noise Filter Stage Select 0 0: Filter out noise of up to 1 IIC ϕ cycle (single-stage filter) 0 1: Filter out noise of up to 2 IIC ϕ cycles (2-stage filter) 1 0: Filter out noise of up to 3 IIC ϕ cycles (3-stage filter) 1 1: Filter out noise of up to 4 IIC ϕ cycles (4-stage filter)	R/W
2	ACKBR	Receive Acknowledge 0: 0 received as the acknowledge bit (ACK reception) 1: 1 received as the acknowledge bit (NACK reception)	R
3	ACKBT	Transmit Acknowledge 0: Send 0 as the acknowledge bit (ACK transmission) 1: Send 1 as the acknowledge bit (NACK transmission)	R/W ^{*1}
4	ACKWP	ACKBT Write Protect 0: Write protect ACKBT bit 1: Write enable ACKBT bit	R/W

Bit	Symbol	Function	R/W
5	RDRFS	RDRF Flag Set Timing Select Low-hold is released by writing to ACKBT. 0: Set the RDRF flag on the rising edge of the 9th SCL clock cycle. The SCLn line is not held low on the falling edge of the 8th clock cycle. 1: Set the RDRF flag on the rising edge of the 8th SCL clock cycle. The SCLn line is held low on the falling edge of the 8th clock cycle.	R/W ²
6	WAIT	Low-hold is released by reading ICDRR. 0: No wait (The SCLn line is not held low during the period between the 9th clock cycle and the 1st clock cycle.) 1: Wait (The SCLn line is held low during the period between the 9th clock cycle and the 1st clock cycle.)	R/W ²
7	SMBS	SMBus/I ² C Bus Select 0: Select I ² C Bus 1: Select SMBus	R/W

Note: S-TYPE3, P-TYPE3

Note 1. Write to the ACKBT bit only while the ACKWP bit is already 1. If the application writes 1 to the ACKWP and ACKBT bits at the same time, the ACKBT bit is not set to 1.

Note 2. The WAIT and RDRFS bits are only valid in receive mode (invalid in transmit mode).

NF[1:0] bits (Noise Filter Stage Select)

The NF[1:0] bits select the number of stages in the digital noise filter. For details on this function, see [section 39.6. Digital Noise Filter Circuits](#)

Note: Set the noise range to be filtered within a range less than the SCLn line high- or low-level period. If the noise range is set to a value of [SCL clock width: high- or low-level period, whichever is shorter] - [1.5 internal reference clock (IIC ϕ) cycles + analog noise filter: 120 ns (reference values)] or more, the SCL clock is regarded as noise, which might prevent the IIC from operating normally.

ACKBR bit (Receive Acknowledge)

The ACKBR bit stores the acknowledge bit information received from the receive device in transmit mode.

[Setting condition]

- When 1 is received as the acknowledge bit with the TRS bit in ICCR2 set to 1.

[Clearing conditions]

- When 0 is received as the acknowledge bit with the TRS bit in ICCR2 set to 1
- When 1 is written to the IICRST bit in ICCR1 while the ICE bit in ICCR1 is 0 (IIC reset).

ACKBT bit (Transmit Acknowledge)

The ACKBT bit sets the acknowledge bit to be sent in receive mode

[Setting condition]

- When 1 is written to this bit with the ACKWP bit set to 1.

[Clearing conditions]

- When 0 is written to this bit with the ACKWP bit set to 1
- When stop condition request is detected with the SP bit in ICCR2 set to 1
- When 1 is written to the IICRST bit in ICCR1 while the ICE bit in ICCR1 is 0 (IIC reset).

ACKWP bit (ACKBT Write Protect)

The ACKWP bit controls write enabling of the ACKBT bit.

RDRFS bit (RDRF Flag Set Timing Select)

The RDRFS bit selects the RDRF flag set timing in receive mode and also selects whether to hold the SCLn line low on the falling edge of the 8th SCL clock cycle.

When the RDRFS bit is 0, the SCLn line is not held low on the falling edge of the 8th SCL clock cycle, and the RDRF flag is set to 1 on the rising edge of the 9th SCL clock cycle.

When the RDRFS bit is 1, the RDRF flag is set to 1 on the rising edge of the 8th SCL clock cycle, and the SCLn line is held low on the falling edge of the 8th SCL clock cycle. The low-hold of the SCLn line is released by a write to the ACKBT bit.

After data is received with this setting, the SCLn line is automatically held low before the acknowledge bit is sent. This enables processing to send ACK (ACKBT = 0) or NACK (ACKBT = 1), based on the receive data.

WAIT bit (WAIT)

The WAIT bit controls whether to force a low-hold between the ninth SCL clock cycle and the first SCL clock cycle, until the receive data buffer (ICDRR) is completely read each time single-byte data is received in receive mode.

When the WAIT bit is 0, the receive operation continues without a low-hold between the ninth and the first SCL clock cycle. When both the RDRFS and WAIT bits are 0, continuous receive operation is enabled with the double buffer.

When the WAIT bit is 1, the SCLn line is held low from the falling edge of the ninth clock cycle until the ICDRR value is read each time single-byte data is received. This enables receive operation in byte units.

Note: When the value of the WAIT bit is to be read, always read ICDRR first.

SMBS bit (SMBus/I²C Bus Select)

Setting the SMBS bit to 1 selects the SMBus and enables the HOAE bit in ICSEI.

39.2.6 ICFER : I²C Bus Function Enable Register

Base address: IICn = 0x4025_E000 + 0x0100 × n (n = 0 to 2)
IICn_NS = 0x5025_E000 + 0x0100 × n (n = 0 to 2)

Offset address: 0x05

Bit position:	7	6	5	4	3	2	1	0
Bit field:	FMPE	SCLE	NFE	NACK E	SALE	NALE	MALE	TMOE
Value after reset:	0	1	1	1	0	0	1	0

Bit	Symbol	Function	R/W
0	TMOE	Timeout Function Enable 0: Disable 1: Enable	R/W
1	MALE	Master Arbitration-Lost Detection Enable 0: Disable the arbitration-lost detection function and disable automatic clearing of the MST and TRS bits in ICCR2 when arbitration is lost 1: Enable the arbitration-lost detection function and enable automatic clearing of the MST and TRS bits in ICCR2 when arbitration is lost	R/W
2	NALE	NACK Transmission Arbitration-Lost Detection Enable 0: Disable 1: Enable	R/W
3	SALE	Slave Arbitration-Lost Detection Enable 0: Disable 1: Enable	R/W
4	NACK E	NACK Reception Transfer Suspension Enable 0: Do not suspend transfer operation during NACK reception (disable transfer suspension) 1: Suspend transfer operation during NACK reception (enable transfer suspension)	R/W
5	NFE	Digital Noise Filter Circuit Enable 0: Do not use the digital noise filter circuit 1: Use the digital noise filter circuit	R/W
6	SCLE	SCL Synchronous Circuit Enable 0: Do not use the SCL synchronous circuit 1: Use the SCL synchronous circuit	R/W

Bit	Symbol	Function	R/W
7	FMPE ^{*1}	Fast-Mode Plus Enable 0: Do not use the Fm+ slope control circuit for the SCLn and SDA _n pins 1: Use the Fm+ slope control circuit for the SCLn and SDA _n pins.	R/W

Note: S-TYPE3, P-TYPE3

Note 1. The Fast-mode Plus enable bit (FMPE) is supported by IIC0 (SCL0_A, SDA0_A), IIC1 (SCL1_A, SDA1_A) and IIC2 (SCL2_A, SDA2_A). Bit [7] is the reserved bit in the not supported channel.

TMOE bit (Timeout Function Enable)

The TMOE bit enables or disables the timeout function. For details on this function, see [section 39.12.1. Timeout Function](#).

MALE bit (Master Arbitration-Lost Detection Enable)

The MALE bit specifies whether to use the arbitration-lost detection function in master mode. Normally, set this bit to 1.

NALE bit (NACK Transmission Arbitration-Lost Detection Enable)

The NALE bit specifies whether to cause arbitration to be lost when ACK is detected during transmission of NACK in receive mode, for example when slaves with the same address exist on the bus or when two or more masters select the same slave device simultaneously with a different number of receive bytes.

SALE bit (Slave Arbitration-Lost Detection Enable)

The SALE bit specifies whether to cause arbitration to be lost when a value different from the value being transmitted is detected on the bus in slave transmit mode, for example when slaves with the same address exist on the bus or when a mismatch with the transmit data occurs because of noise.

NACKE bit (NACK Reception Transfer Suspension Enable)

The NACKE bit specifies whether to continue or discontinue the transfer operation when NACK is received in transmit mode. Normally, set this bit to 1.

When NACK is received with the NACKE bit set to 1, the next transfer operation is suspended. When the NACKE bit is 0, the next transfer operation continues regardless of the received acknowledge content.

For details, see [section 39.9.2. NACK Reception Transfer Suspension Function](#).

SCLE bit (SCL Synchronous Circuit Enable)

The SCLE bit specifies whether to synchronize the SCL clock with the SCL input clock. Normally, set this bit to 1.

When the SCLE bit is set to 0 (no SCL synchronous circuit used), the IIC does not synchronize the SCL clock with the SCL input clock. With this setting, the IIC outputs the SCL clock at the transfer rate set in ICBRH and ICBRL, regardless of the SCLn line state. For this reason, if the bus load of the I²C bus line is much larger than the specification value, or if the SCL clock output overlaps in multiple masters, a short-cycle SCL clock that does not meet the specification might be output. When no SCL synchronous circuit is used, it also affects the issuing of the start, restart, and stop conditions, and the continuous output of extra SCL clock cycles.

Do not set this bit to 0 except when checking the output of the set transfer rate.

FMPE bit (Fast-Mode Plus Enable)

The FMPE bit specifies whether to use a slope control circuit for Fast-mode Plus (Fm+).

When this bit is set to 1, a slope control circuit conforming to the I²C bus Fast-mode Plus (Fm+) standard (tof) is selected. When this bit is set to 0, a slope control circuit conforming to the I²C bus Standard-mode (Sm) and Fast-mode (Fm) standards (tof) is selected.

Set this bit to 1 when using transmission rates up to 1 Mbps (Fast-mode Plus (Fm+) standard). Set it to 0 when using other transmission rates (up to 100 kbps (Sm) or up to 400 kbps (Fm)) or for SMBus (10 to 100 kbps).

39.2.7 ICSESR : I²C Bus Status Enable Register

Base address: IICn = 0x4025_E000 + 0x0100 × n (n = 0 to 2)
IICn_NS = 0x5025_E000 + 0x0100 × n (n = 0 to 2)

Offset address: 0x06

Bit position:	7	6	5	4	3	2	1	0
Bit field:	HOAE	—	DIDE	—	GCAE	SAR2E	SAR1E	SAR0E
Value after reset:	0	0	0	0	1	0	0	1

Bit	Symbol	Function	R/W
0	SAR0E	Slave Address Register 0 Enable 0: Disable slave address in SARL0 and SARU0 1: Enable slave address in SARL0 and SARU0	R/W
1	SAR1E	Slave Address Register 1 Enable 0: Disable slave address in SARL1 and SARU1 1: Enable slave address in SARL1 and SARU1	R/W
2	SAR2E	Slave Address Register 2 Enable 0: Disable slave address in SARL2 and SARU2 1: Enable slave address in SARL2 and SARU2	R/W
3	GCAE	General Call Address Enable 0: Disable general call address detection 1: Enable general call address detection	R/W
4	—	This bit is read as 0. The write value should be 0.	R/W
5	DIDE	Device-ID Address Detection Enable 0: Disable device-ID address detection 1: Enable device-ID address detection	R/W
6	—	This bit is read as 0. The write value should be 0.	R/W
7	HOAE	Host Address Enable 0: Disable host address detection 1: Enable host address detection	R/W

Note: S-TYPE3, P-TYPE3

SARyE bit (Slave Address Register y Enable) (y = 0 to 2)

The SARyE bit enables or disables the received slave address and the slave address set in SARLy and SARUy.

When this bit is set to 1, the slave address set in SARLy and SARUy is enabled and is compared with the received slave address. When this bit is set to 0, the slave address set in SARLy and SARUy is disabled and is ignored even if it matches the received slave address.

GCAE bit (General Call Address Enable)

The GCAE bit specifies whether to ignore the general call address (0000 000b + 0 [W]: All 0) when it is received.

When this bit is set to 1, if the received slave address matches the general call address, the IIC recognizes the received slave address as the general call address independently of the slave addresses set in SARLy and SARUy (y = 0 to 2) and performs the data receive operation. When this bit is set to 0, the received slave address is ignored even if it matches the general call address.

DIDE bit (Device-ID Address Detection Enable)

The DIDE bit specifies whether to recognize and execute the device-ID address when a device ID (1111 100b) is received in the first frame after a start or restart condition is detected.

When this bit is set to 1, if the received first frame matches the device ID, the IIC recognizes that the device-ID address was received. When the next R/W# bit is 0 (W), the IIC recognizes the second and the subsequent frames as slave addresses and continues the receive operation. When this bit is set to 0, the IIC ignores the received first frame even if it matches the device-ID address, and it recognizes the first frame as a normal slave address.

For details on this function, see [section 39.7.3. Device-ID Address Detection](#).

HOAE bit (Host Address Enable)

The HOAE bit specifies whether to ignore the received host address (0001 000b) when the SMBS bit in ICMR3 is 1.

When this bit is set to 1 while the SMBS bit in ICMR3 is 1, if the received slave address matches the host address, the IIC recognizes the received slave address as the host address independently of the slave addresses set in SARLy and SARUy (y = 0 to 2) and performs the receive operation.

When the SMBS bit in ICMR3 or the HOAE bit is set to 0, the received slave address is ignored even if it matches the host address.

39.2.8 ICIER : I²C Bus Interrupt Enable Register

Base address: IICn = 0x4025_E000 + 0x0100 × n (n = 0 to 2)
IICn_NS = 0x5025_E000 + 0x0100 × n (n = 0 to 2)

Offset address: 0x07

Bit position:	7	6	5	4	3	2	1	0
Bit field:	TIE	TEIE	RIE	NAKIE	SPIE	STIE	ALIE	TMOIE

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	TMOIE	Timeout Interrupt Request Enable 0: Disable timeout interrupt (TMOI) request 1: Enable timeout interrupt (TMOI) request	R/W
1	ALIE	Arbitration-Lost Interrupt Request Enable 0: Disable arbitration-lost interrupt (ALI) request 1: Enable arbitration-lost interrupt (ALI) request	R/W
2	STIE	Start Condition Detection Interrupt Request Enable 0: Disable start condition detection interrupt (STI) request 1: Enable start condition detection interrupt (STI) request	R/W
3	SPIE	Stop Condition Detection Interrupt Request Enable 0: Disable stop condition detection interrupt (SPI) request 1: Enable stop condition detection interrupt (SPI) request	R/W
4	NAKIE	NACK Reception Interrupt Request Enable 0: Disable NACK reception interrupt (NAKI) request 1: Enable NACK reception interrupt (NAKI) request	R/W
5	RIE	Receive Data Full Interrupt Request Enable 0: Disable receive data full interrupt (IICn_RXI) request 1: Enable receive data full interrupt (IICn_RXI) request	R/W
6	TEIE	Transmit End Interrupt Request Enable 0: Disable transmit end interrupt (IICn_TEI) request 1: Enable transmit end interrupt (IICn_TEI) request	R/W
7	TIE	Transmit Data Empty Interrupt Request Enable 0: Disable transmit data empty interrupt (IICn_TXI) request 1: Enable transmit data empty interrupt (IICn_TXI) request	R/W

Note: S-TYPE3, P-TYPE3

TMOIE bit (Timeout Interrupt Request Enable)

The TMOIE bit enables or disables timeout interrupt (TMOI) requests when the TMOF flag in ICSR2 is 1. To cancel a TMOI interrupt request, set the TMOF flag or the TMOIE bit to 0.

ALIE bit (Arbitration-Lost Interrupt Request Enable)

The ALIE bit enables or disables arbitration-lost interrupt (ALI) requests when the AL flag in ICSR2 is 1. To cancel an ALI interrupt request, set the AL flag or the ALIE bit to 0.

STIE bit (Start Condition Detection Interrupt Request Enable)

The STIE bit enables or disables start condition detection interrupt (STI) requests when the START flag in ICSR2 is 1. To cancel an STI interrupt request, set the START flag or the STIE bit to 0.

SPIE bit (Stop Condition Detection Interrupt Request Enable)

The SPIE bit enables or disables stop condition detection interrupt (SPI) requests when the STOP flag in ICSR2 is 1. To cancel an SPI interrupt request, set the STOP flag or the SPIE bit to 0.

NAKIE bit (NACK Reception Interrupt Request Enable)

The NAKIE bit enables or disables NACK reception interrupt (NAKI) requests when the NACKF flag in ICSR2 is 1. To cancel an NAKI interrupt request, set the NACKF flag or the NAKIE bit to 0.

RIE bit (Receive Data Full Interrupt Request Enable)

The RIE bit enables or disables receive data full interrupt (IICn_RXI) requests when the RDRF flag in ICSR2 is 1.

TEIE bit (Transmit End Interrupt Request Enable)

The TEIE bit enables or disables transmit end interrupt (IICn_TEI) requests when the TEND flag in ICSR2 is 1. To cancel an IICn_TEI interrupt request, set the TEND flag or the TEIE bit to 0.

TIE bit (Transmit Data Empty Interrupt Request Enable)

The TIE bit enables or disables transmit data empty interrupt (IICn_TXI) requests when the TDRE flag in ICSR2 is 1.

39.2.9 ICSR1 : I²C Bus Status Register 1

Base address: IICn = 0x4025_E000 + 0x0100 × n (n = 0 to 2)
IICn_NS = 0x5025_E000 + 0x0100 × n (n = 0 to 2)

Offset address: 0x08

Bit position:	7	6	5	4	3	2	1	0
Bit field:	HOA	—	DID	—	GCA	AAS2	AAS1	AAS0

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	AAS0	Slave Address 0 Detection Flag 0: Slave address 0 not detected 1: Slave address 0 detected	R/(W) ^{*1}
1	AAS1	Slave Address 1 Detection Flag 0: Slave address 1 not detected 1: Slave address 1 detected	R/(W) ^{*1}
2	AAS2	Slave Address 2 Detection Flag 0: Slave address 2 not detected 1: Slave address 2 detected	R/(W) ^{*1}
3	GCA	General Call Address Detection Flag 0: General call address not detected 1: General call address detected	R/(W) ^{*1}
4	—	This bit is read as 0. The write value should be 0.	R/W
5	DID	Device-ID Address Detection Flag This bit is set to 1 when the first frame received immediately after a start condition is detected matches a value of (device ID (1111 100b) + 0[W]). 0: Device-ID command not detected 1: Device-ID command detected	R/(W) ^{*1}
6	—	This bit is read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
7	HOA	Host Address Detection Flag This bit is set to 1 when the received slave address matches the host address (0001 000b). 0: Host address not detected 1: Host address detected	R/(W)*1

Note: S-TYPE3, P-TYPE3

Note 1. Only 0 can be written to clear the flag.

AASy flag (Slave Address y Detection flag) (y = 0 to 2)

The AASy flag indicates whether slave address y was detected.

[Setting conditions]

For 7-bit address format (SARUy.FS = 0):

- When the received slave address matches the SVA[6:0] value in SARLy, with the SARyE bit in ICSEr set to 1 (slave address y detection enabled).

The AASy flag is set to 1 on the rising edge of the ninth SCL clock cycle in the frame.

For 10-bit address format (SARUy.FS = 1):

- When the received slave address matches a value of (11110b + SVA[1:0] in SARUy), and the subsequent address matches the SARLy value, with the SARyE bit in ICSEr set to 1 (slave address y detection enabled).

The AASy flag is set to 1 on the rising edge of the ninth SCL clock cycle in the frame.

[Clearing conditions]

- When 0 is written to the AASy flag after reading AASy = 1
- When a stop condition is detected
- When 1 is written to the IICRST bit in ICCR1 to apply an IIC reset or an internal reset.

For 7-bit address format (SARUy.FS = 0):

- When the received slave address does not match the SVA[6:0] value in SARLy, with the SARyE bit in ICSEr set to 1 (slave address y detection enabled).

The AASy flag is set to 0 on the rising edge of the ninth SCL clock cycle in the frame.

For 10-bit address format (SARUy.FS = 1):

- When the received slave address does not match a value of (11110b + SVA[1:0] in SARUy), with the SARyE bit in ICSEr set to 1 (slave address y detection enabled)
The AASy flag is set to 0 on the rising edge of the ninth SCL clock cycle in the frame.
- When the received slave address matches a value of (11110b + SVA[1:0] in SARUy), and the subsequent address does not match the SARLy value, with the SARyE bit in ICSEr set to 1 (slave address y detection enabled).
The AASy flag is set to 0 on the rising edge of the ninth SCL clock cycle in the frame.

GCA flag (General Call Address Detection Flag)

The GCA flag indicates whether the general call address was detected.

[Setting condition]

- When the received slave address matches the general call address (0000 000b + 0 [W]), with the GCAE bit in ICSEr set to 1 (general call address detection enabled).

The GCA flag is set to 1 on the rising edge of the ninth SCL clock cycle in the frame.

[Clearing conditions]

- When 0 is written to the GCA flag after reading GCA = 1
- When a stop condition is detected
- When the received slave address does not match the general call address (0000 000b + 0 [W]), with the GCAE bit in ICSEr set to 1 (general call address detection enabled)
The GCA flag is set to 0 on the rising edge of the ninth SCL clock cycle in the frame.

- When 1 is written to the IICRST bit in ICCR1 to apply an IIC reset or an internal reset.

DID flag (Device-ID Address Detection Flag)

The DID flag indicates whether the device-ID address was detected.

[Setting condition]

- When the first frame received immediately after a start or restart condition is detected matches a value of (device ID (1111 100b) + 0 [W]), with the DIDE bit in IC SER set to 1 (device-ID address detection enabled).
The DID flag is set to 1 on the rising edge of the ninth SCL clock cycle in the frame.

[Clearing conditions]

- When 0 is written to the DID flag after reading DID = 1
- When a stop condition is detected
- When the first frame received immediately after a start or restart condition is detected does not match a value of (device ID (1111 100b)), with the DIDE bit in IC SER set to 1 (device-ID address detection enabled)
The DID flag is set to 0 on the rising edge of the ninth SCL clock cycle in the frame.
- When the first frame received immediately after a start or restart condition is detected matches a value of (device ID (1111 100b) + 0 [W]), and the second frame does not match any slave address from 0 to 2, with the DIDE bit in IC SER set to 1 (device-ID address detection enabled)
The DID flag is set to 0 on the rising edge of the ninth SCL clock cycle in the frame.
- When 1 is written to the IICRST bit in ICCR1 to apply an IIC reset or an internal reset.

HOA flag (Host Address Detection Flag)

The HOA flag indicates whether the host address was detected.

[Setting condition]

- When the received slave address matches the host address (0001 000b), with the HOAE bit in IC SER set to 1 (host address detection enabled).
The HOA flag is set to 1 on the rising edge of the ninth SCL clock cycle in the frame.

[Clearing conditions]

- When 0 is written to the HOA flag after reading HOA = 1
- When a stop condition is detected
- When the received slave address does not match the host address (0001 000b), with the HOAE bit in IC SER set to 1 (host address detection enabled)
The HOA flag is set to 0 on the rising edge of the ninth SCL clock cycle in the frame.
- When 1 is written to the IICRST bit in ICCR1 to apply an IIC reset or an internal reset.

39.2.10 ICSR2 : I²C Bus Status Register 2

Base address: IICn = 0x4025_E000 + 0x0100 × n (n = 0 to 2)
IICn_NS = 0x5025_E000 + 0x0100 × n (n = 0 to 2)

Offset address: 0x09

Bit position:	7	6	5	4	3	2	1	0
Bit field:	TDRE	TEND	RDRF	NACK F	STOP	START	AL	TMOF
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	TMOF	Timeout Detection Flag 0: Timeout not detected 1: Timeout detected	R/(W) ^{*1}

Bit	Symbol	Function	R/W
1	AL	Arbitration-Lost Flag 0: Arbitration not lost 1: Arbitration lost	R/(W) ^{*1}
2	START	Start Condition Detection Flag 0: Start condition not detected 1: Start condition detected	R/(W) ^{*1}
3	STOP	Stop Condition Detection Flag 0: Stop condition not detected 1: Stop condition detected	R/(W) ^{*1}
4	NACKF	NACK Detection Flag 0: NACK not detected 1: NACK detected	R/(W) ^{*1}
5	RDRF	Receive Data Full Flag 0: ICDRR contains no receive data 1: ICDRR contains receive data	R/(W) ^{*1}
6	TEND	Transmit End Flag 0: Data being transmitted 1: Data transmit complete	R/(W) ^{*1}
7	TDRE	Transmit Data Empty Flag 0: ICDRT contains transmit data 1: ICDRT contains no transmit data	R

Note: S-TYPE3, P-TYPE3

Note 1. Only 0 can be written, to clear the flag.

TMOF flag (Timeout Detection Flag)

The TMOF flag is set to 1 when the IIC detects a timeout because the SCLn line state remains unchanged for the set period.

[Setting condition]

- When the SCLn line state remains unchanged for the period specified in the ICMR2.TMOH, TMOL, and TMOS bits while the ICFER.TMOE bit is 1 (timeout function enabled) in master or in slave mode and the received slave address matches.

[Clearing conditions]

- When 0 is written to the TMOF flag after reading TMOF = 1
- When 1 is written to the IICRST bit in ICCR1 to apply an IIC reset or an internal reset.

AL flag (Arbitration-Lost Flag)

The AL flag indicates that bus mastership was lost in arbitration because of a bus conflict or some other reason when a start condition was issued or an address and data was transmitted. The IIC monitors the level on the SDAn line during transmission and, if the level on the line does not match the value of the bit being output, is set the value of the AL flag to 1 to indicate that the bus is occupied by another device.

The IIC can also set the AL flag to indicate the detection of arbitration loss during NACK transmission or during data transmission.

[Setting conditions]

When master arbitration-lost detection is enabled (ICFER.MALE = 1):

- When the internal SDA output state does not match the SDAn line level on the rising edge of the SCL clock except for the ACK period during data transmission in master transmit mode
- When a start condition is detected while the ST bit in ICCR2 is 1 (start condition requested) or the internal SDA output state does not match the SDAn line level
- When the ST bit in ICCR2 is 1 (start condition requested), with the BBSY flag in ICCR2 set to 1.

When NACK arbitration-lost detection is enabled (ICFER.NALE = 1):

- When the internal SDA output state does not match the SDAn line level on the rising edge of the SCL clock in the ACK period during NACK transmission in receive mode.

When slave arbitration-lost detection is enabled (ICFER.SALE = 1):

- When the internal SDA output state does not match the SDAn line level on the rising edge of the SCL clock, except for the ACK period during data transmission in slave transmit mode.

[Clearing conditions]

- When 0 is written to the AL flag after reading AL = 1
- When 1 is written to the IICRST bit in ICCR1 to apply an IIC reset or an internal reset.

Table 39.4 Relationship between arbitration-lost generation sources and arbitration-lost enable functions

ICFER			ICSR2	Error	Arbitration-lost generation source
MALE	NALE	SALE	AL		
1	x	x	1	Start condition issuance error	When internal SDA output state does not match SDAn line level when a start condition is detected, while the ST bit in ICCR2 is 1
			1	Transmit data mismatch	When ST in ICCR2 is set to 1 while BBSY in ICCR2 is 1
x	1	x	1	NACK transmission mismatch	When ACK is detected during transmission of NACK in master or slave receive mode
x	x	1	1	Transmit data mismatch	When transmit data does not match the bus state in slave transmit mode

x: Don't care

START flag (Start Condition Detection Flag)

The START flag indicates whether a start or restart condition was detected.

[Setting condition]

- When a start or restart condition is detected.

[Clearing conditions]

- When 0 is written to the START flag after reading START = 1
- When a stop condition is detected
- When 1 is written to the IICRST bit in ICCR1 to apply an IIC reset or an internal reset.

STOP flag (Stop Condition Detection Flag)

The STOP flag indicates whether a stop condition was detected.

[Setting condition]

- When a stop condition is detected.

[Clearing conditions]

- When 0 is written to the STOP flag after reading STOP = 1
- When 1 is written to the IICRST bit in ICCR1 to apply an IIC reset or an internal reset.

NACKF flag (NACK Detection Flag)

The NACKF flag indicates whether a NACK was detected.

[Setting condition]

- When acknowledge is not received (NACK received) from the receive device in transmit mode, with the NACKE bit in ICFER set to 1 (transfer suspension enabled).

[Clearing conditions]

- When 0 is written to the NACKF flag after reading NACKF = 1
- When 1 is written to the IICRST bit in ICCR1 to apply an IIC reset or an internal reset.

Note: When the NACKF flag is set to 1, the IIC suspends data transmission and reception. Writing to ICDRT in transmit mode or reading from ICDRR in receive mode with the NACKF flag set to 1 does not enable data transmit or receive operation. To restart data transmission or reception, set the NACKF flag to 0.

RDRF flag (Receive Data Full Flag)

The RDRF flag indicates whether the ICDRR contains receive data.

[Setting conditions]

- When receive data is transferred from ICDRS to ICDRR
The RDRF flag is set to 1 on the rising edge of the eighth or ninth SCL clock cycle (selected in the RDRFS bit in ICMR3).
- When the received slave address matches after a start or restart condition is detected with the TRS bit in ICCR2 set to 0.

[Clearing conditions]

- When 0 is written to the RDRF flag after reading RDRF = 1
- When data is read from ICDRR
- When 1 is written to the IICRST bit in ICCR1 to apply an IIC reset or an internal reset.

TEND flag (Transmit End Flag)

The TEND flag indicates completion of transmission.

[Setting condition]

- On the rising edge of the ninth SCL clock cycle while the TDRE flag is 1.

[Clearing conditions]

- When 0 is written to the TEND flag after reading TEND = 1
- When data is written to ICDRT
- When a stop condition is detected
- When 1 is written to the IICRST bit in ICCR1 to apply an IIC reset or an internal reset.

TDRE flag (Transmit Data Empty Flag)

The TDRE flag indicates whether the ICDRT contains transmit data.

[Setting conditions]

- When data is transferred from ICDRT to ICDRS and ICDRT becomes empty
- When the TRS bit in ICCR2 is set to 1
- When the received slave address matches while the TRS bit is 1.

[Clearing conditions]

- When data is written to ICDRT
- When the TRS bit in ICCR2 is set to 0
- When 1 is written to the IICRST bit in ICCR1 to apply an IIC reset or an internal reset.

Note: When the NACKF flag is set to 1 while the NACK bit in ICFER is 1, the IIC suspends data transmission and reception. In this case, if the TDRE flag is 0 (next transmit data written), data is transferred to the ICDRS register and the ICDRT register becomes empty on the rising edge of the 9th clock cycle, but the TDRE flag does not set to 1.

39.2.11 ICWUR : I²C Bus Wakeup Unit Register

Base address: IIC0WU = 0x4025_E014
IIC0WU_NS = 0x5025_E014

Offset address: 0x02

Bit position:	7	6	5	4	3	2	1	0
Bit field:	WUE	WUIE	WUF	WUACK	—	—	—	WUAF A

Value after reset: 0 0 0 1 0 0 0 0

Bit	Symbol	Function	R/W
0	WUAF A	Wakeup Analog Filter Additional Selection 0: Do not add the wakeup analog filter 1: Add the wakeup analog filter	R/W
3:1	—	These bits are read as 0. The write value should be 0.	R/W
4	WUACK	ACK Bit for Wakeup Mode Choice of four response modes in combination with ICCR1.IICRST and WUACK. See Table 39.5 .	R/W
5	WUF	Wakeup Event Occurrence Flag 0: Slave address not matching during wakeup 1: Slave address matching during wakeup	R/W
6	WUIE	Wakeup Interrupt Request Enable 0: Disable wakeup interrupt request (IIC0_WUI) 1: Enable wakeup interrupt request (IIC0_WUI)	R/W
7	WUE	Wakeup Function Enable 0: Disable wakeup function 1: Enable wakeup function	R/W

Note: S-TYPE3, P-TYPE3

Table 39.5 Wakeup mode

IICRST	WUACK	Operation mode	Description
0	0	Normal wakeup mode 1	ACK response on 9th SCL, and SCL low-hold after 9th SCL.
0	1	Normal wakeup mode 2	No ACK response immediately and SCL low-hold between 8th and 9th SCL. SCL low-hold release and ACK response on 9th SCL.
1	0	Command recovery mode	ACK response on 9th SCL and no SCL low-hold.
1	1	EEP response mode	NACK response on 9th SCL and no SCL low-hold.

WUF flag (Wakeup Event Occurrence Flag)

The WUF flag indicates whether the slave address is matching during wakeup.

[Setting condition]

- When PCLKB is supplied after the slave address of I²C bus is matched with the address whose enable bits of IC SER (except Device ID address) are set to 1 (enable), the bit is set.

[Clearing conditions]

- When 0 is written to the WUF flag after reading WUF = 1 (when WUSYF flag is set to 1).
- When ICCR1.ICE = 0 and IICRST = 1.

39.2.12 ICWUR2 : I²C Bus Wakeup Unit Register 2

Base address: IIC0WU = 0x4025_E014
IIC0WU_NS = 0x5025_E014

Offset address: 0x03

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	WUSY F	WUAS YF	WUSE N
Value after reset:	1	1	1	1	1	1	0	1

Bit	Symbol	Function	R/W
0	WUSEN	Wakeup Function Synchronous Enable 0: IIC asynchronous circuit enable 1: IIC synchronous circuit enable	R/W
1	WUASYF	Wakeup Function Asynchronous Operation Status Flag 0: IIC synchronous circuit enable condition 1: IIC asynchronous circuit enable condition	R
2	WUSYF	Wakeup Function Synchronous Operation Status Flag 0: IIC asynchronous circuit enable condition 1: IIC synchronous circuit enable condition	R
7:3	—	These bits are read as 1. The write value should be 1.	R/W

Note: S-TYPE3, P-TYPE3

WUSEN bit (Wakeup Function Synchronous Enable)

The WUSEN bit is used in combination with the WUASYF flag (or WUSYF flag) to switch between the PCLKB synchronous and asynchronous operation, when the wakeup effective function is enabled (ICWUR.WUE = 1).

The PCLKB operation switches from synchronous to asynchronous operation:

When the ICCR2.BBSY flag is 0, if 0 is written to the WUSEN bit while the WUASYF flag is 0, the reception occurs independently of the operation of PCLKB (with PCLKB stopped) after it switches to the PCLKB asynchronous operation, on wakeup event detection.

The PCLKB operation switches from asynchronous to synchronous operation:

- When 1 is written to the WUSEN bit, with the WUASYF flag at 1, when a wakeup event is detected. After writing 1, the WUASYF flag immediately becomes 0.
- When the stop condition is detected with a wakeup event undetected.

[Setting condition]

- When 1 is written to the WUSEN bit.
- ICCR1.ICE = 0 and IICRST = 1 (IIC reset)
- ICWUR.WUE = 0

[Clearing conditions]

- When 0 is written to the WUSEN bit.

WUASYF flag (Wakeup Function Asynchronous Operation Status Flag)

The WUASYF flag can place the IIC in PCLKB asynchronous operation when the wakeup effective function is enabled (ICWUR.WUE = 1).

[Setting condition]

- When the ICCR2.BBSY flag is 0, and the WUSEN bit is set to 0 with the ICWUR.WUE bit set to 1.

[Clearing conditions]

- When 1 is written to the WUSEN bit after detecting the wake-up event with ICWUR.WUE bit set to 1.

- When a stop condition is detected with WUSEN bit set to 1 before detecting the wake-up event with WUASY flag set to 1 with ICWUR.WUE bit set to 1.
- When you write 1 in the WUSEN bit with the WUASYF flag detected 1 and the wake-up event in the state of ICWUR.WUE = 1.
- ICCR1.ICE = 0 and ICCRST = 1 (ICC reset)
- ICWUR.WUE = 0.

WUSYF flag (Wakeup Function Synchronous Operation Status Flag)

It is shown that IIC is in the PCLKB synchronous operation at wake-up effective function (ICWUR.WUE = 1). This flag is a value in which the WUASYF flag is always reserved.

[Setting conditions]

- When 1 is written to the WUSEN bit after detecting the wake-up event with ICWUR.WUE bit set to 1 with WUSYF flag set to 0 with ICWUR.WUE bit set to 1.
- When a stop condition is detected with WUSEN bit set to 1 before detecting the wake-up event with WUSYF flag set to 0 with ICWUR.WUE bit set to 1.
- ICCR1.ICE = 0 and ICCRST = 1 (ICC reset)
- ICWUR.WUE = 0.

[Clearing condition]

- When the ICCR2.BBSY flag is 0 with the ICWUR.WUE bit set to 1 after writing 0 to the WUSEN bit.

39.2.13 SARLy : Slave Address Register Ly (y = 0 to 2)

Base address: IICn = 0x4025_E000 + 0x0100 × n (n = 0 to 2)
IICn_NS = 0x5025_E000 + 0x0100 × n (n = 0 to 2)

Offset address: 0x0A + 0x02 × y

Bit position:	7	6	5	4	3	2	1	0
Bit field:	SVA[6:0]							SVA0

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	SVA0	10-bit Address LSB Slave address setting.	R/W
7:1	SVA[6:0]	7-bit Address/10-bit Address Lower Bits Slave address setting.	R/W

Note: S-TYPE3, P-TYPE3

SVA0 bit (10-bit Address LSB)

When the 10-bit address format is selected (SARUy.FS = 1), the SVA0 bit functions as the LSB of a 10-bit address and is combined with the SVA[6:0] bits to form the lower 8 bits of a 10-bit address.

This bit is valid when the SARyE bit in ICSEr is set to 1 (SARLy and SARUy enabled) and the SARUy.FS bit is 1. When the SARUy.FS or SARyE bit is 0, the setting in this bit is ignored.

SVA[6:0] bits (7-bit Address/10-bit Address Lower Bits)

When the 7-bit address format is selected (SARUy.FS = 0), the SVA[6:0] bits function as a 7-bit address. When the 10-bit address format is selected (SARUy.FS = 1), these bits combine with the SVA0 bit to form the lower 8 bits of a 10-bit address.

When the SARyE bit in ICSEr is 0, the setting in these bits is ignored.

39.2.14 SARUy : Slave Address Register Uy (y = 0 to 2)

Base address: IICn = 0x4025_E000 + 0x0100 × n (n = 0 to 2)
IICn_NS = 0x5025_E000 + 0x0100 × n (n = 0 to 2)

Offset address: 0x0B + 0x02 × y

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	SVA[1:0]	FS	
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	FS	7-bit/10-bit Address Format Select 0: Select 7-bit address format 1: Select 10-bit address format	R/W
2:1	SVA[1:0]	10-bit Address Upper Bits Slave address setting.	R/W
7:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

FS bit (7-bit/10-bit Address Format Select)

The FS bit selects 7- or 10-bit format for slave address y (in SARLy and SARUy).

When the SARyE bit in ICSEr is set to 1 (SARLy and SARUy enabled) and the SARUy.FS bit is 0, the 7-bit address format is selected for slave address y, the SVA[6:0] setting in SARLy is valid, and the SVA[1:0] and SVA0 settings in SARLy are ignored.

When the SARyE bit in ICSEr is set to 1 (SARLy and SARUy enabled) and the SARUy.FS bit is 1, the 10-bit address format is selected for slave address y and the SVA[1:0] and SARLy settings are valid.

When the SARyE bit in ICSEr is 0 (SARLy and SARUy disabled), the SARUy.FS setting is invalid.

SVA[1:0] bits (10-bit Address Upper Bits)

When the 10-bit address format is selected (FS = 1), the SVA[1:0] bits function as the upper 2 bits of a 10-bit address.

These bits are valid when the SARyE bit in ICSEr is set to 1 (SARLy and SARUy enabled) and the SARUy.FS bit is 1.

When the SARUy.FS or SARyE bit is 0, the setting in these bits is ignored.

39.2.15 ICBRL : I²C Bus Bit Rate Low-Level Register

Base address: IICn = 0x4025_E000 + 0x0100 × n (n = 0 to 2)
IICn_NS = 0x5025_E000 + 0x0100 × n (n = 0 to 2)

Offset address: 0x10

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	BRL[4:0]				
Value after reset:	1	1	1	1	1	1	1	1

Bit	Symbol	Function	R/W
4:0	BRL[4:0]	Bit Rate Low-Level Period Low-level period of SCL clock.	R/W
7:5	—	These bits are read as 1. The write value should be 1.	R/W

Note: S-TYPE3, P-TYPE3

BRL[4:0] bits (Bit Rate Low-Level Period)

The BRL[4:0] bits set the low-level period of the SCL clock. ICBRL counts the low-level period with the internal reference clock source (IICφ) specified by the CKS[2:0] bits in ICMR1. ICBRL also generates the data setup time for automatic SCL.

low-hold operation, see [section 39.9. Automatic Low-Hold Function for SCL](#). When the IIC is used in slave mode, the BRL[4:0] bits must be set to a value longer than the data setup time*¹.

If the digital noise filter is enabled (NFE bit in ICFER is 1), set the BRL[4:0] bits to a value at least one greater than the number of stages in the noise filter. For details on the number of stages, see the description of the NF[1:0] bits in [section 39.2.5. ICMR3 : I²C Bus Mode Register 3](#).

Note 1. Data setup time (t_{SU}: DAT)

250 ns for up to 100 kbps: Standard-mode (Sm)

100 ns for up to 400 kbps: Fast-mode (Fm)

50 ns for up to 1 Mbps: Fast-mode Plus (Fm+)

39.2.16 ICBRH : I²C Bus Bit Rate High-Level Register

Base address: IICn = 0x4025_E000 + 0x0100 × n (n = 0 to 2)
IICn_NS = 0x5025_E000 + 0x0100 × n (n = 0 to 2)

Offset address: 0x11

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	BRH[4:0]				
Value after reset:	1	1	1	1	1	1	1	1

Bit	Symbol	Function	R/W
4:0	BRH[4:0]	Bit Rate High-Level Period High-level period of SCL clock.	R/W
7:5	—	These bits are read as 1. The write value should be 1.	R/W

Note: S-TYPE3, P-TYPE3

BRH[4:0] bits (Bit Rate High-Level Period)

The BRH[4:0] bits set the high-level period of SCL clock. BRH[4:0] bits are valid in master mode. If the IIC is used only in slave mode, do not set the BRH[4:0] bits.

ICBRH counts the high-level period with the internal reference clock source (IICφ) specified in the CKS[2:0] bits in ICMR1.

If the digital noise filter is enabled (the NFE bit in ICFER is 1), set these bits to a value at least one greater than the number of stages in the noise filter. For the number of stages in the noise filter, see the description of the NF[1:0] bits in [section 39.2.5. ICMR3 : I²C Bus Mode Register 3](#).

The IIC transfer rate and the SCL clock duty are calculated using the following expressions (1) to (5):

1. ICFER.SCLE = 0

$$\text{Transfer rate} = 1 / [\{ (BRH + 1) + (BRL + 1) \} / IIC\phi^{*1} + tr^{*2} + tf^{*2}]$$

$$\text{Duty cycle} = [tr + \{ (BRH + 1) / IIC\phi \}] / [tr + tf + \{ (BRH + 1) + (BRL + 1) \} / IIC\phi]$$

2. ICFER.SCLE = 1 and ICFER.NFE = 0 and CKS[2:0] = 000b (IICφ = PCLKB)

$$\text{Transfer rate} = 1 / [\{ (BRH + 3) + (BRL + 3) \} / IIC\phi + tr + tf]$$

$$\text{Duty cycle} = [tr + \{ (BRH + 3) / IIC\phi \}] / [tr + tf + \{ (BRH + 3) + (BRL + 3) \} / IIC\phi]$$

3. ICFER.SCLE = 1 and ICFER.NFE = 1 and CKS[2:0] = 000b (IICφ = PCLKB)

$$\text{Transfer rate} = 1 / [\{ (BRH + 3 + nf^{*3}) + (BRL + 3 + nf) \} / IIC\phi + tr + tf]$$

$$\text{Duty cycle} = [tr + \{ (BRH + 3 + nf) / IIC\phi \}] / [tr + tf + \{ (BRH + 3 + nf) + (BRL + 3 + nf) \} / IIC\phi]$$

4. ICFER.SCLE = 1 and ICFER.NFE = 0 and CKS[2:0] ≠ 000b

$$\text{Transfer rate} = 1 / [\{ (BRH + 2) + (BRL + 2) \} / IIC\phi + tr + tf]$$

$$\text{Duty cycle} = [tr + \{ (BRH + 2) / IIC\phi \}] / [tr + tf + \{ (BRH + 2) + (BRL + 2) \} / IIC\phi]$$

5. ICFER.SCLE = 1 and ICFER.NFE = 1 and CKS[2:0] ≠ 000b

$$\text{Transfer rate} = 1 / [\{ (BRH + 2 + nf) + (BRL + 2 + nf) \} / IIC\phi + tr + tf]$$

$$\text{Duty cycle} = [tr + \{ (BRH + 2 + nf) / IIC\phi \}] / [tr + tf + \{ (BRH + 2 + nf) + (BRL + 2 + nf) \} / IIC\phi]$$

Note 1. $IIC\phi = PCLKB \times \text{division ratio}$

Note 2. The SCLn line rise time (t_r) and SCLn line fall time (t_f) depend on the total bus line capacitance (C_b) and the pull-up resistor (R_p). For details, see the I²C bus standard from NXP Semiconductors.

Note 3. nf = Number of digital noise filters selected in the ICMR3.NF bit.

Table 39.6 Example of ICBRH/ICBRL Settings for Transfer Rate IIC when SCLE = 0

Transfer rate (kbps)	CKS[2:0] (ICMR1)	BRH[4:0] (ICBRH)	BRL[4:0] (ICBRL)	PCLKB (MHz)	NF[1:0]	Computation expression
100	011b	24 (0xF8)	30 (0xFE)	50	—	(1)
400	010b	7 (0xE7)	15 (0xEF)	50	—	(1)
1000	000b	12 (0xEC)	24 (0xF8)	50	—	(1)

Table 39.7 Example of ICBRH/ICBRL Settings for Transfer Rate when SCLE = 1 and NFE = 0

Transfer rate (kbps)	CKS[2:0] (ICMR1)	BRH[4:0] (ICBRH)	BRL[4:0] (ICBRL)	PCLKB (MHz)	NF[1:0]	Computation expression
100	100b	11 (0xEB)	13 (0xED)	50	—	(4)
400	001b	13 (0xED)	31 (0xFF)	50	—	(4)
1000	000b	10 (0xEA)	22 (0xF6)	50	—	(2)

Table 39.8 Example of ICBRH/ICBRL Settings for Transfer Rate when SCLE = 1 and NFE = 1

Transfer rate (kbps)	CKS[2:0] (ICMR1)	BRH[4:0] (ICBRH)	BRL[4:0] (ICBRL)	PCLKB (MHz)	NF[1:0]	Computation expression
100	011b	21 (0xF5)	26 (0xFA)	50	01b	(5)
400	001b	11 (0xEB)	29 (0xFD)	50	01b	(5)
1000	000b	8 (0xE8)	20 (0xF4)	50	01b	(3)

39.2.17 ICDRT : I²C Bus Transmit Data Register

Base address: IICn = 0x4025_E000 + 0x0100 × n (n = 0 to 2)
IICn_NS = 0x5025_E000 + 0x0100 × n (n = 0 to 2)

Offset address: 0x12

Bit position: 7 6 5 4 3 2 1 0

Bit field:

Value after reset: 1 1 1 1 1 1 1 1

When ICDRT detects a space in the I²C Bus Shift Register (ICDRS), it transfers the transmit data that was written to ICDRT to ICDRS and starts transmitting data in transmit mode. The double-buffer structure of ICDRT and ICDRS allows continuous transmit operation if the next transmit data is written to ICDRT while the ICDRS data is being transmitted.

ICDRT can always be read and written to. Write transmit data to ICDRT once when a transmit data empty interrupt (IICn_TXI) request is generated.

39.2.18 ICDRR : I²C Bus Receive Data Register

Base address: IICn = 0x4025_E000 + 0x0100 × n (n = 0 to 2)
IICn_NS = 0x5025_E000 + 0x0100 × n (n = 0 to 2)

Offset address: 0x13

Bit position: 7 6 5 4 3 2 1 0

Bit field:

Value after reset: 0 0 0 0 0 0 0 0

When 1 byte of data is received, the received data is transferred from the I²C Bus Shift Register (ICDRS) to ICDRR to enable the next data to be received. The double-buffer structure of ICDRS and ICDRR allows continuous receive operation if the received data is read from ICDRR while ICDRS is receiving data. ICDRR cannot be written to. Read data from ICDRR once when a receive data full interrupt (IICn_RXI) request is generated.

If ICDRR receives the next receive data before the current data is read from ICDRR (while the RDRF flag in ICSR2 is 1), the IIC automatically holds the SCL clock low 1 cycle before the RDRF flag is set to 1 next.

39.2.19 ICDRS : I²C Bus Shift Register

Base address: n/a

Offset address: n/a

Bit position:	7	6	5	4	3	2	1	0
Bit field:								
Value after reset:	—	—	—	—	—	—	—	—

ICDRS is an 8-bit shift register for data transmit and receive. During transmission, transmit data is transferred from ICDRT to ICDRS and is transmitted from the SDA_n pin. During reception, data is transferred from ICDRS to ICDRR after 1 byte of data is received. ICDRS cannot be accessed directly.

39.3 Operation

39.3.1 Communication Data Format

The I²C bus format consists of 8-bit data and 1-bit acknowledge. The frame following a start or restart condition is an address frame that specifies a slave device with which the master device communicates. The specified slave is valid until a new slave is specified or a stop condition is issued.

Figure 39.3 shows the I²C bus format, and Figure 39.4 shows the I²C bus timing.

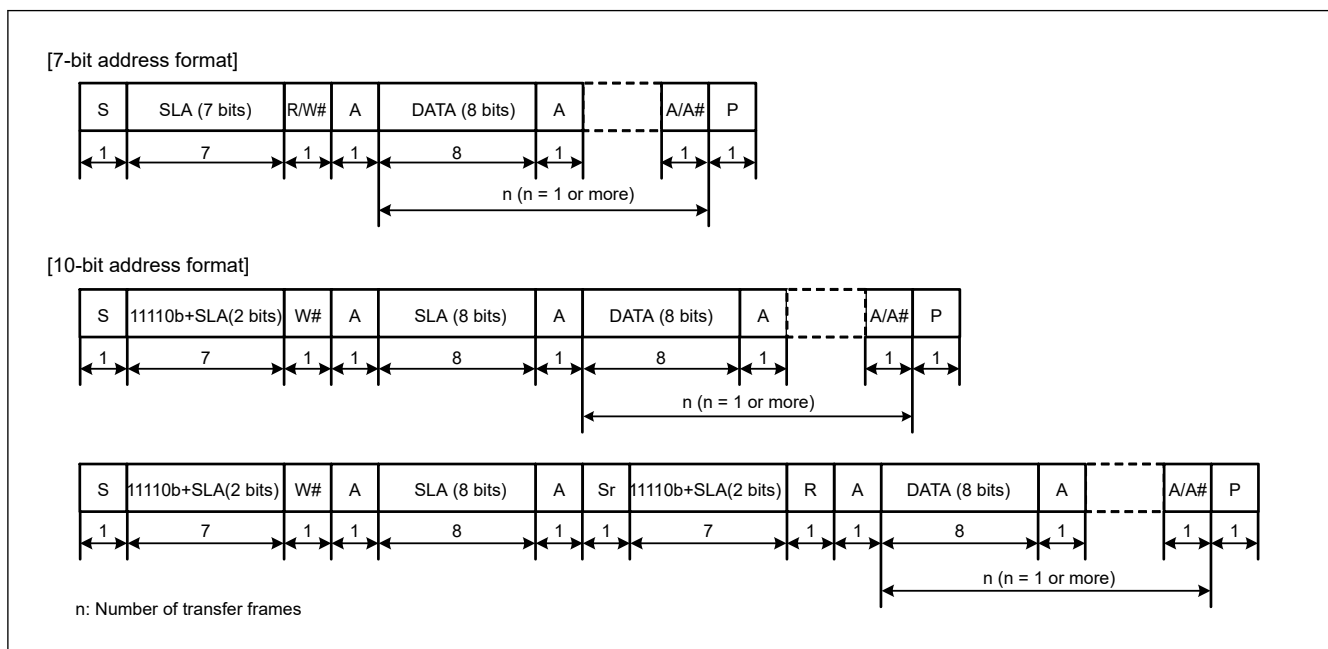


Figure 39.3 I²C bus format

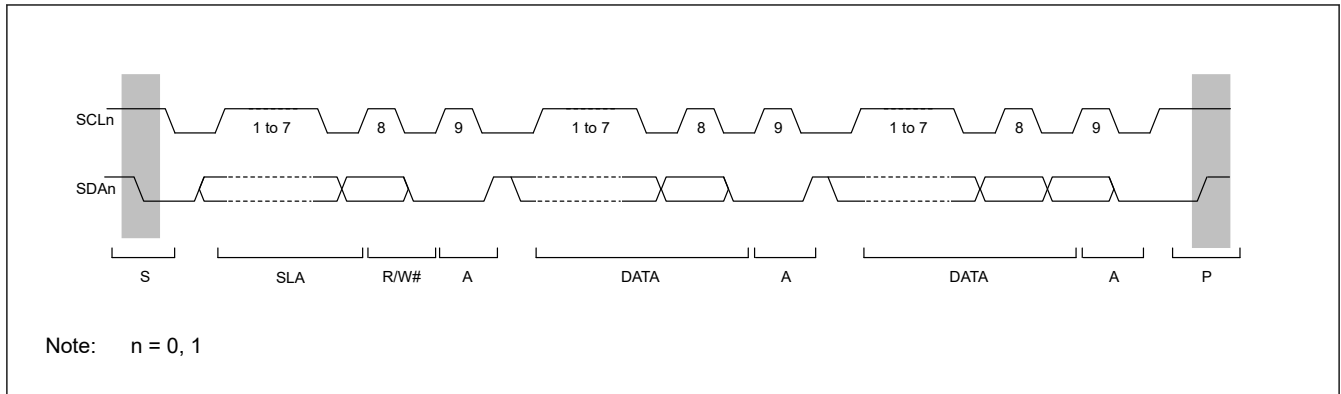


Figure 39.4 I²C bus timing when the SLA setting = 7 bits

- S: Start condition. The master device drives the SDAn line low from high while the SCLn line is high.
- SLA: Slave address, by which the master device selects a slave device.
- R/W#: Indicates the direction of data transfer: from the slave device to the master device when R/W# is 1, or from the master device to the slave device when R/W# is 0.
- A: Acknowledge. The receive device drives the SDAn line low. In master transmit mode, the slave device returns acknowledge. In master receive mode, the master device returns acknowledge.
- A#: Not Acknowledge. The receive device drives the SDAn line high.
- Sr: Restart condition. The master device drives the SDAn line low from the high level after the setup time has elapsed with the SCLn line high.
- DATA: Transmitted or received data.
- P: Stop condition. The master device drives the SDAn line high from low while the SCLn line is high.

39.3.2 Initial Settings

Before starting data transmission or reception, initialize the IIC using the procedure shown in [Figure 39.5](#).

1. Set the ICCR1.ICE bit to 0 to set the SCLn and SDAn pins to the inactive state.
2. Set the ICCR1.IICRST bit to 1 to initiate IIC reset.
3. Set the ICCR1.ICE bit to 1 to initiate internal reset.
4. Set the SARLy, SARUy, ICSEr, ICMR1, ICBRH, and ICBRL registers (y = 0 to 2), and set the other registers as required. For initial settings of the IIC, see [Figure 39.5](#).
5. When the required register settings are complete, set the ICCR1.IICRST bit to 0 to release the IIC reset.

This procedure is not required if the IIC initialization is already complete.

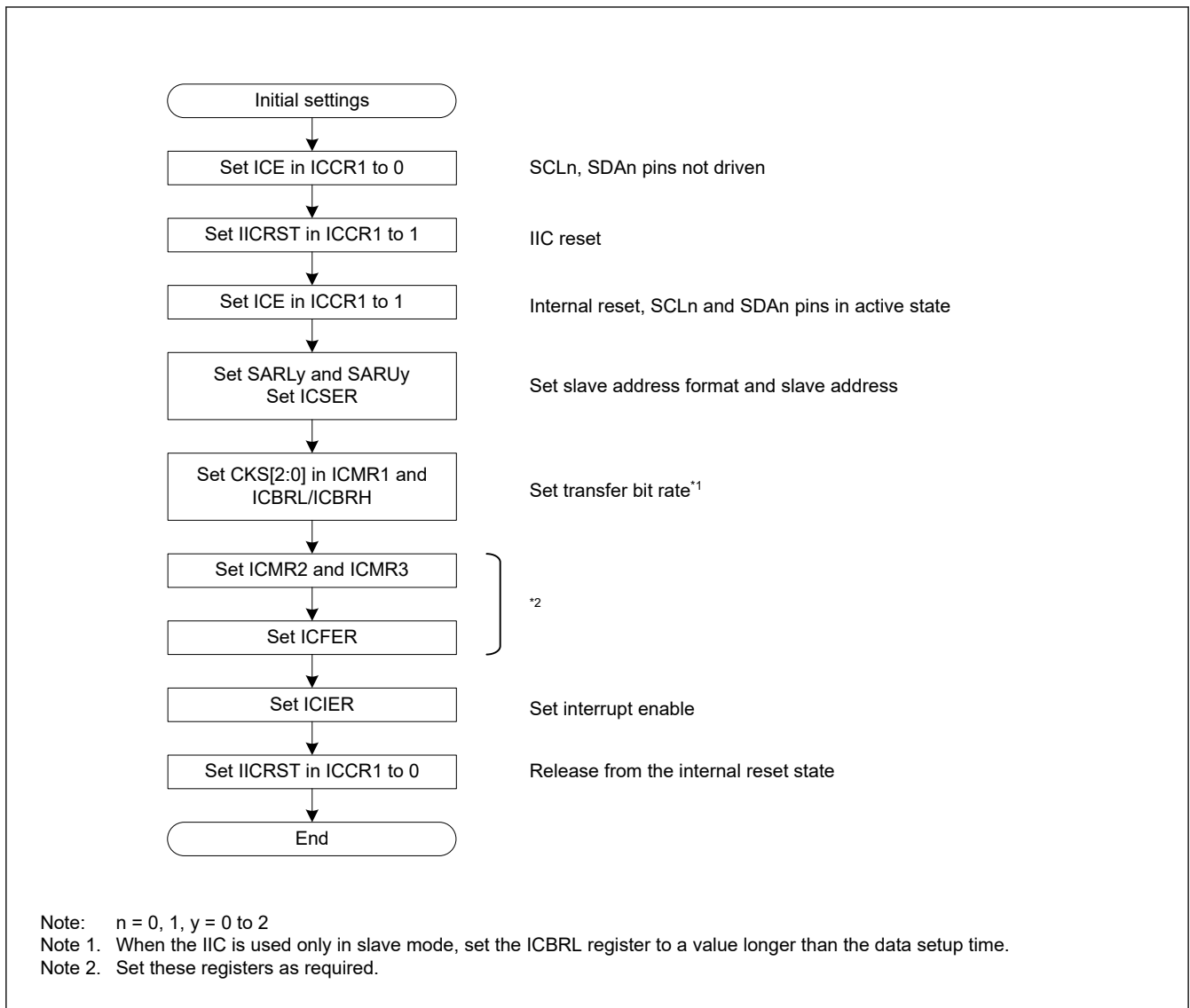


Figure 39.5 Example IIC initialization flow

39.3.3 Master Transmit Operation

In master transmit operation, the IIC outputs the SCL clock and transmitted data signals as the master device, and the slave device returns acknowledgments. Figure 39.6 shows an example of master transmission, and Figure 39.7 to Figure 39.9 show the operation timing in master transmission.

To set up and perform master transmission:

1. Process initial settings. For details, see [section 39.3.2. Initial Settings](#).
2. Read the BBSY flag in ICCR2 to check that the bus is free, and then set the ST bit in ICCR2 to 1 (start condition request). On receiving the request, the IIC issues a start condition. At the same time, the BBSY and START flags in ICSR2 automatically set to 1 and the ST bit automatically is set to 0. At this time, if the start condition is detected and the internal levels for the SDA output state and the levels on the SDAn line match while the ST bit is 1, the IIC recognizes that issuance of the start condition as requested by the ST bit is successfully complete, and the MST and TRS bits in ICCR2 automatically set to 1, placing the IIC in master transmit mode. The TDRE flag in ICSR2 also automatically is set to 1 in response to the setting of the TRS bit to 1.
3. Check that the TDRE flag in ICSR2 is 1, and then write the value for transmission (the slave address and the R/W# bit) to ICDRT. When the transmit data is written to ICDRT, the TDRE flag automatically is set to 0, the data is transferred from ICDRT to ICDRS, and the TDRE flag again is set to 1. After the byte containing the slave address and R/W# bit is transmitted, the value of the TRS bit automatically updates to select master transmit or master receive mode based on the value of the transmitted R/W# bit. If the value of the R/W# bit was 0, the IIC continues in master transmit mode.

Because the ICSR2.NACKF flag being 1 at this time indicates that no slave device recognized the address or there was an error in communications, write 1 to the ICCR2.SP bit to issue a stop condition.

For data transmission with an address in the 10-bit format, start by writing 11110b, the 2 higher-order bits of the slave address, and W to ICDRT as the first address transmission. For the second address transmission, write the 8 lower-order bits of the slave address to ICDRT.

4. Check that the TDRE flag in ICSR2 is 1, and then write the transmit data to the ICDRT register. The IIC automatically holds the SCLn line low until the transmit data is ready or a stop condition is issued.
5. After all bytes of transmit data are written to the ICDRT register, wait until the value in the TEND flag in ICSR2 returns to 1, and then set the SP bit in ICCR2 to 1 (stop condition requested). On receiving a stop condition request, the IIC issues the stop condition. Regarding issuing a stop condition, see [section 39.11.3. Issuing a Stop Condition](#).
6. On detecting the stop condition, the IIC automatically sets the MST and TRS bits in ICCR2 to 00b and enters slave receive mode. Additionally, it automatically sets the TDRE and TEND flags to 0, and sets the STOP flag in ICSR2 to 1.
7. Check that the ICSR2.STOP flag is 1, and then set the ICSR2.NACKF and STOP flags to 0 for the next transfer operation.

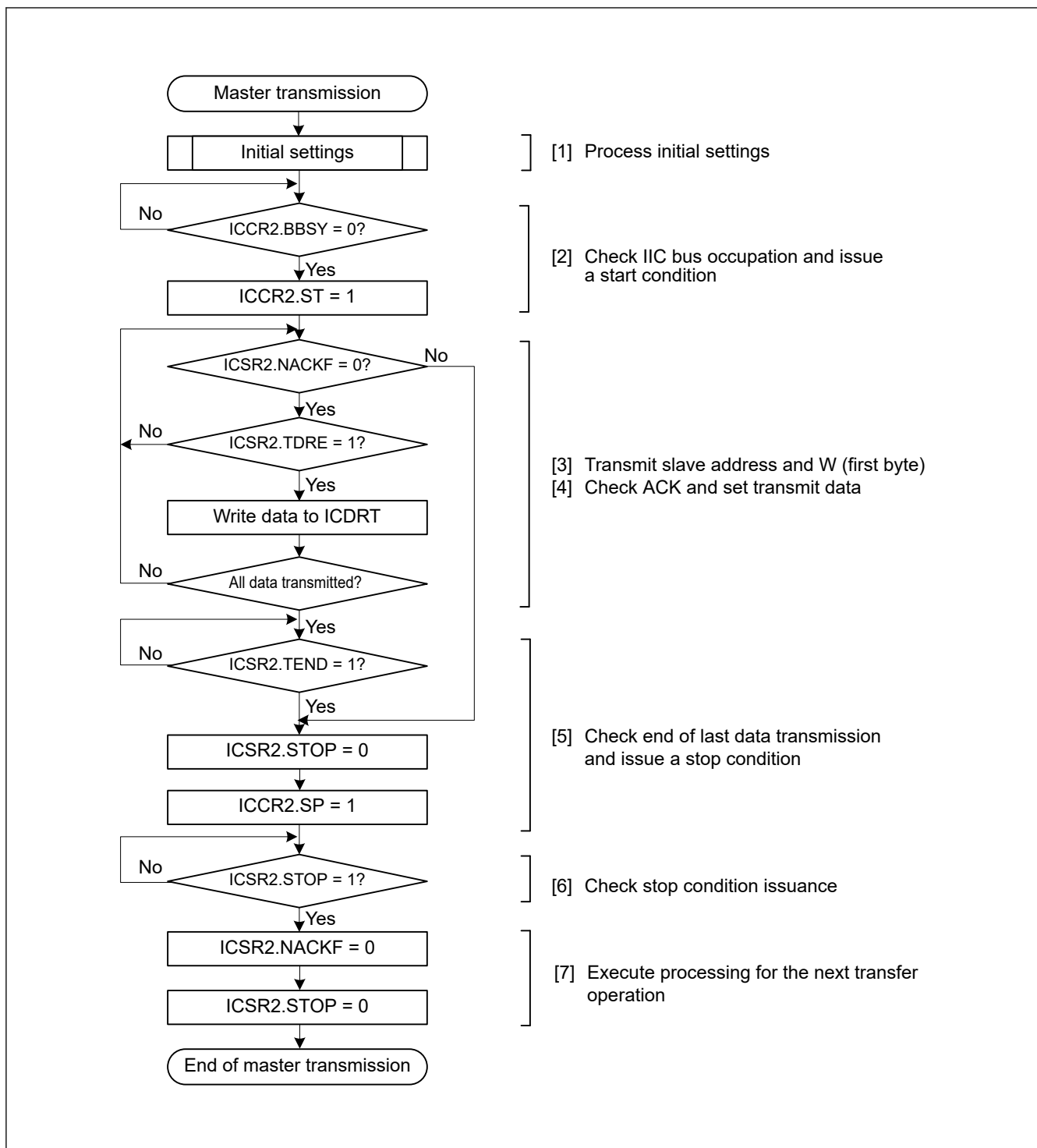


Figure 39.6 Example master transmission flow

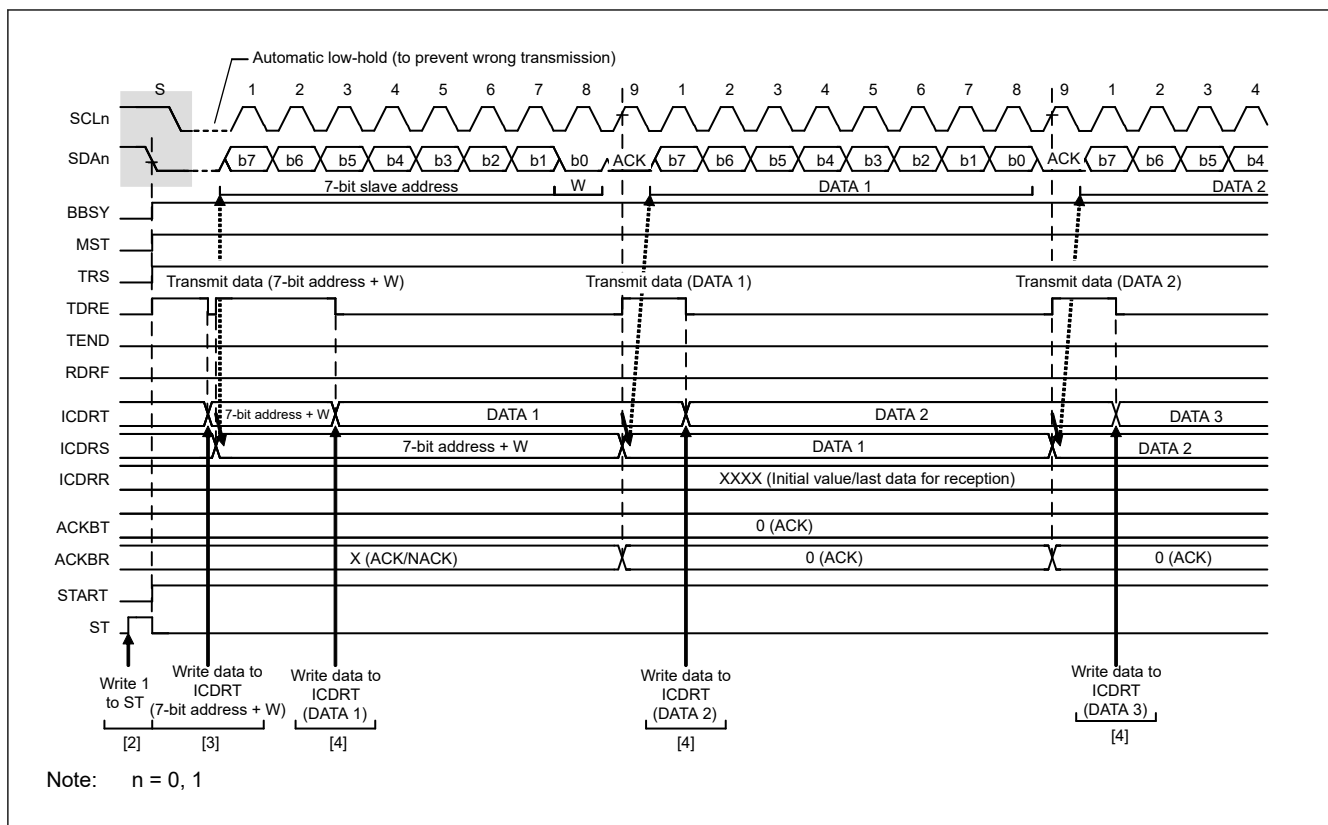


Figure 39.7 Master transmit operation timing (1) with 7-bit address format

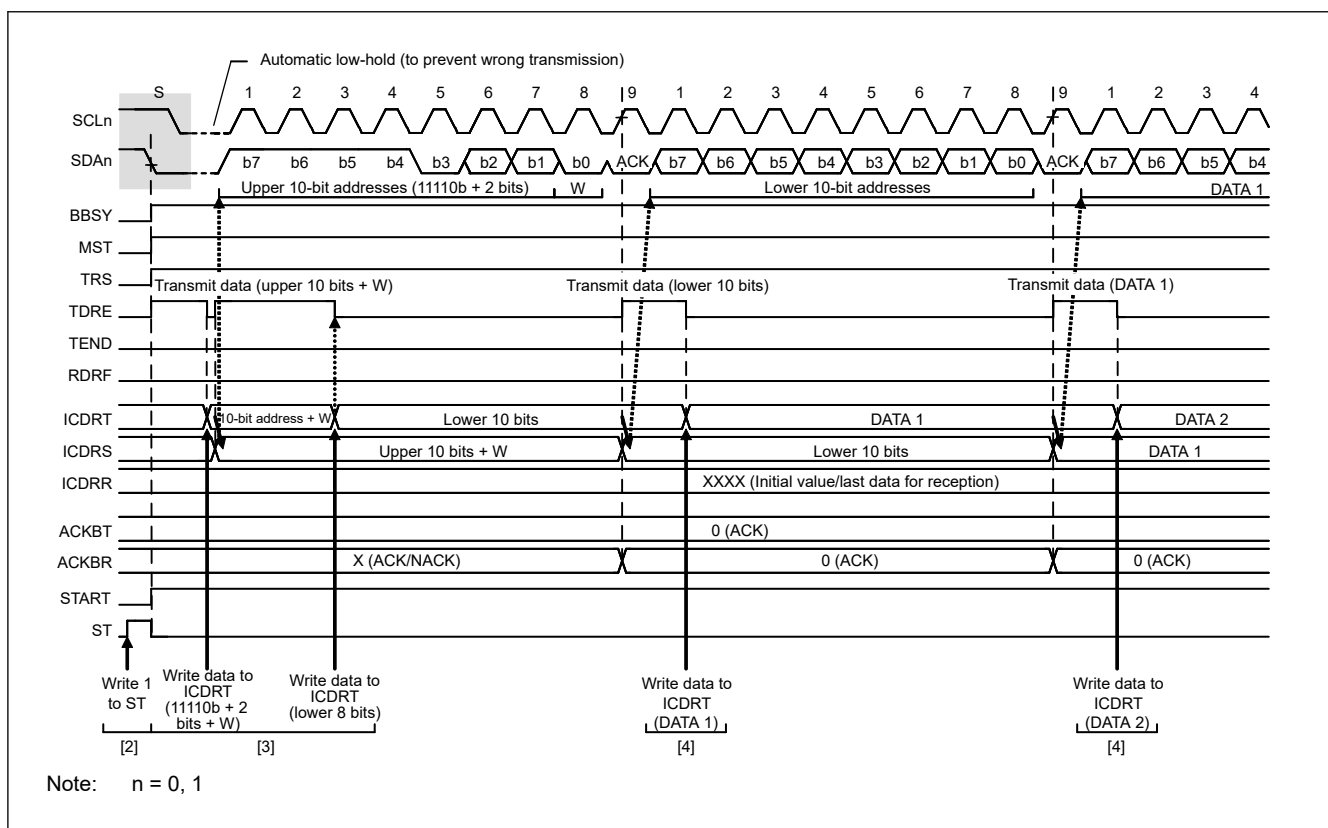


Figure 39.8 Master transmit operation timing (2) with 10-bit address format

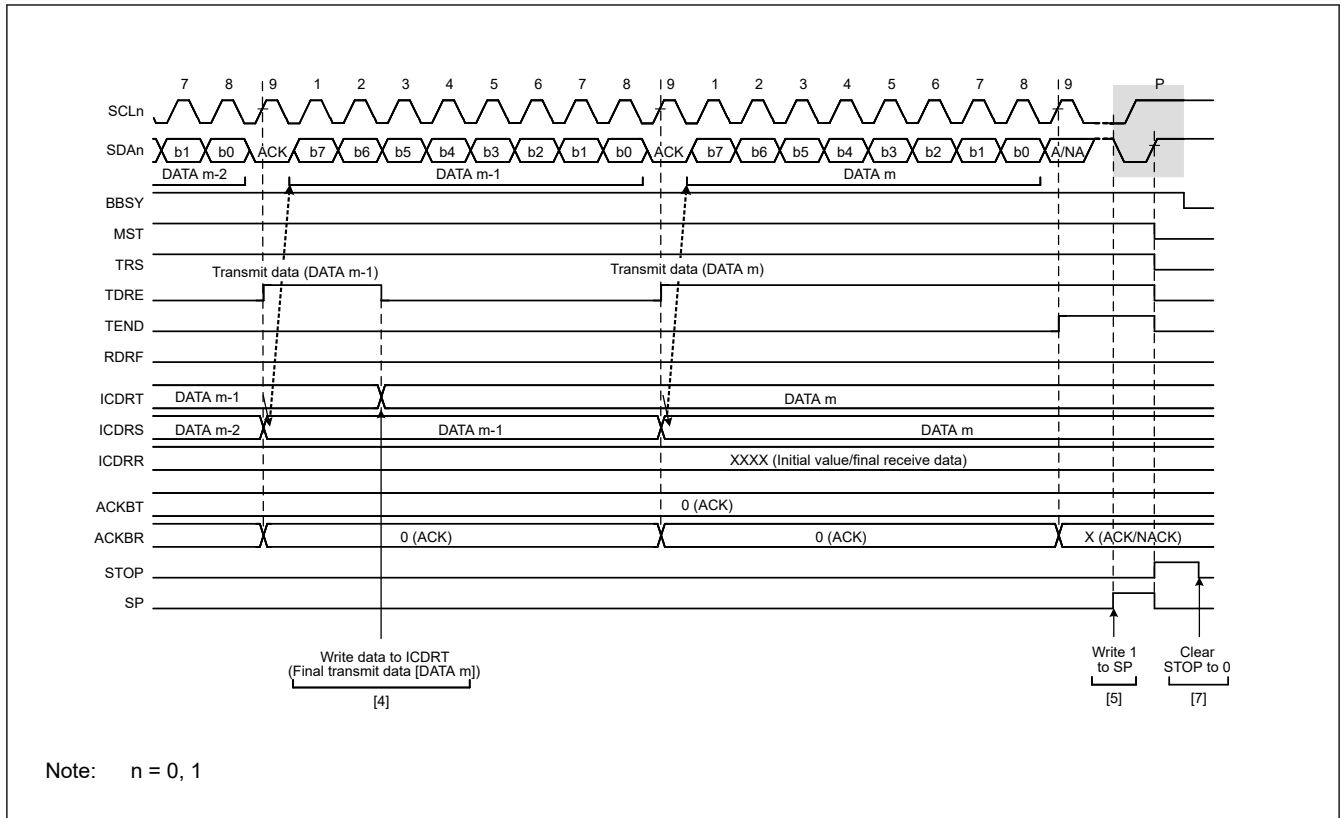


Figure 39.9 Master transmit operation timing (3)

39.3.4 Master Receive Operation

In master receive operation, the IIC as a master device outputs the SCL clock, receives data from the slave device, and returns acknowledgments. Because the IIC must start by sending a slave address to the associated slave device, this part of the procedure is performed in master transmit mode, but the subsequent steps are in master receive mode.

Figure 39.10 and Figure 39.11 show examples of master reception (7-bit address format), and Figure 39.12 to Figure 39.14 show the operation timing in master reception.

To set up and perform master reception:

1. Process initial settings. For details, see [section 39.3.2. Initial Settings](#).
2. Read the BBSY flag in ICCR2 to check that the bus is free, and then set the ST bit in ICCR2 to 1 (start condition request). On receiving the request, the IIC issues a start condition. When the IIC detects the start condition, the BBSY and START flags in ICSR2 automatically set to 1, and the ST bit automatically is set to 0. At this time, if the start condition is detected and the levels for the SDA output and the levels on the SDA line match while the ST bit is 1, the IIC recognizes that issuance of the start condition as requested by the ST bit is successfully complete, and the MST and TRS bits in ICCR2 automatically set to 1, placing the IIC in master transmit mode. The TDRE flag in ICSR2 also automatically is set to 1 in response to the setting of the TRS bit to 1.
3. Check that the TDRE flag in ICSR2 is 1, and then write the value for transmission (the first byte indicates the slave address and value of the R/W# bit) to ICDRT. When the transmit data is written to ICDRT, the TDRE flag automatically is set to 0, the data is transferred from ICDRT to ICDRS, and the TDRE flag again is set to 1. When the byte containing the slave address and R/W# bit is transmitted, the value of the ICCR2.TRS bit automatically updates to select transmit or receive mode based on the value of the transmitted R/W# bit. If the value of the R/W# bit is 1, the TRS bit is set to 0 on the rising edge of the ninth cycle of the SCL clock, placing the IIC in master receive mode. At this time, the TDRE flag is set to 0 and the ICSR2.RDRF flag automatically is set to 1. Because the ICSR2.NACKF flag being 1 at this time indicates that no slave device recognized the address or there was an error in communications, write 1 to the ICCR2.SP bit to issue a stop condition. For master reception from a device with a 10-bit address, start by using master transmission to issue the 10-bit address, and then issue a restart condition. After that, transmitting 11110b, the two higher-order bits of the slave address, and the R bit places the IIC in master receive mode.

4. Dummy read ICDRR after confirming that the RDRF flag in ICSR2 is 1. This makes the IIC start output of the SCL clock and start data reception.
5. After 1 byte of data is received, the RDRF flag in ICSR2 is set to 1 on the rising edge of the 8th or 9th cycle of the SCL clock, as selected in the RDRFS bit in ICMR3. Reading ICDRR at this time produces the received data, and the RDRF flag is automatically set to 0 at the same time. Additionally, the value of the acknowledgment field received during the ninth cycle of the SCL clock is returned as the value set in the ICMR3.ACKBT bit. If the next byte to be received is the second-to-last byte, set the ICMR3.WAIT bit to 1 for wait insertion before reading ICDRR, containing the second-to-last byte. In addition to enabling NACK output, even when interrupts or other operations result in delays in setting the ICMR3.ACKBT bit to 1 (NACK) in step (6), this fixes the SCLn line to the low level on the rising edge of the ninth clock cycle in reception of the last byte, which enables the issuing of a stop condition.
6. When the ICMR3.RDRFS bit is 0, and the slave device must be notified that it is to end transfer for data reception after transfer of the next and final byte, set the ICMR3.ACKBT bit to 1 (NACK).
7. After reading the second-to-last byte from the ICDRR register, if the value of the ICSR2.RDRF flag is 1, write 1 to the SP bit in ICCR2 (stop condition requested), and then read the last byte from ICDRR. When ICDRR is read, the IIC is released from the wait state and issues the stop condition after low-level output in the ninth clock cycle is complete or the SCLn line is released from the low-hold state.
8. On detecting the stop condition, the IIC automatically sets the MST and TRS bits in ICCR2 to 00b and enters slave receive mode. Additionally, detection of the stop condition sets the ICSR2.STOP flag to 1.
9. Check that the ICSR2.STOP flag is 1, then set the ICSR2.NACKF and STOP flags to 0 for the next transfer operation.

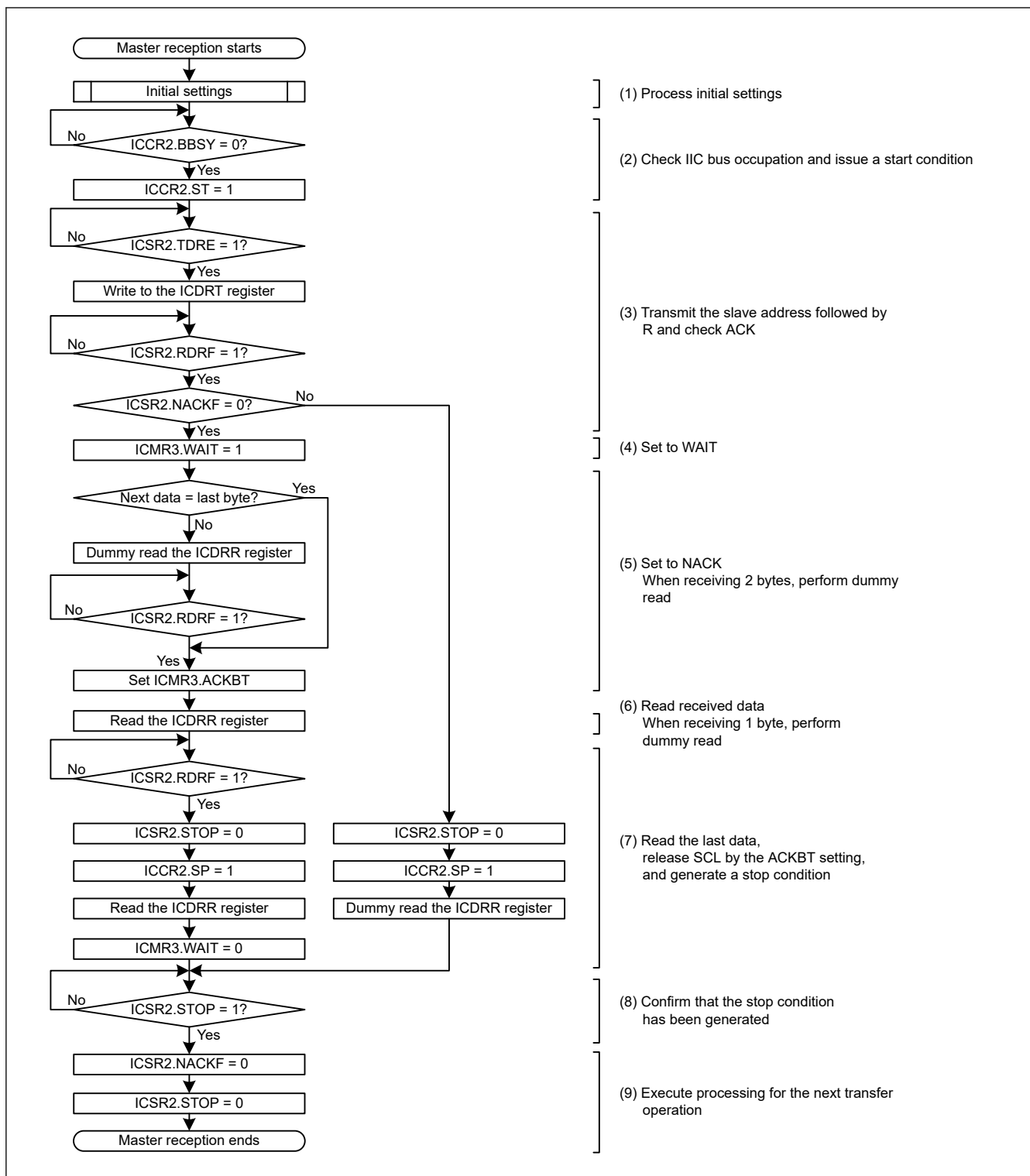


Figure 39.10 Example master reception flow with 7-bit address format of 1 byte or 2 bytes

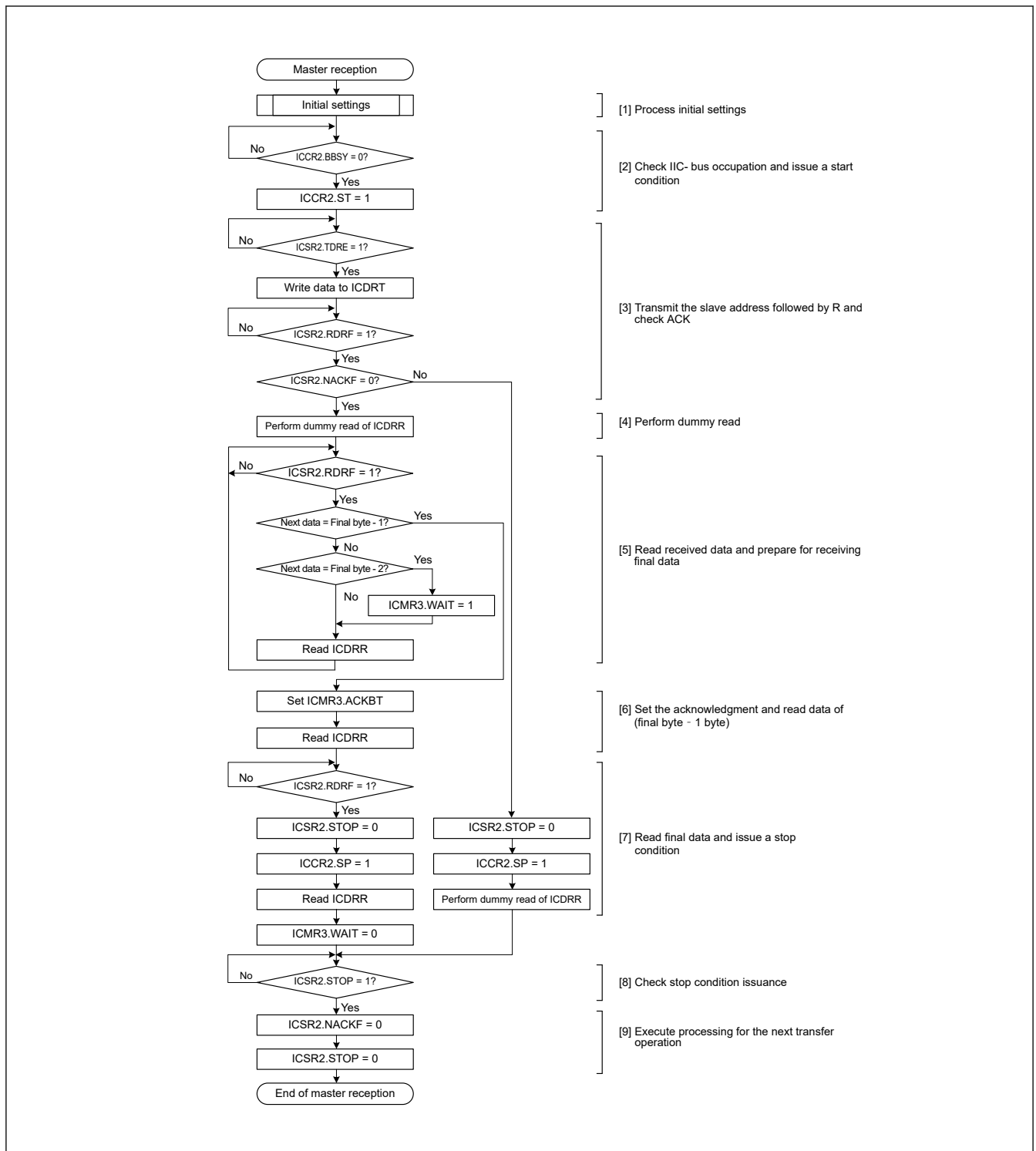
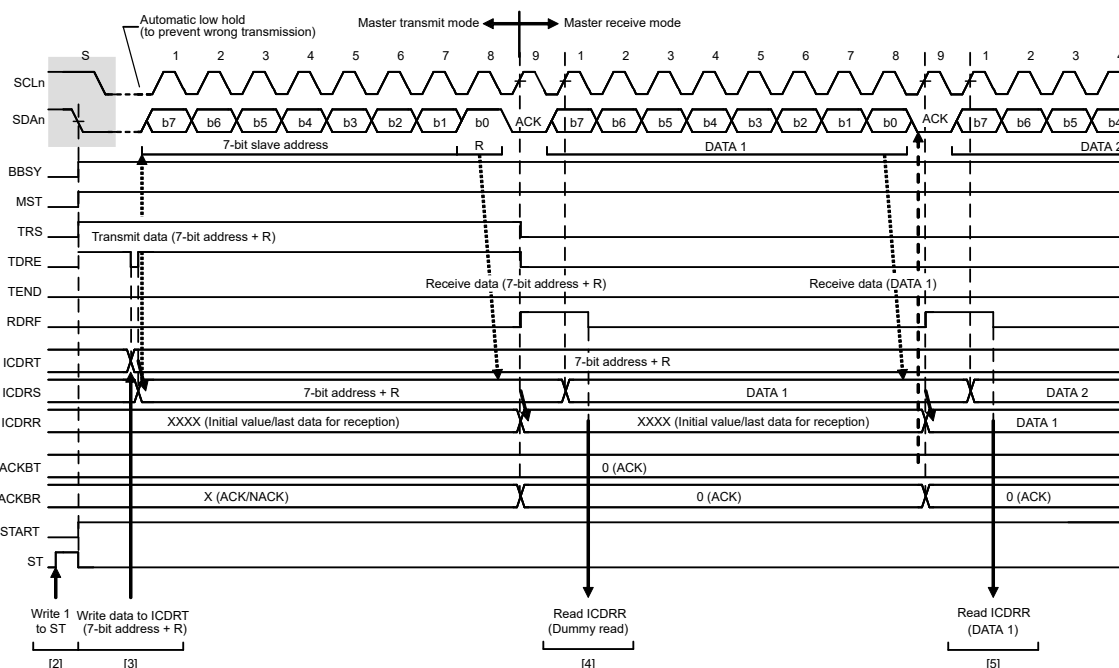
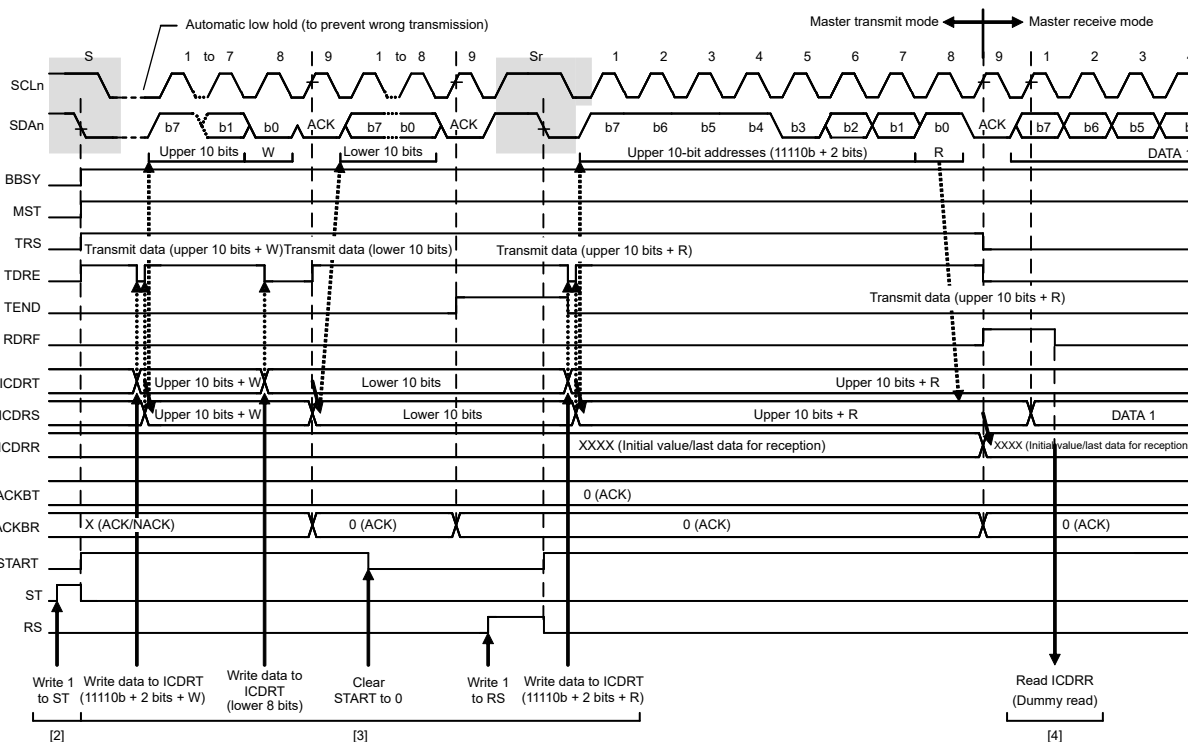


Figure 39.11 Example master reception flow with 7-bit address format of 3 or more bytes



Note: $n = 0, 1$

Figure 39.12 Master receive operation timing (1) with 7-bit address format when RDRFS = 0



Note: $n = 0, 1$

Figure 39.13 Master receive operation timing (2) with 10-bit address format when RDRFS = 0

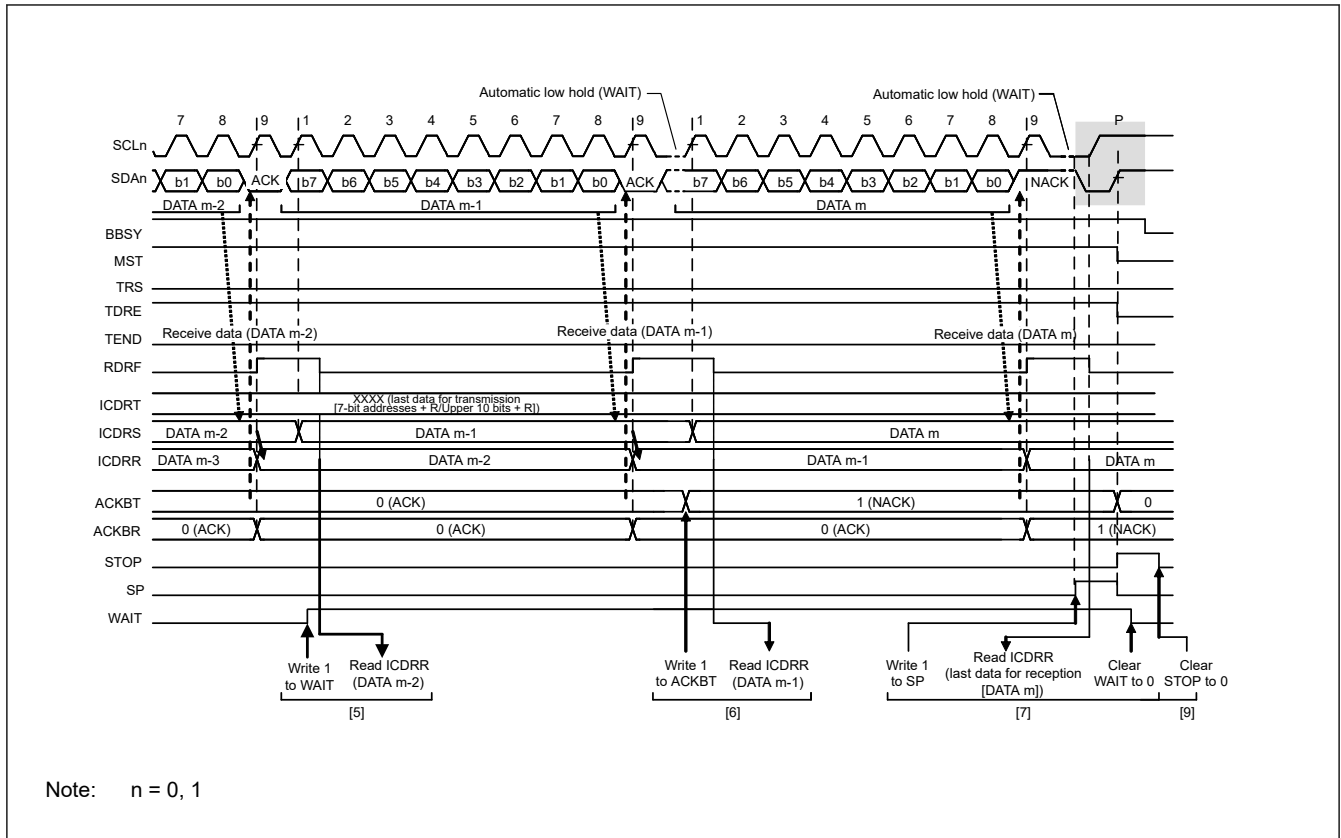


Figure 39.14 Master receive operation timing (3) when RDRFS = 0

39.3.5 Slave Transmit Operation

In slave transmit operation, the master device outputs the SCL clock, the IIC transmits data as a slave device, and the master device returns acknowledgments.

Figure 39.15 shows an example of slave transmission, and Figure 39.16 and Figure 39.17 show the operation timing in slave transmission.

To set up and perform slave transmission:

1. Initialize the IIC using the procedure in [section 39.3.2. Initial Settings](#).
After initialization, the IIC stays in the standby state until it receives a slave address that matches.
2. After receiving a matching slave address, the IIC sets one of the associated ICSR1.HOA, GCA, and AASy flags ($y = 0$ to 2) to 1 on the rising edge of the ninth cycle of the SCL clock and outputs the value set in the ICMR3.ACKBT bit to the acknowledge bit on the ninth cycle of the SCL clock. If the value of the R/W# bit is 1, the IIC automatically places itself in slave transmit mode by setting both the ICCR2.TRS bit and the ICSR2.TDRE flag to 1.
3. Check that the ICSR2.TDRE flag is 1, then write the transmit data to the ICDRT register. If the IIC receives no acknowledge from the master device (receives an NACK signal) while the ICFER.NACKF bit is 1, the IIC suspends transfer of the next data.
4. Wait until the ICSR2.TEND flag is set to 1 while the ICSR2.TDRE flag is 1, after the ICSR2.NACKF flag is set to 1 or the last byte for transmission is written to the ICDRT register. When the ICSR2.NACKF flag or the TEND flag is 1, the IIC drives the SCLn line low on the ninth falling edge of the SCL clock.
5. When the ICSR2.NACKF or ICSR2.TEND flag is 1, dummy read ICDRR to complete the processing. This releases the SCLn line.
6. On detecting the stop condition, the IIC automatically sets the ICSR1.HOA, GCA, and AASy flags ($y = 0$ to 2), the ICSR2.TDRE and TEND flags, and the ICCR2.TRS bit to 0, and enters slave receive mode.
7. Check that the ICSR2.STOP flag is 1, then set the ICSR2.NACKF and STOP flags to 0 for the next transfer operation.

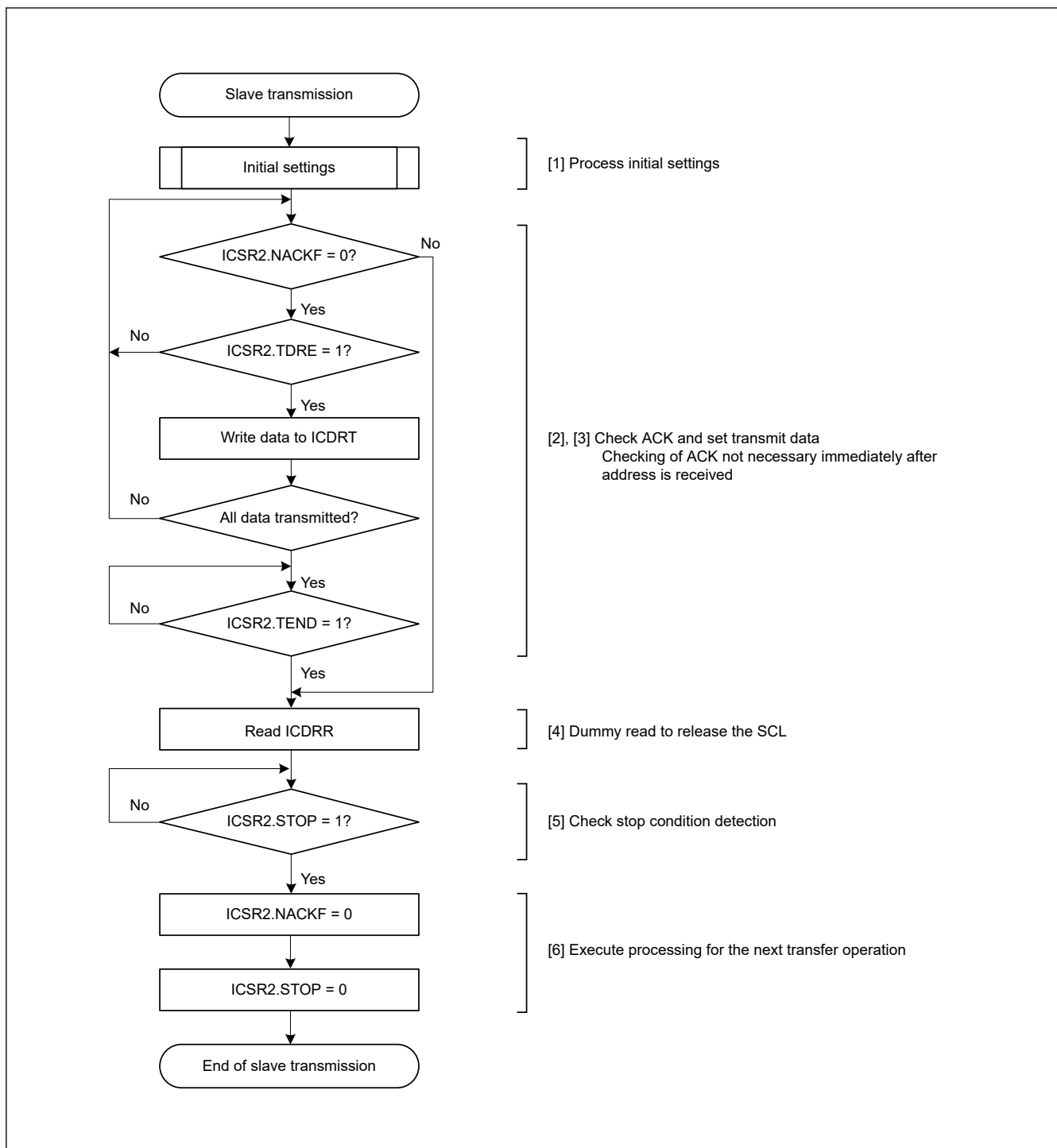


Figure 39.15 Example slave transmission flow

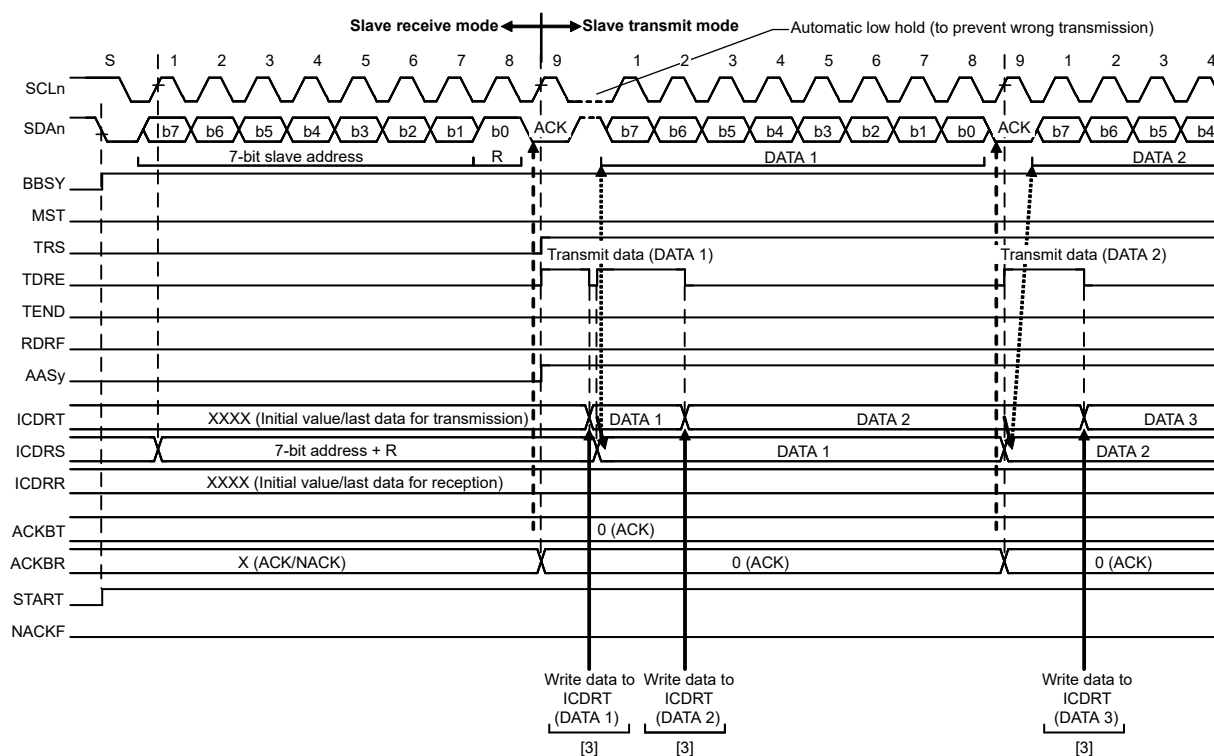


Figure 39.16 Slave transmit operation timing (1) with 7-bit address format

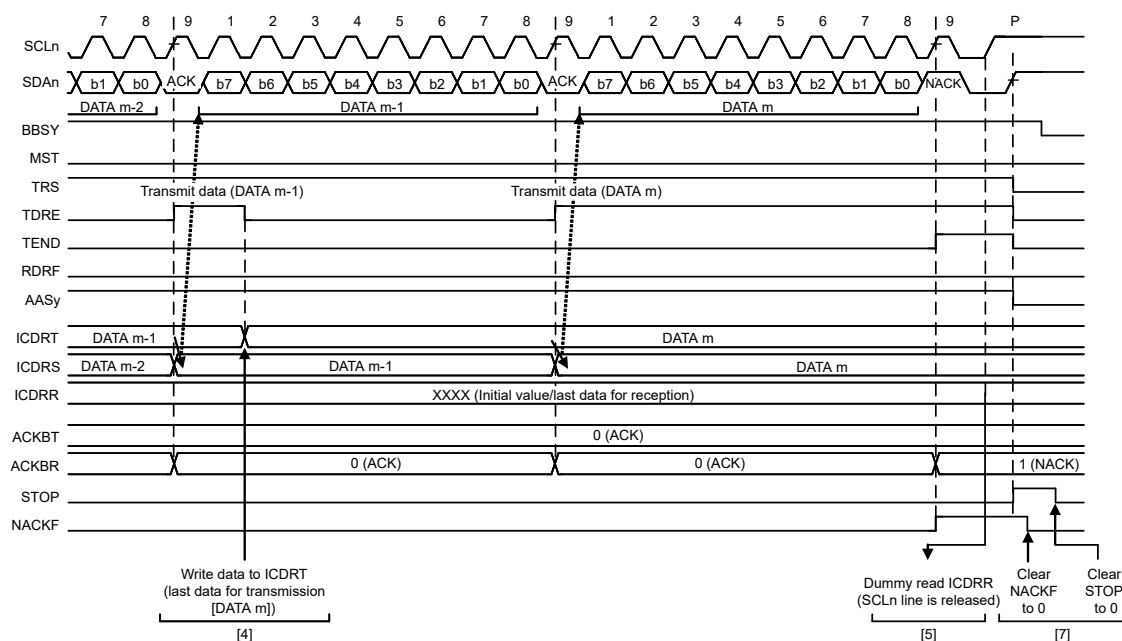


Figure 39.17 Slave transmit operation timing (2)

39.3.6 Slave Receive Operation

In slave receive operation, the master device outputs the SCL clock and transmit data, and the IIC returns acknowledgments as a slave device.

Figure 39.18 shows an example of slave reception, and Figure 39.19 and Figure 39.20 show the operation timing in slave reception.

To set up and perform slave reception:

1. Initialize the IIC using the procedure in [section 39.3.2. Initial Settings](#).
After initialization, the IIC stays in the standby state until it receives a slave address that matches.
2. After receiving a matching slave address, the IIC sets one of the associated ICSR1.HOA, GCA, and AASy flags (y = 0 to 2) to 1 on the rising edge of the ninth cycle of the SCL clock and outputs the value set in the ICMR3.ACKBT bit to the acknowledged bit on the ninth cycle of the SCL clock. If the value of the R/W# bit is 0, the IIC continues to place itself in slave receive mode and sets the RDRF flag in ICSR2 to 1.
3. Check that the ICSR2.STOP flag is 0 and the ICSR2.RDRF flag is 1, then dummy read ICDRR. The dummy value consists of the slave address and R/W# bit when the 7-bit address format is selected, or the lower 8 bits when the 10-bit address format is selected.
4. When ICDRR is read, the IIC automatically sets the ICSR2.RDRF flag to 0. If reading of ICDRR is delayed and a next byte is received while the RDRF flag is still set to 1, the IIC holds the SCLn line low until 1 SCL cycle before the point where RDRF must be set. In this case, reading ICDRR releases the SCLn line from being held low.
When the ICSR2.STOP flag is 1 and the ICSR2.RDRF flag is also 1, read ICDRR until all the data is completely received.
5. On detecting the stop condition, the IIC automatically clears the ICSR1.HOA, GCA, and AASy flags (y = 0 to 2) to 0.
6. Check that the ICSR2.STOP flag is 1, then set the ICSR2.STOP flag to 0 for the next transfer operation.

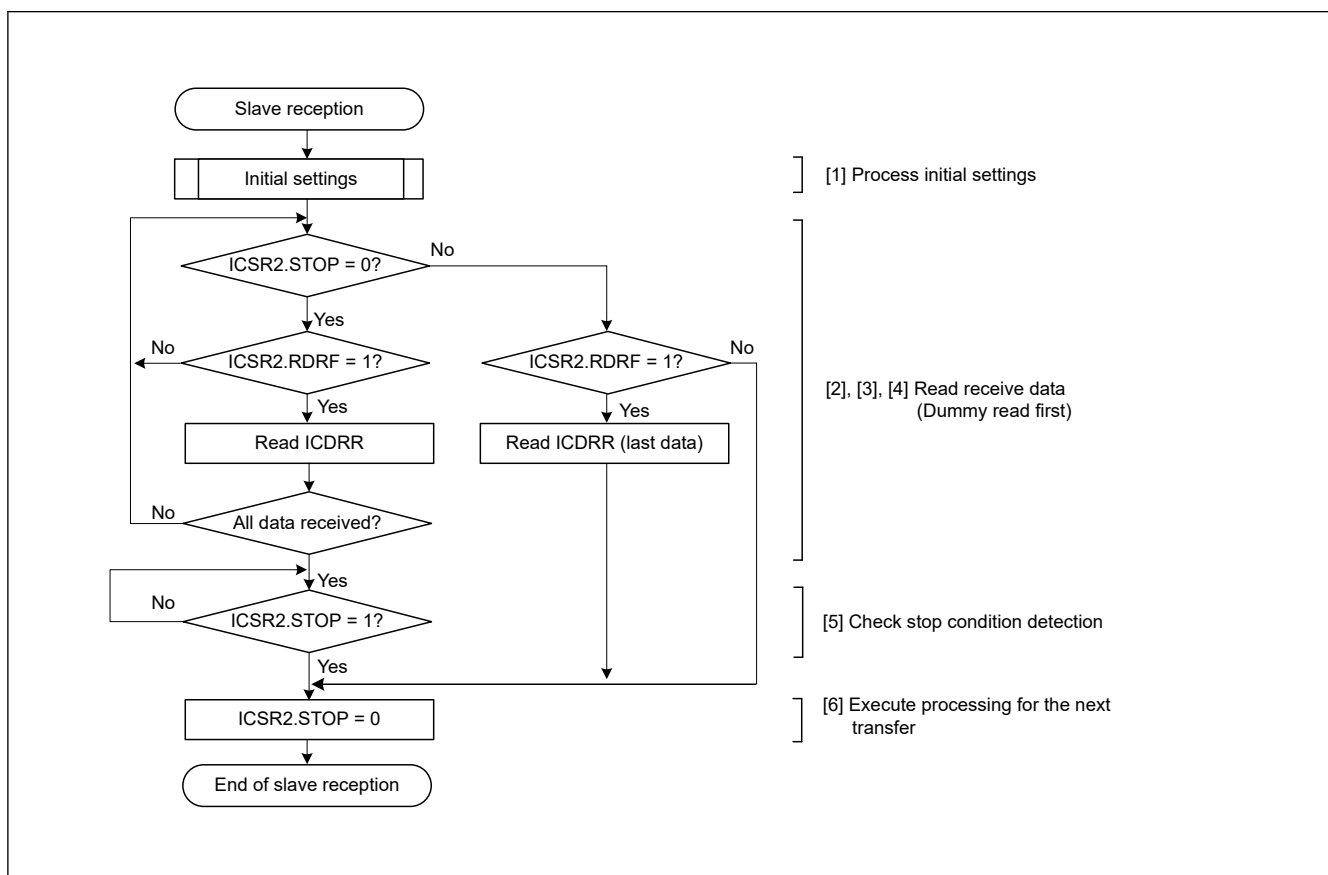
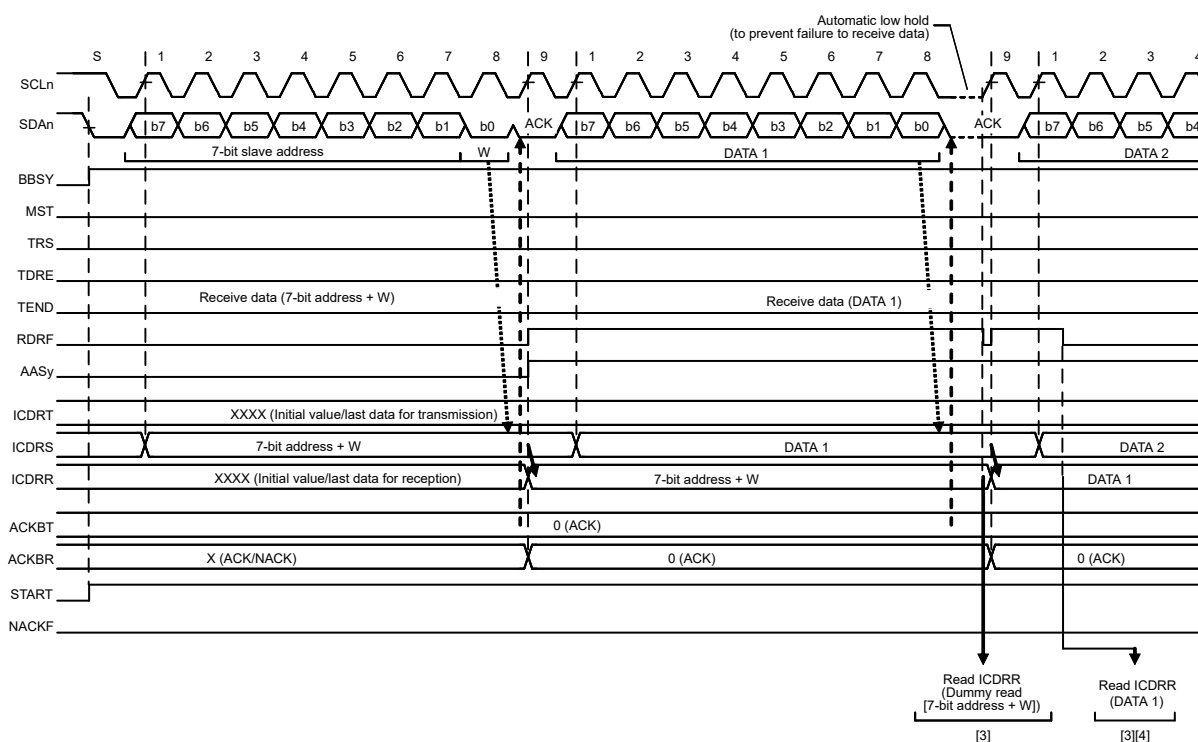
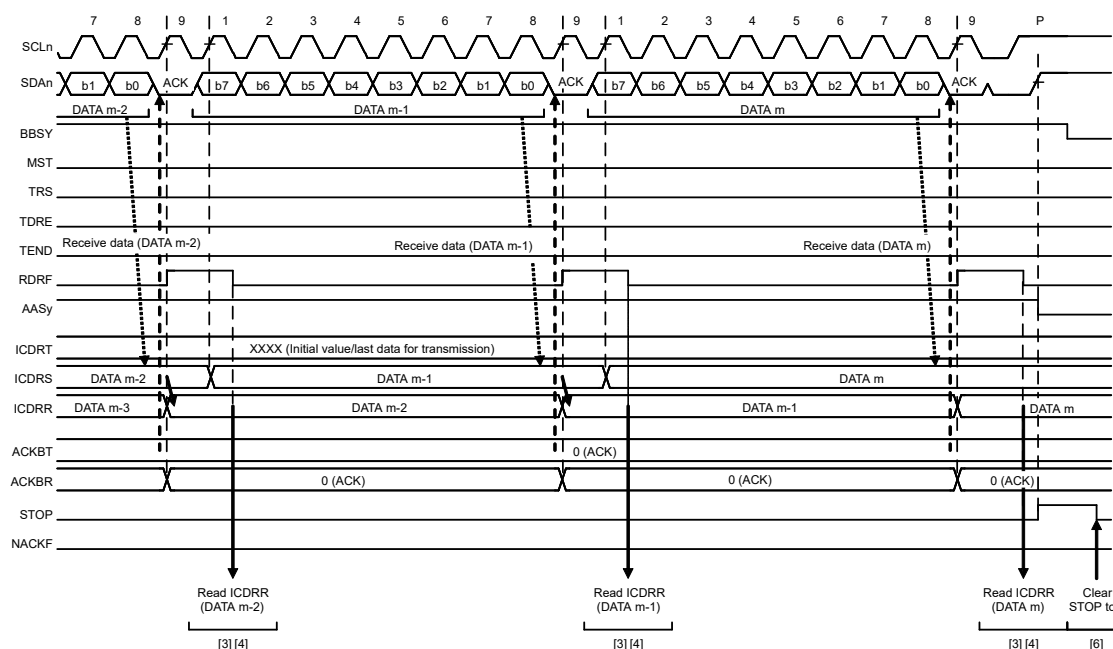


Figure 39.18 **Example slave reception flow**



Note: $n = 0, 1, y = 0 \text{ to } 2$

Figure 39.19 Slave receive operation timing (1) with 7-bit address format when RDRFS = 0



Note: $n = 0, 1, y = 0 \text{ to } 2$

Figure 39.20 Slave receive operation timing (2) when RDRFS = 0

39.4 SCL Synchronization Circuit

For generation of the SCL clock, the IIC starts counting the value for the high-level period specified in ICBRH when it detects a rising edge on the SCLn line, and it drives the SCLn line low when it completes counting. When the IIC detects the falling edge of the SCLn line, it starts counting the value for the low-level period specified in ICBRL, and then it stops driving the SCLn line, releasing the line, when it completes counting. The IIC repeats this process to generate the SCL clock.

If multiple master devices are connected to the I²C bus, a collision of SCL signals might arise because of contention with another master device. In such cases, the master devices must synchronize their SCL signals. Because this synchronization of SCL signals must be bit-by-bit, the IIC is equipped with an SCL synchronization circuit to obtain bit-by-bit synchronization of the SCL clock signals by monitoring the SCLn line while in master mode.

When the IIC detects a rising edge on the SCLn line and starts counting the high-level period specified in ICBRH.BRH[4:0], and the level on the SCLn line falls because an SCL signal is being generated by another master device, the IIC performs the following:

1. Stops counting when it detects the falling edge.
2. Drives the level on the SCLn line low.
3. Starts counting the low-level period specified in ICBRL.BRL[4:0].

When the IIC finishes counting the low-level period, it stops driving the SCLn line low to release the line. If the low-level period of the SCL clock signal from the other master device is longer than the low-level period set in the IIC, the low-level period of the SCL signal is extended. When the low-level period for the other master device ends, the SCL signal rises because the SCLn line is released.

When the IIC finishes outputting the low-level period of the SCL clock, the SCLn line is released and the SCL clock rises. That is, when SCL signals from more than one master are contending, the high-level period of the SCL signal is synchronized with that of the clock with the narrower period, and the low-level period of the SCL signal is synchronized with that of the clock with the broader period. However, such synchronization of the SCL signal is only enabled when the SCL bit in ICFER is set to 1.

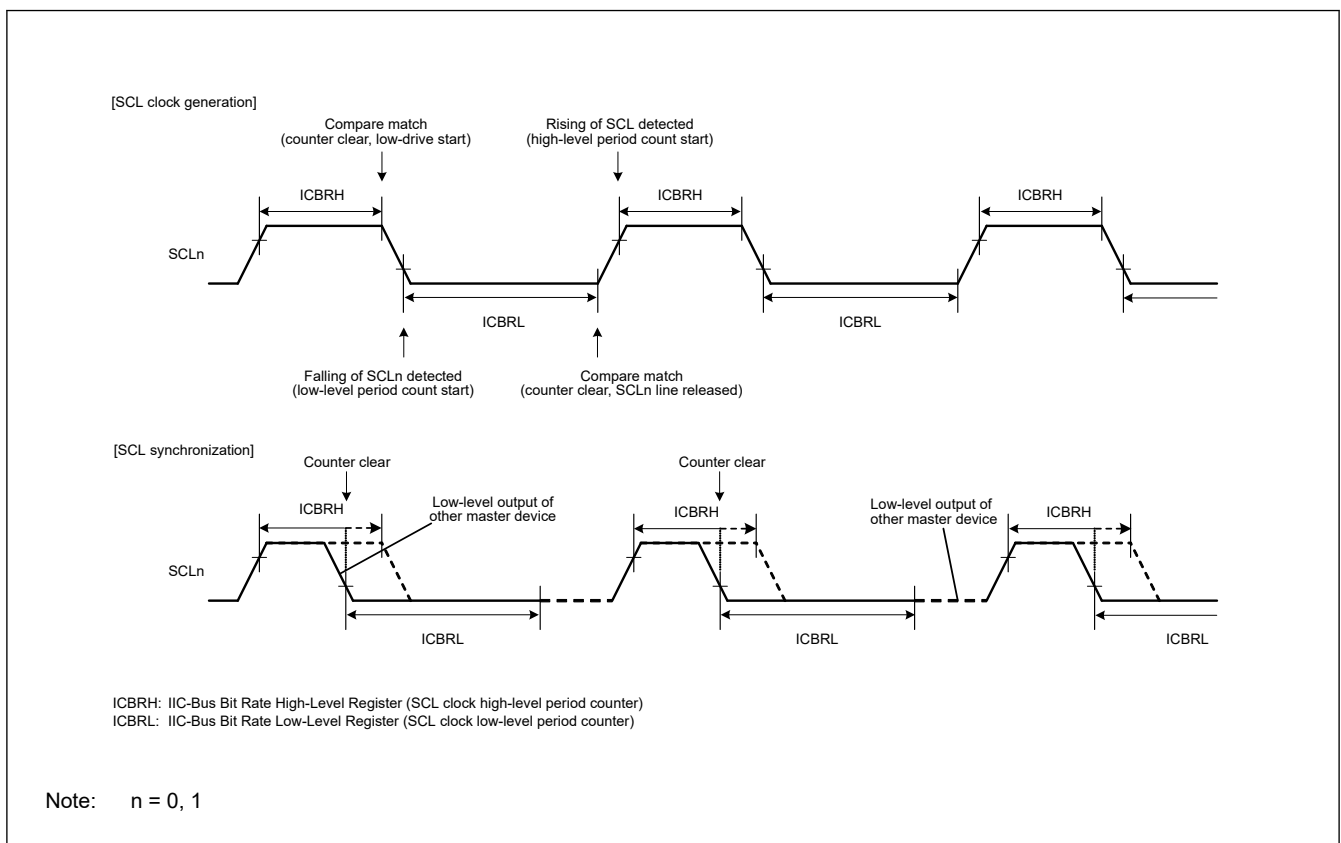


Figure 39.21 Generation and synchronization of SCL signal from IIC

39.5 SDA Output Delay Function

The IIC module incorporates a function for delaying output on the SDA line. The delay can be applied to all output on the SDA line, including issuing of the start, restart, and stop conditions, data, and the ACK and NACK signals.

With this function, SDA output is delayed from the detection of a falling edge of the SCL signal to ensure that the SDA signal is output within the interval during which the SCL clock is low. This approach helps prevent erroneous operation of communications devices, with the aim of satisfying the 300-ns minimum data-hold time requirement of the SMBus specification. The output delay function is enabled by setting the SDDL[2:0] bits in ICMR2 to a value other than 000b, and disabled by setting the same bits to 000b.

When the SDA output delay function is enabled, for example, the DLCS bit in ICMR2 selects the clock source for the SDA output delay counter, either as the internal base clock (IIC ϕ) for the IIC module or as the internal base clock divided by 2 (IIC ϕ /2). The counter counts the number of cycles set in the SDDL[2:0] bits in ICMR2. When the delay cycles count is reached, the IIC module places the required output (start, restart, or stop condition, data, or an ACK or NACK signal) on the SDA line.

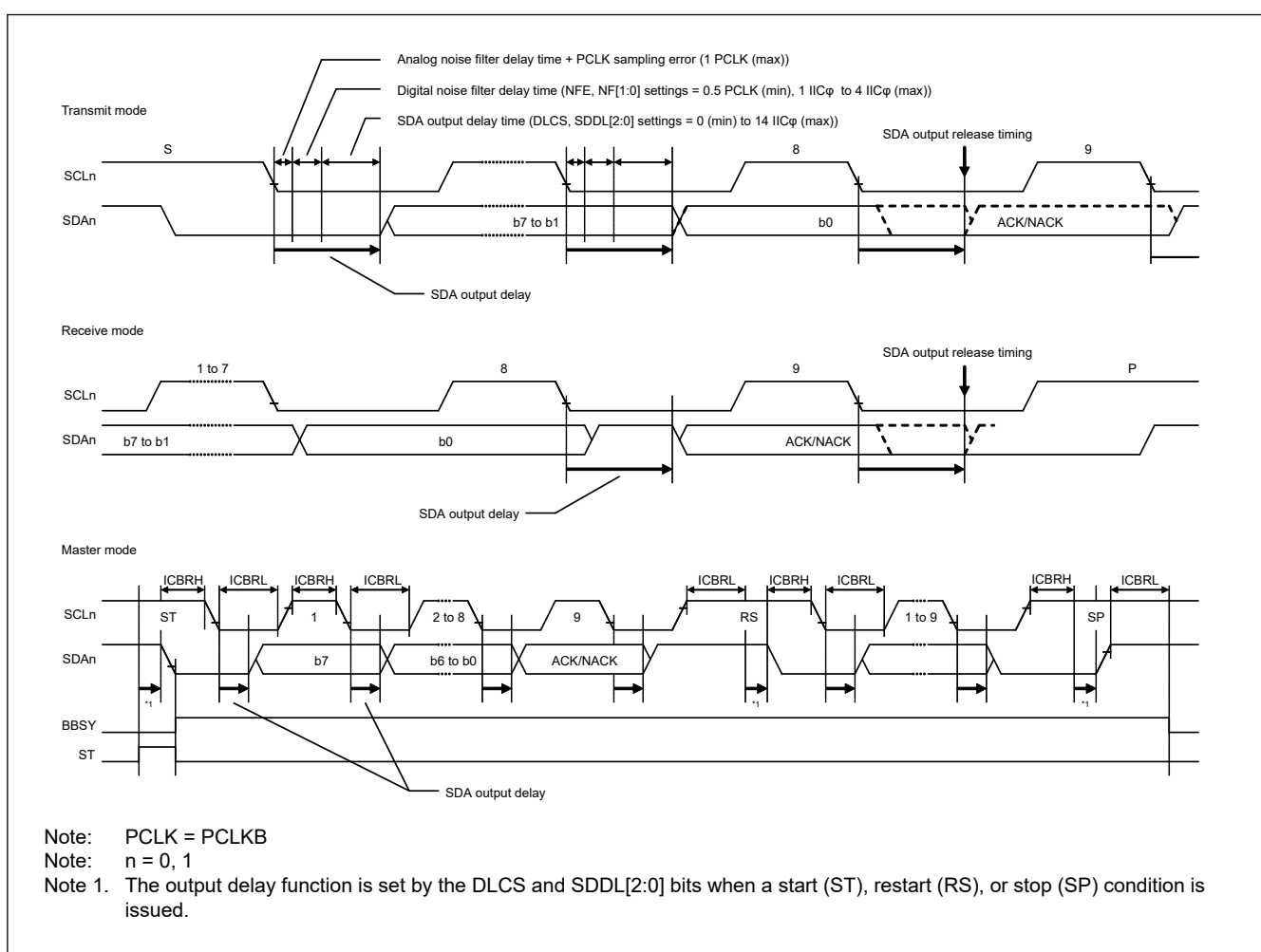


Figure 39.22 SDA output delay function

39.6 Digital Noise Filter Circuits

The internal circuitry sees the states of the SCLn and SDA pins through analog and digital noise-filter circuits. [Figure 39.23](#) shows a block diagram of the digital noise-filter circuit.

The on-chip digital noise-filter circuit of the IIC consists of four flip-flop circuit stages connected in series and a match-detection circuit. The number of valid stages in the digital noise filter is selected in the NF[1:0] bits in ICMR3. The selected number of valid stages determines the noise-filtering capability as a period from 1 to 4 IIC ϕ cycles.

The input signal to the SCLn pin (or SDAn pin) is sampled on falling edges of the IIC ϕ signal. When the input signal level matches the output level of the number of valid flip-flop circuit stages as selected in the NF[1:0] bits in ICMR3, the signal level is seen in the subsequent stage. If the signal levels do not match, the previous value is saved.

If the ratio between the frequency of the internal operating clock (PCLKB) and the transfer rate is small, for example, if data transfer is 400 kbps with PCLKB = 4 MHz, the characteristics of the digital noise filter might lead to the elimination of required signals as noise. In such cases, it is possible to disable the digital noise-filter circuit by setting the ICFER.NFE bit to 0, and use only the analog noise filter circuit.

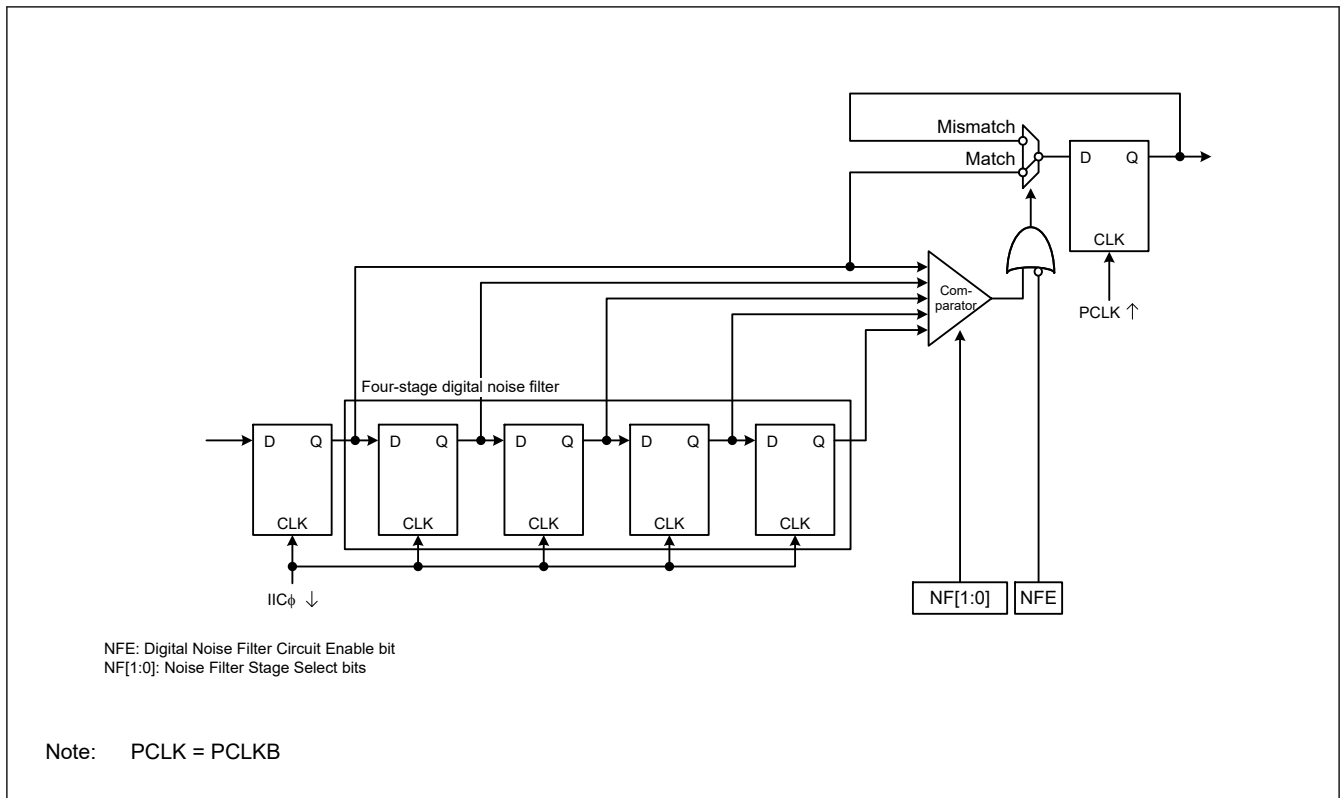


Figure 39.23 Digital noise filter circuit block diagram

39.7 Address Match Detection

The IIC can set three unique slave addresses in addition to the general call address and host address. The slave addresses can be 7-bit or 10-bit slave addresses.

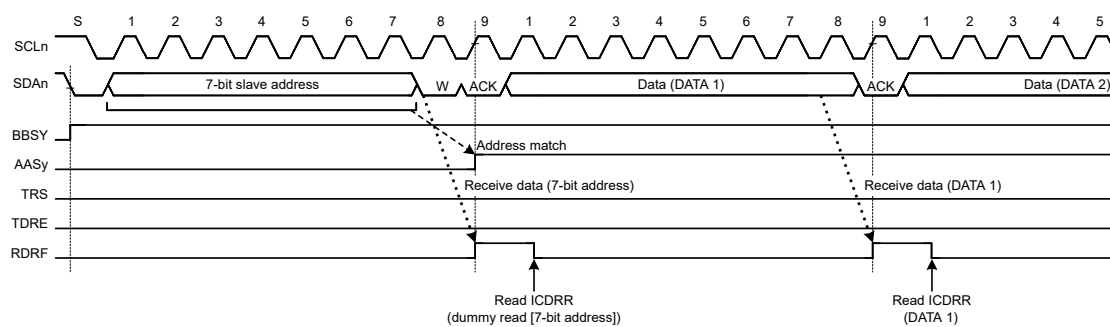
39.7.1 Slave-Address Match Detection

The IIC can set three unique slave addresses and has a slave address detection function for each unique slave address. When the SARyE bit (y = 0 to 2) in ICSE is set to 1, the slave addresses set in SARUy and SARLy (y = 0 to 2) can be detected.

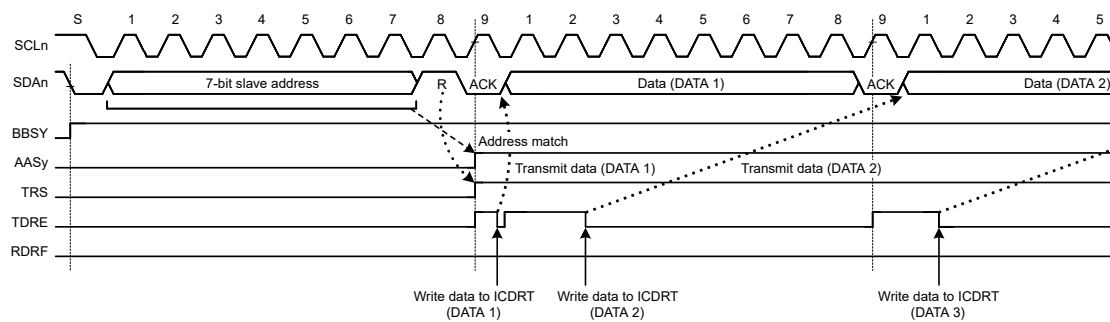
When the IIC detects a match of the set slave address, the associated AASy flag (y = 0 to 2) in ICSR1 is set to 1 on the rising edge of the ninth SCL clock cycle, and the RDRF flag in ICSR2 or the TDRE flag in ICSR2 is set to 1 by the subsequent R/W# bit. This causes a receive data full interrupt (IICn_RXI) or transmit data empty interrupt (IICn_TXI) to be generated. The AASy flag identifies which slave address is specified.

Figure 39.24 to Figure 39.26 show the AASy flag set timing in three cases.

[7-bit address format: slave reception]



[7-bit address format: slave transmission]

Note: $n = 0, 1, y = 0 \text{ to } 2$ **Figure 39.24 AASy flag set timing with 7-bit address format**

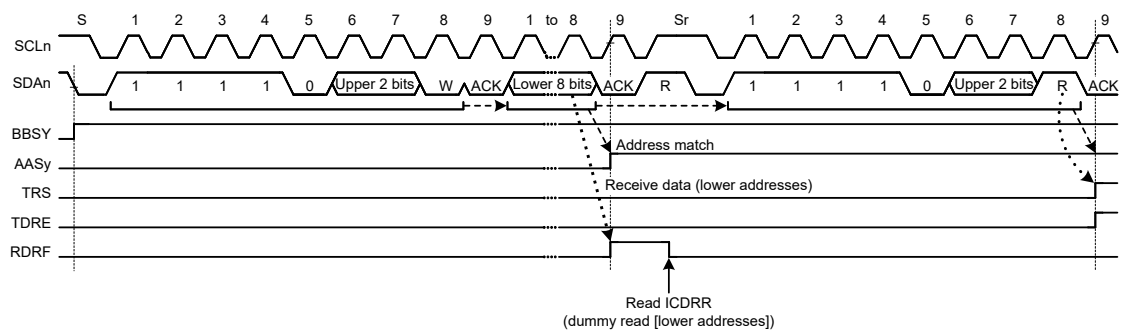
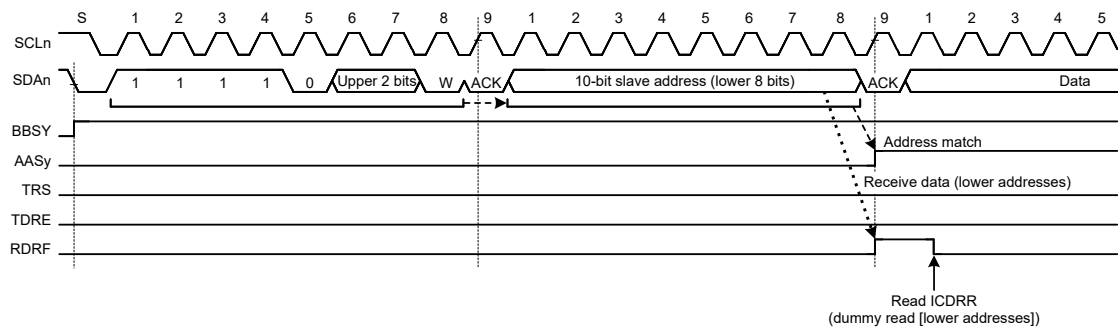


Figure 39.25 AASy flag set timing with 10-bit address format

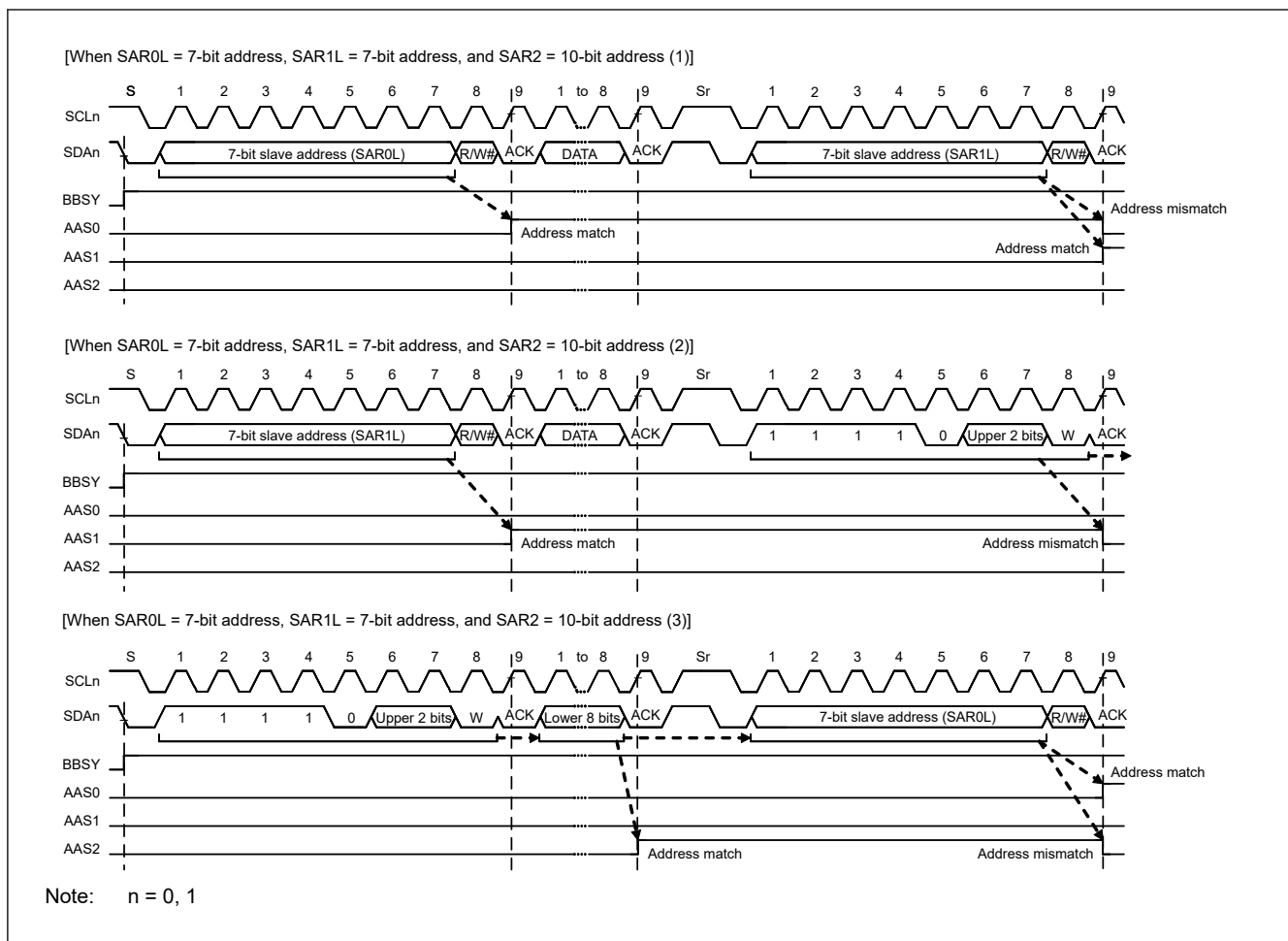


Figure 39.26 AASy flag set and clear timing with mixed 7-bit and 10-bit address formats

39.7.2 Detection of General Call Address

The IIC provides detection of the general call address (0000 000b + 0 [W]). This is enabled by setting the GCAE bit in IC SER to 1.

If the address received after a start or restart condition is issued is 0000 000b + 1[R] (start byte), the IIC recognizes this as the address of a slave device with an all-zero address, but not as the general call address.

When the IIC detects the general call address, both the GCA flag in ICSR1 and the RDRF flag in ICSR2 set to 1 on the rising edge of the ninth cycle of the SCL clock. This leads to the generation of a receive data full interrupt (IICn_RXI). The value of the GCA flag can be checked to confirm that the general call address was transmitted.

Operation after detection of the general call address is the same as normal slave receive operation.

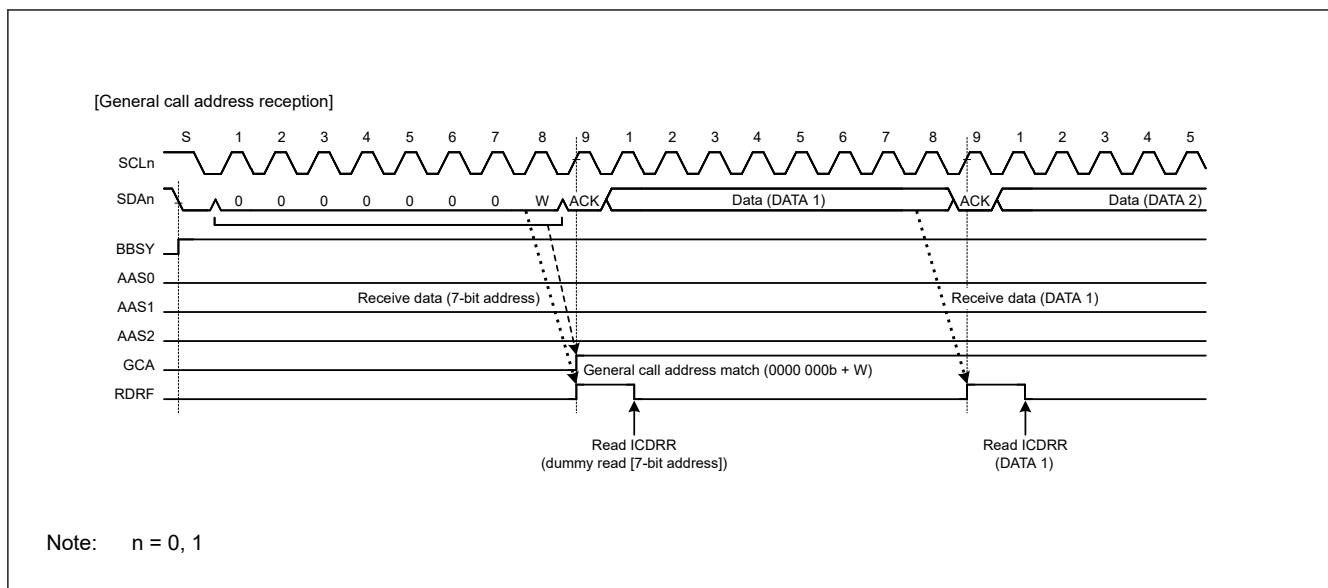


Figure 39.27 Timing of GCA flag setting during reception of general call address

39.7.3 Device-ID Address Detection

The IIC module provides detection of device-ID address compliant with the I²C bus specification (revision 03). When the IIC receives 1111 100b as the first byte after a start or restart condition is issued with the DIDE bit in ICSER set to 1, it recognizes the address as a device ID, sets the DID flag in ICSR1 to 1 on the rising edge of the ninth SCL clock cycle when the subsequent R/W# bit is 0, then compares the second and subsequent bytes with its own slave address. If the address matches the value in the slave address register, the IIC sets the associated AASy flag ($y = 0$ to 2) in ICSR1 to 1.

When the first byte received after the issue of a start or restart condition matches the device ID address (1111 100b) again and the subsequent R/W# bit is 1, the IIC does not compare the second and subsequent bytes and sets the ICSR2.TDRE flag to 1.

In the device ID address detection function, the IIC sets the DID flag to 0 if a match with the IIC slave address is not obtained or a match with the device ID address is not obtained after a match with the IIC slave address and a restart condition is not detected. If the first byte after detection of a start or restart condition matches the device ID address (1111 100b), and the R/W# bit is 0, the IIC sets the DID flag to 1 and compares the second and subsequent bytes with the slave address of the IIC. If the R/W# bit is 1, the DID flag holds the previous value and the IIC does not compare the second and subsequent bytes. Therefore, the reception of a device ID address can be checked by reading the DID flag after confirming that TDRE = 1.

Additionally, prepare the device ID fields (3 bytes: 12 bits indicating the manufacturer + 9 bits identifying the part + 3 bits indicating the revision) that must be sent to the host after reception of a continuous device-ID field as normal transmit data. For details on the information that must be included in device ID fields, contact NXP Semiconductors.

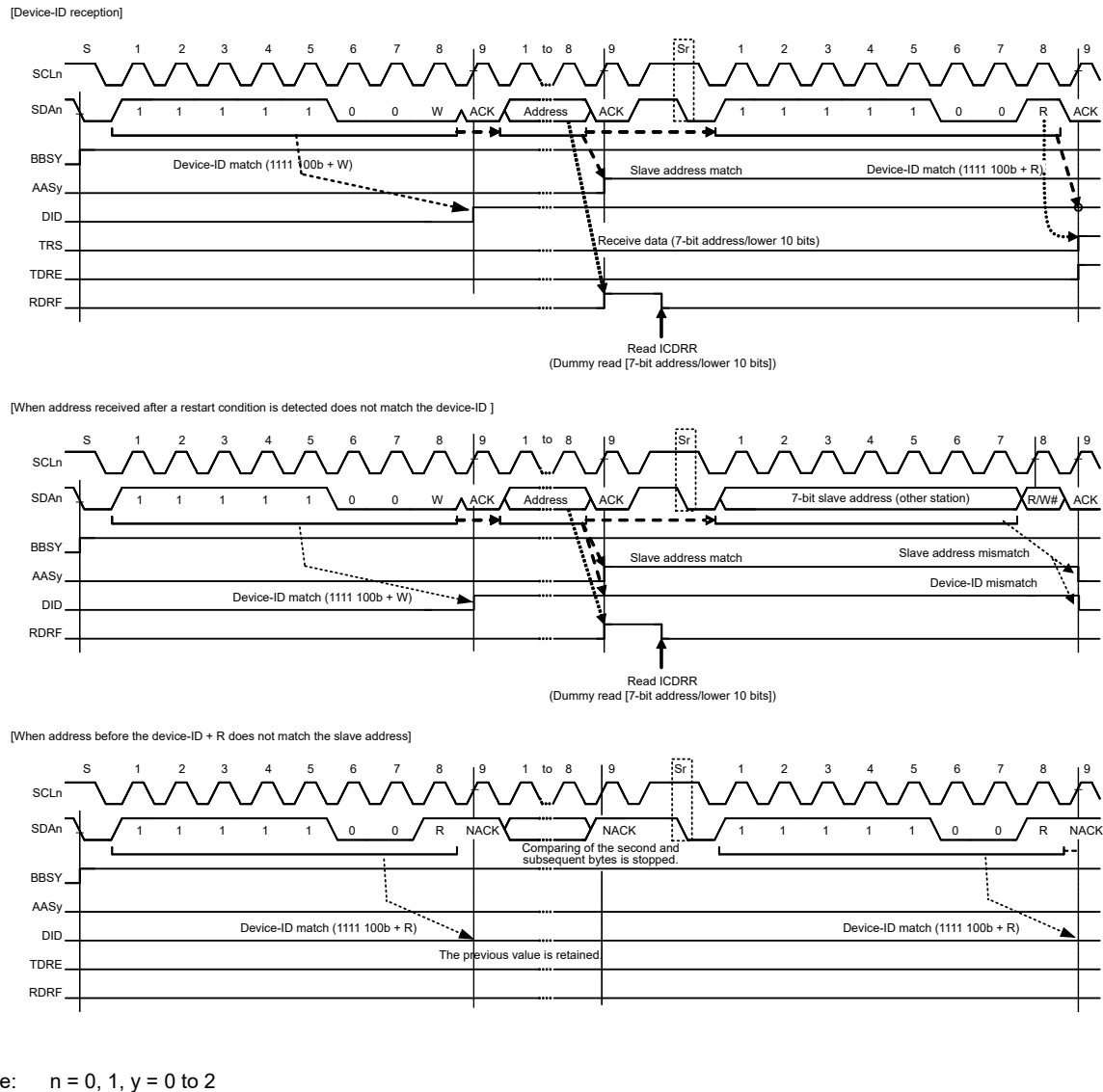


Figure 39.28 AASy and DID flag set and clear timing during reception of device ID

39.7.4 Host Address Detection

The IIC provides host address detection when operating in SMBus. When the HOAE bit in ICSER is set to 1 while the SMBS bit in ICMR3 is 1, the IIC can detect the host address (0001 000b) in slave receive mode (MST and TRS bits = 00b in ICCR2).

When the IIC detects the host address, the HOA flag in ICSR1 is set to 1 on the rising edge of the 9th SCL clock cycle, and at the same time, the RDRF flag in ICSR2 is set to 1 when the R/W# bit is 0 (Wr bit). This causes a receive data full interrupt (IICn_RXI) to be generated. The HOA flag indicates that the host address was sent from another device.

If the bit following the host address (0001 000b) is an Rd bit (R/W# bit = 1), the IIC can also detect the host address. After the host address is detected, the IIC operates in the same manner as in normal slave operation.

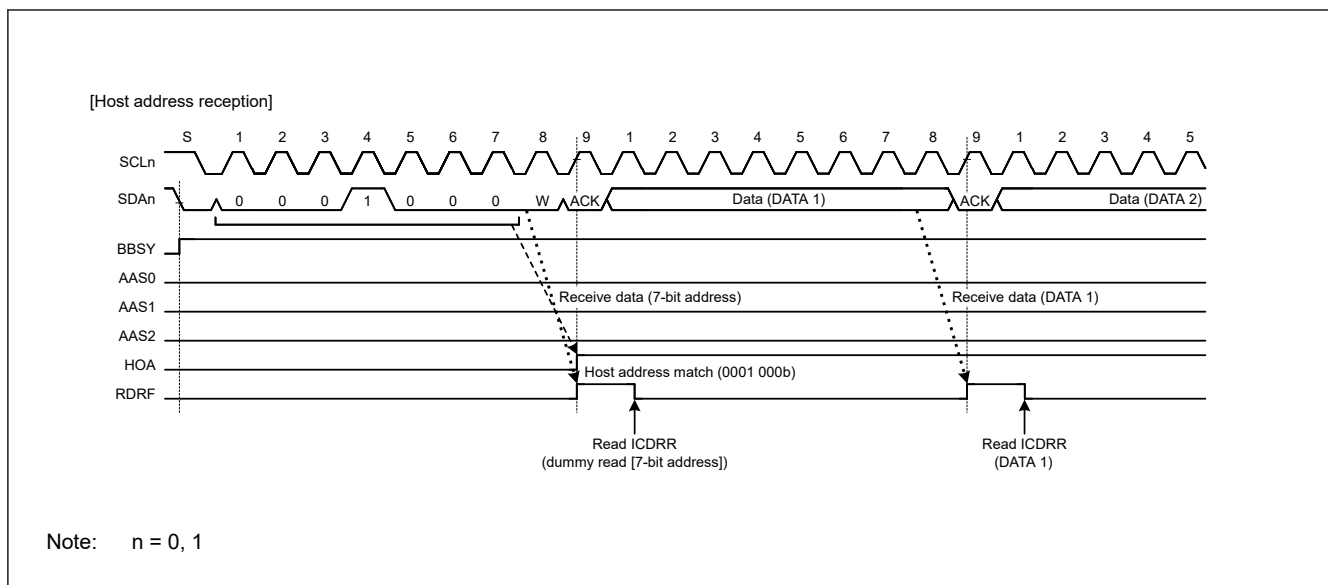


Figure 39.29 HOA flag set timing during reception of host address

39.8 Wakeup Function

The IIC provides a wakeup function that causes the MCU to transition from Software Standby mode to normal operation. The wakeup function enables the reception of data when the peripheral module clock (PCLKB) is stopped, and generates a wakeup interrupt signal on a match of the slave address of the received data. This wakeup interrupt signal triggers the return to normal operation. After the wakeup interrupt occurs, switch the IIC to PCLKB synchronous operation so that communication can continue.

The wakeup function has four operation modes:

- Normal wakeup mode 1
- Normal wakeup mode 2
- Command recovery mode
- EEP response mode

Table 39.9 describes the behavior in these modes.

Table 39.9 Wakeup operation modes

Operation mode	ACK response timing	ACK response before wakeup to PCLKB synchronous operation	SCL state during wakeup to PCLKB synchronous operation
Normal wakeup mode 1	Before wakeup to PCLKB synchronous operation ^{*1}	ACK	Fixed low
Normal wakeup mode 2	After wakeup to PCLKB synchronous operation ^{*2}	Before wakeup: no response (NACK level retained) After wakeup: ACK response	Fixed low
Command recovery mode	Before wakeup to PCLKB synchronous operation ^{*1}	ACK	Open
EEP response mode	Before recovery to PCLKB synchronous operation ^{*1}	NACK	Open

Note 1. Switching timing from PCLKB asynchronous operation to PCLKB synchronous operation is the falling edge of the 9th clock of the SCL.

Note 2. Switching timing from PCLKB asynchronous operation to PCLKB synchronous operation is the falling edge of the 8th clock of the SCL.

The following can be selected as wakeup interrupt sources:

- Host address detection (valid when `ICSER.HOAE = 1`)
- General call address detection (valid when `ICSER.GCAE = 1`)

- Slave address 0*1 detection (valid when IC SER.SAR0E = 1)
- Slave address 1*1 detection (valid when IC SER.SAR1E = 1)
- Slave address 2*1 detection (valid when IC SER.SAR2E = 1)

Note 1. Only 7-bit address can be set. Set the FS bit in SARUy (y = 0 to 2) to 0.

Precautions on the use of the wakeup function

- Do not change the content of the IIC registers except the WUSEN bit in ICWUR2 while the WUASYF flag in ICWUR2 is 1 (during PCLKB asynchronous operation).
- Set ICWUR.WUE and ICWUR.WUIE to 1, and ICCR2.MST and ICCR2.TRS to 0 (slave reception mode) before switching to PCLKB asynchronous mode.
- The device ID and the 10-bit slave address cannot be selected for the wakeup interrupt source. Set the DIDE bit in IC SER and FS bit in SARUy (y = 0 to 2) to 0.
- Set bits TIE, TEIE, RIE, NAKIE, SPIE, STIE, ALIE, and TMOIE in the ICIER register to 0 (interrupt disabled) before switching to the asynchronous operation.
- When the wakeup function is enabled, do not use the timeout function (ICWUR.WUE = 1)
- Even when a wakeup interrupt is generated during PCLKB asynchronous operation (when ICWUR2.WUASYF = 1), if the slave addresses match in PCLKB synchronous mode (ICWUR2.WUASYF = 0), the wakeup interrupt does not occur and the WUF flag is not set.
- If the timing of writing 0 to the ICWUR2.WUSEN bit and the timing of detecting a start condition conflict, the IIC might start the next reception in PCLKB synchronous operation mode. In this case, ICWUR2.WUASYF flag becomes 1 (switch to PCLKB asynchronous mode) when data communication is complete, a stop condition is detected, and detection of a wakeup event starts.
- If you want to switch from PCLKB asynchronous operation to PCLKB synchronous operation without address match detection, it will switch in the stop condition detection. When the ICWUR2.WUSEN bit was set to 1 in a bus free state, it is continued PCLKB asynchronous operation (Reception operation: waiting communication frame). ICWUR2.WUASYF flag becomes to 1 when IIC detect the stop condition of the next communication frame, and IIC switches to PCLKB synchronous operation.
- After writing 0 to the WUSEN bit in ICWUR2, do not change registers relate to the IIC operation mode setting (ICMR3, IC SER, and SARLy) until the mode is switched to PCLKB asynchronous operation from PCLKB synchronous operation (while the ICWUR2.WUASYF flag is 1). If the register value changes during this period by an interrupt handling or another factor, the IIC might malfunction before switching to the asynchronous operation.
- During PCLKB asynchronous operation (ICWUR2.WUASYF = 0 (or WUASYF = 1)), do not refer to each flag of IC SR1, IC SR2 register and ICCR2.BBSY flag.

39.8.1 Normal Wakeup Mode 1

This section describes the behavior, the timing, and an example operation of normal wakeup mode 1.

In normal wakeup mode 1, a wakeup interrupt triggered by the match of the slave address initiates the transition to normal operation as follows:

- Before wakeup: ACK is sent in response to the data received with its own slave address of the IIC.
- During wakeup: ACK response is made on the 9th clock cycle of SCL, after which SCL is held low*1.
- After wakeup: Normal operation continues.

Note 1. Between the 9th clock cycle and 1st clock cycle during wakeup, ICMR1.WAIT = 1 is invalid.

If the slave address does not match, the SCL line is not held low after the 9th clock cycle of SCL, and the slave operation continues. [Figure 39.30](#) shows an operation example, and [Figure 39.32](#) shows the detailed timing.

If the transition from Software Standby mode is triggered by an interrupt other than a wakeup interrupt, for example the IRQn, the WUF flag is not set to 1. [Figure 39.31](#) shows an operation example.

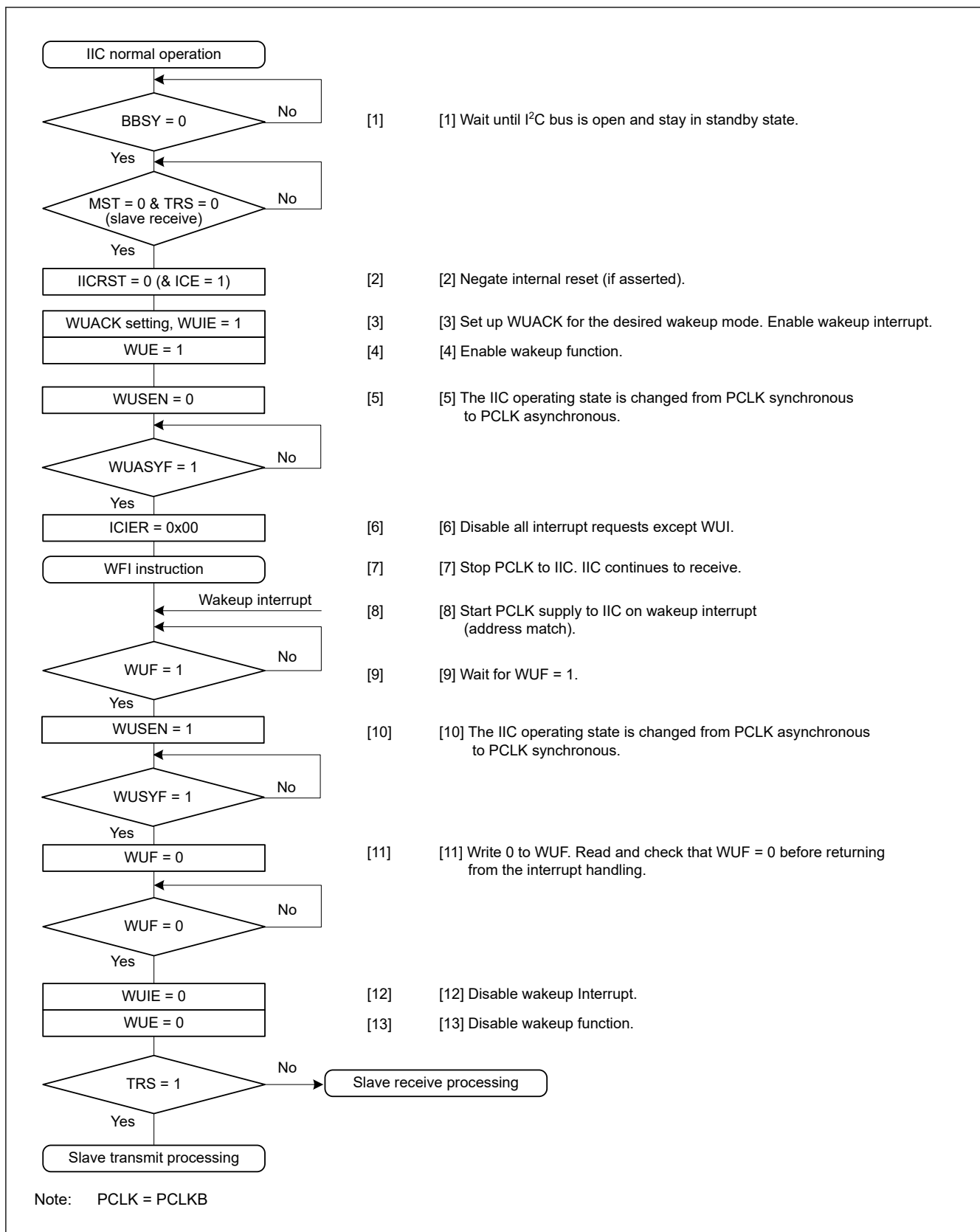


Figure 39.30 Example operation of normal wakeup mode 1 when wakeup is triggered by a wakeup interrupt on match of the slave address

Note: See [Precautions on the use of the wakeup function](#).

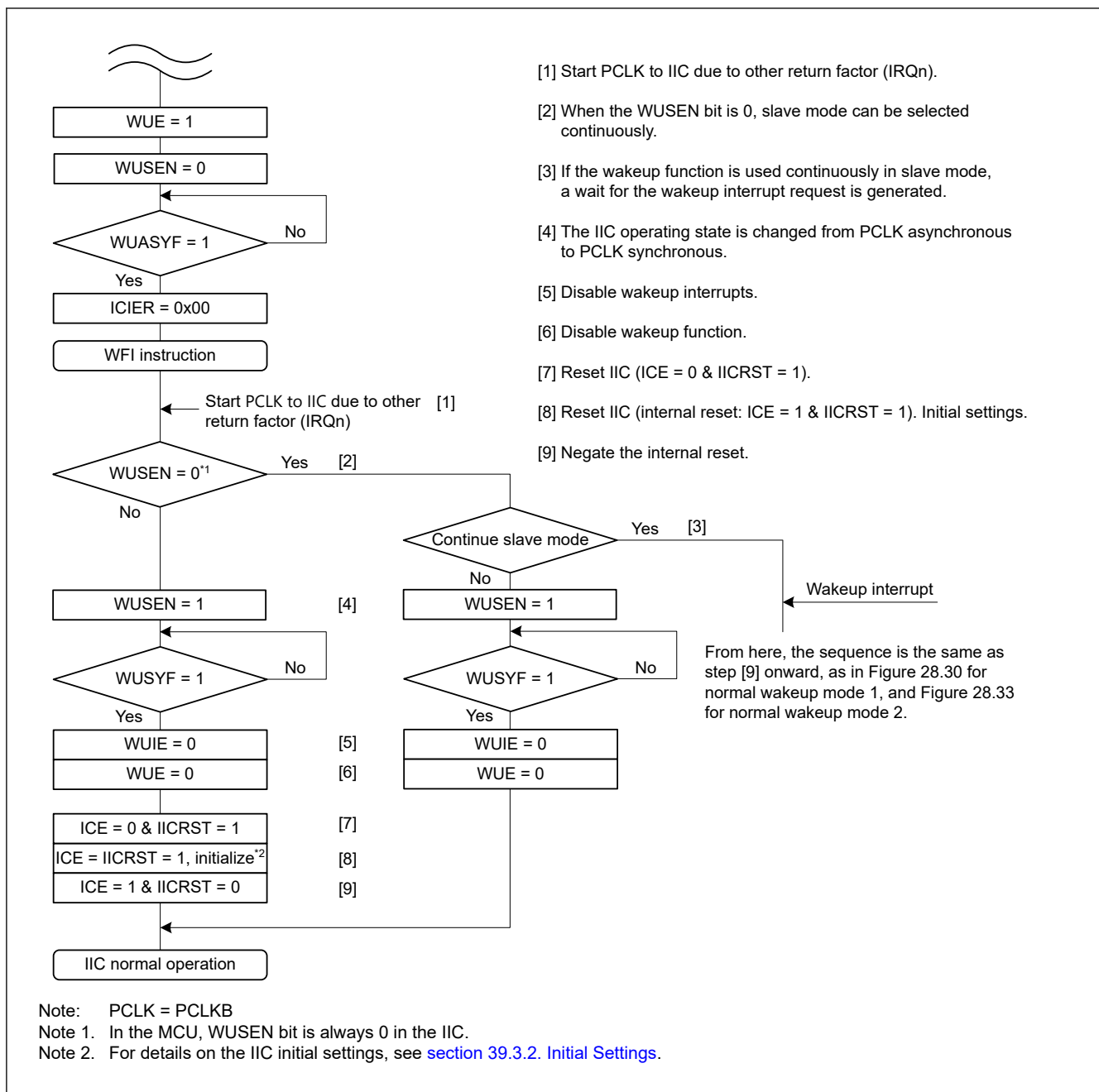


Figure 39.31 Example operation of normal wakeup modes 1 and 2 when wakeup is triggered by an interrupt other than IIC wakeup interrupt, for example, the IRQn

Note: For details on the IIC initial settings, see [section 39.3.2. Initial Settings](#).

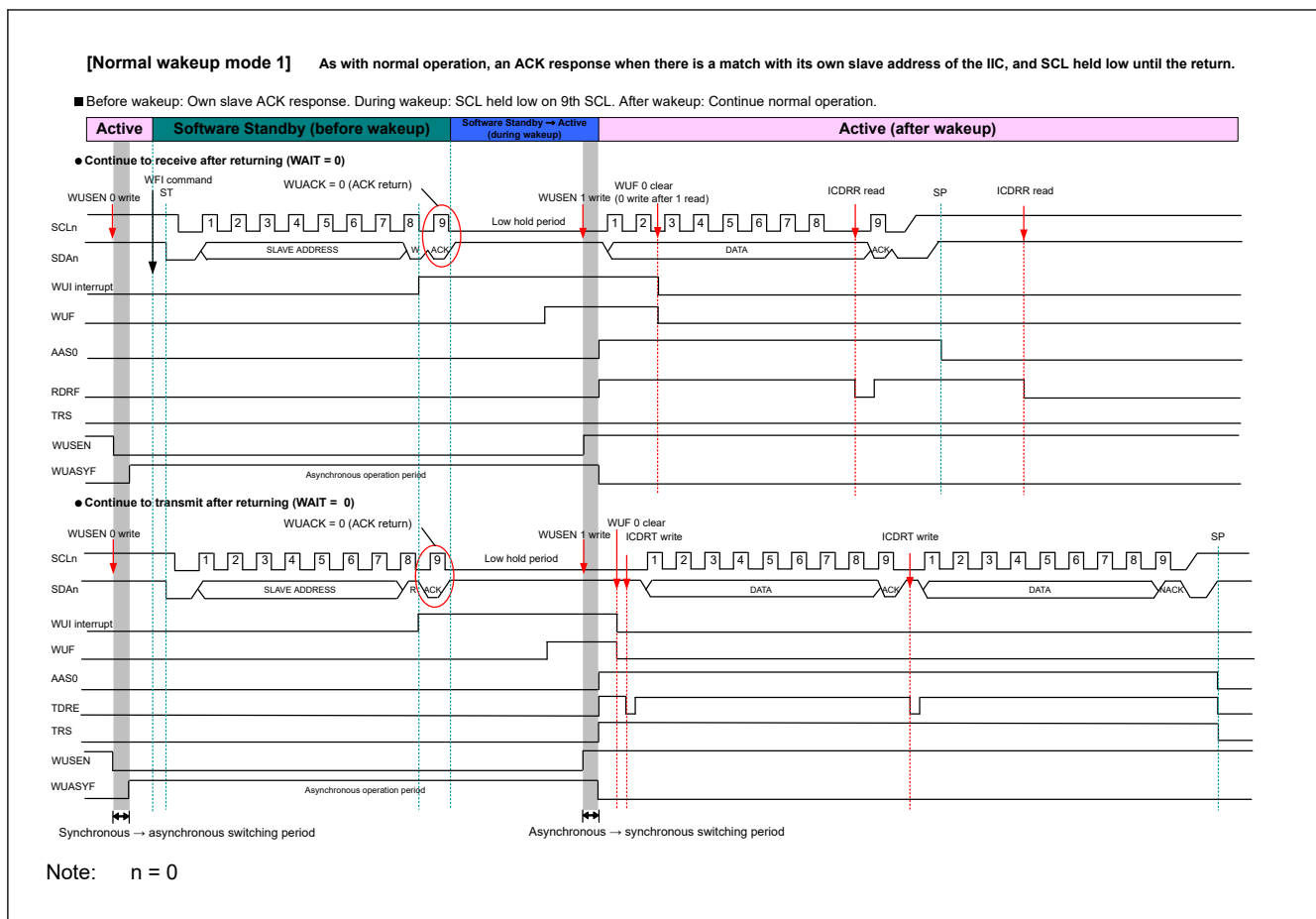


Figure 39.32 Timing of normal wakeup mode 1

39.8.2 Normal Wakeup Mode 2

This section describes the behavior, the timing, and an example operation of normal wakeup mode 2.

In normal wakeup mode 2, a wakeup interrupt triggered by a match of the slave address initiates the transition to normal operation as follows:

- Before wakeup: No response to data received with its own slave address until the end of the 8th SCL cycle.
- During wakeup: SCL line held low during the 8th and 9th clock cycles.
- After wakeup: ACK returns on the 9th clock cycle of SCL, and normal operation continues.

If the slave address does not match, the SCL line is not held low after the 8th SCL clock cycle, and the slave operation continues. Figure 39.33 shows an example operation, and Figure 39.34 shows the detailed timing.

If the transition from Software Standby mode is triggered by an interrupt other than a wakeup interrupt, such as the IRQn, for example, the WUF flag is not set to 1. Figure 39.31 shows an operation example.

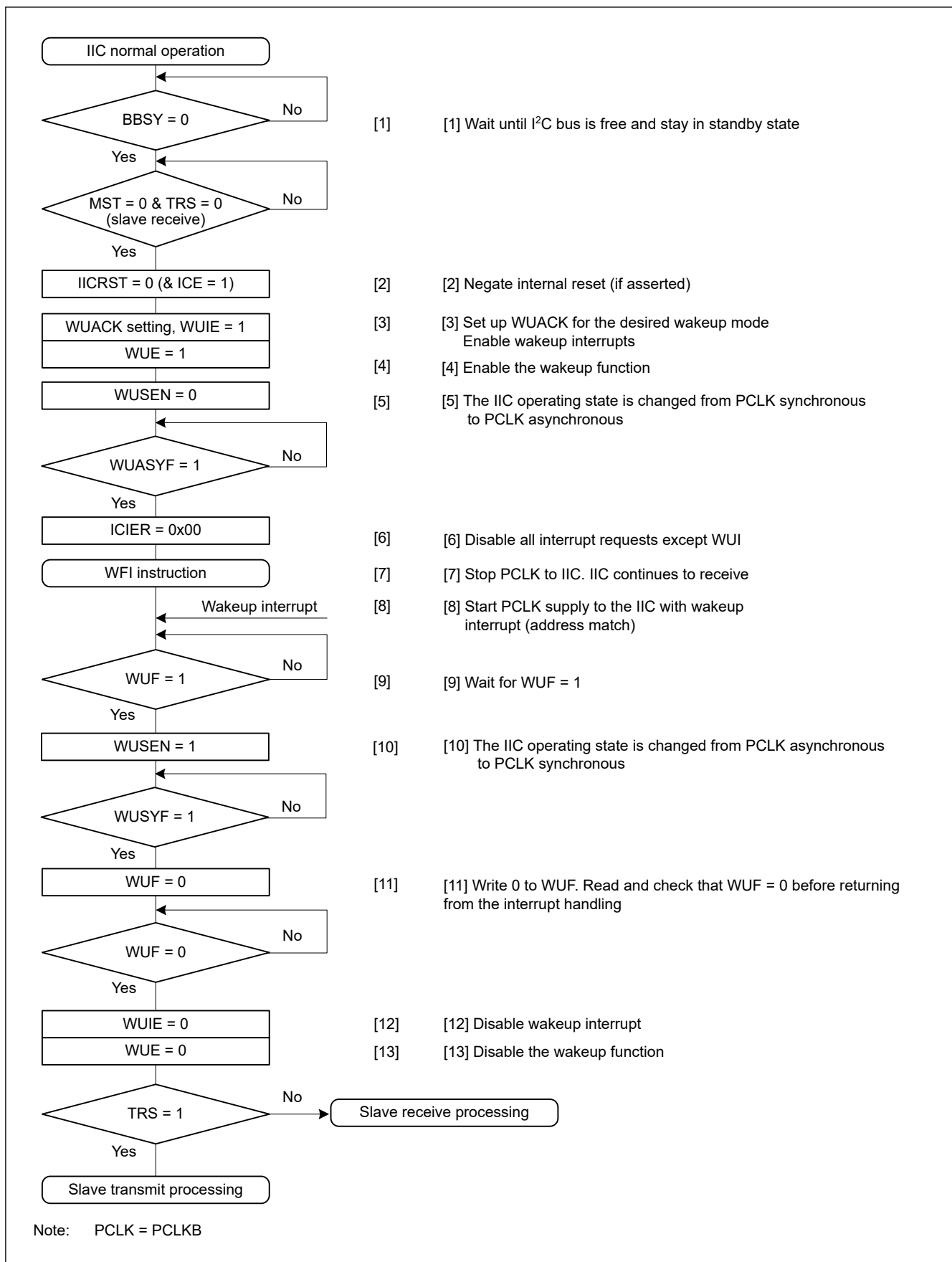


Figure 39.33 Example operation of normal wakeup mode 2 when wakeup is triggered by a wakeup interrupt on match of the slave address

Note: See [Precautions on the use of the wakeup function](#).

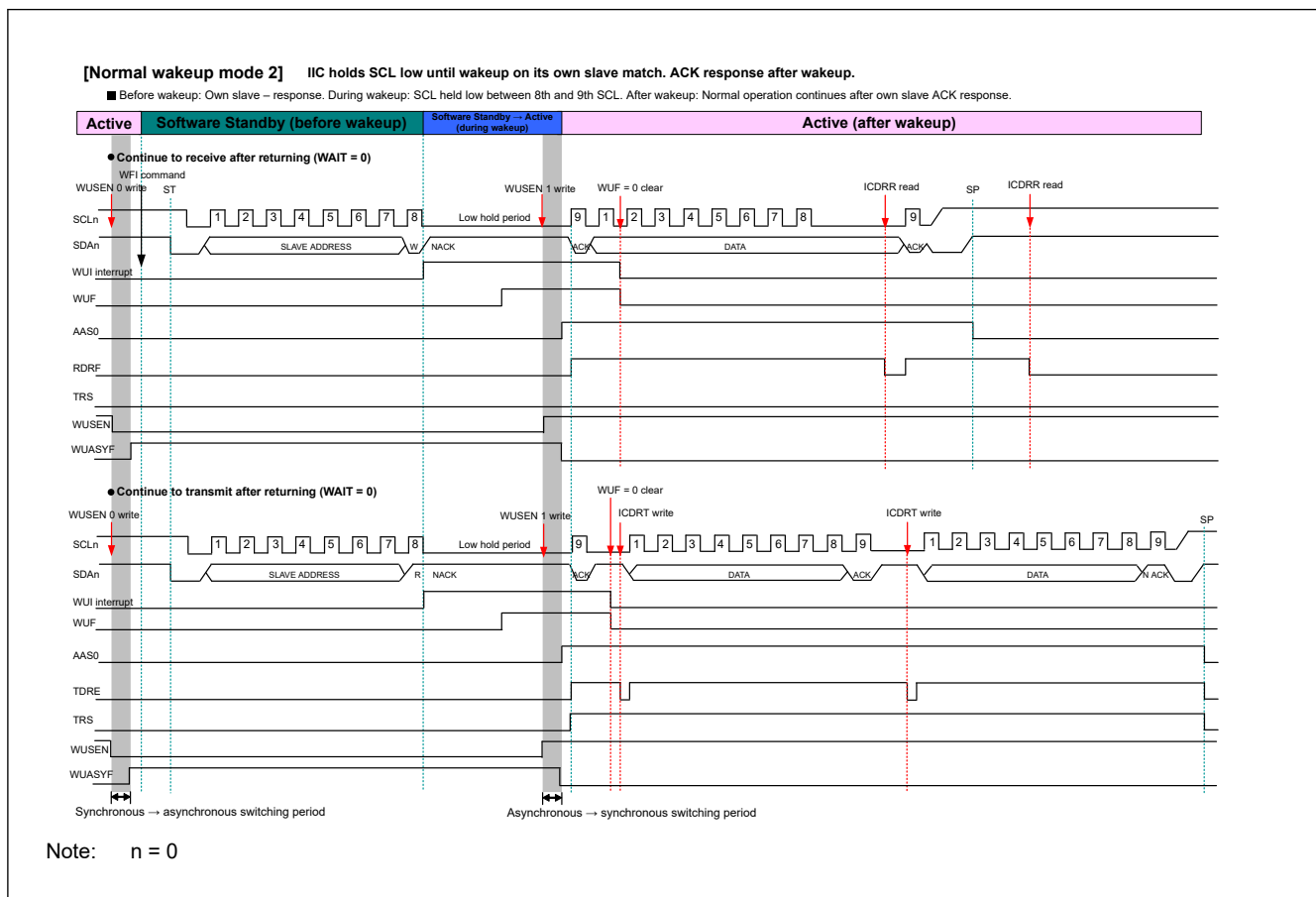


Figure 39.34 Timing of normal wakeup mode 2

39.8.3 Command Recovery Mode and EEP Response Mode (Special Wakeup Modes)

This section describes the behavior, the timing, and example operations of the command recovery and EEP response modes. In the command recovery and EEP response modes, the SCL line is not held low during the wakeup period (after the rise of the 9th clock cycle of SCL). Therefore, other I2C devices can use the I2C bus during this period.

A wakeup interrupt triggered by a match of the slave address initiates the transition to normal operation as follows:

- Before wakeup: In response to the data received with its own slave address, the IIC returns ACK (command recovery mode) or NACK (EEP response mode).
- During wakeup: The SCL line is not held low.
- After wakeup: Normal operation continues after IIC initialization.

If the slave address does not match, the slave operation continues.

Note: Because the SCL line is not held low during wakeup, transmission or reception of the data that follows the slave address is not possible.

Note: The command recovery and EEP response modes are internal reset states (ICE = IICRST = 1). Therefore, the match of the slave address does not set the flags, HOA, GCA, ASS0, ASS1, and ASS2 in the ICSR1 register.

Figure 39.35 shows an example operation in recovery and EEP response modes. Figure 39.37 shows the detailed timing.

If the transition from Software Standby mode is triggered by an interrupt other than a wakeup interrupt, such as IRQn for example, the WUF flag is not set to 1. Follow the processing shown in Figure 39.36.

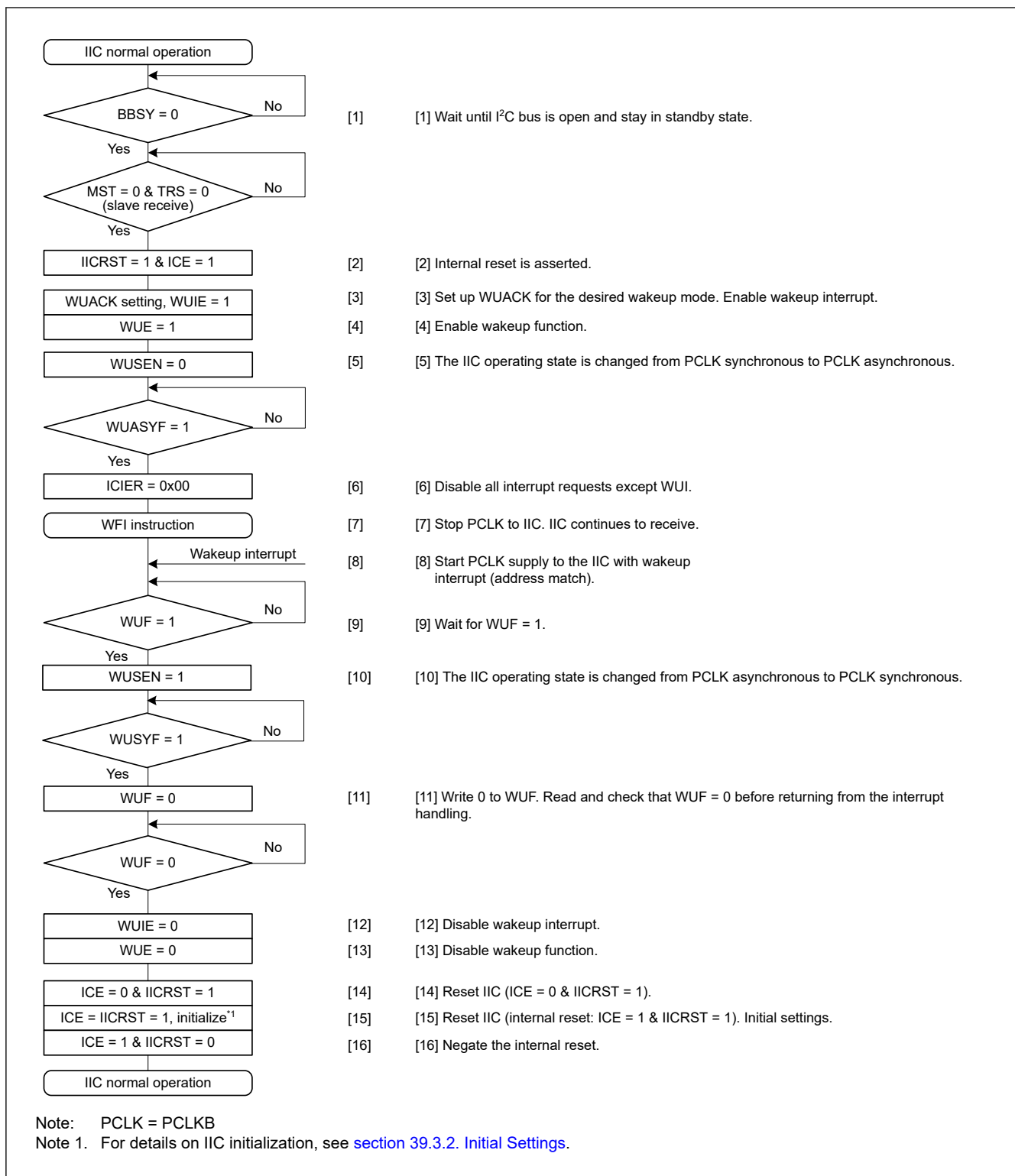


Figure 39.35 Example operation of command recovery mode and EEP response mode when wakeup is triggered by a wakeup interrupt on match of the slave address

Note: See [Precautions on the use of the wakeup function](#).

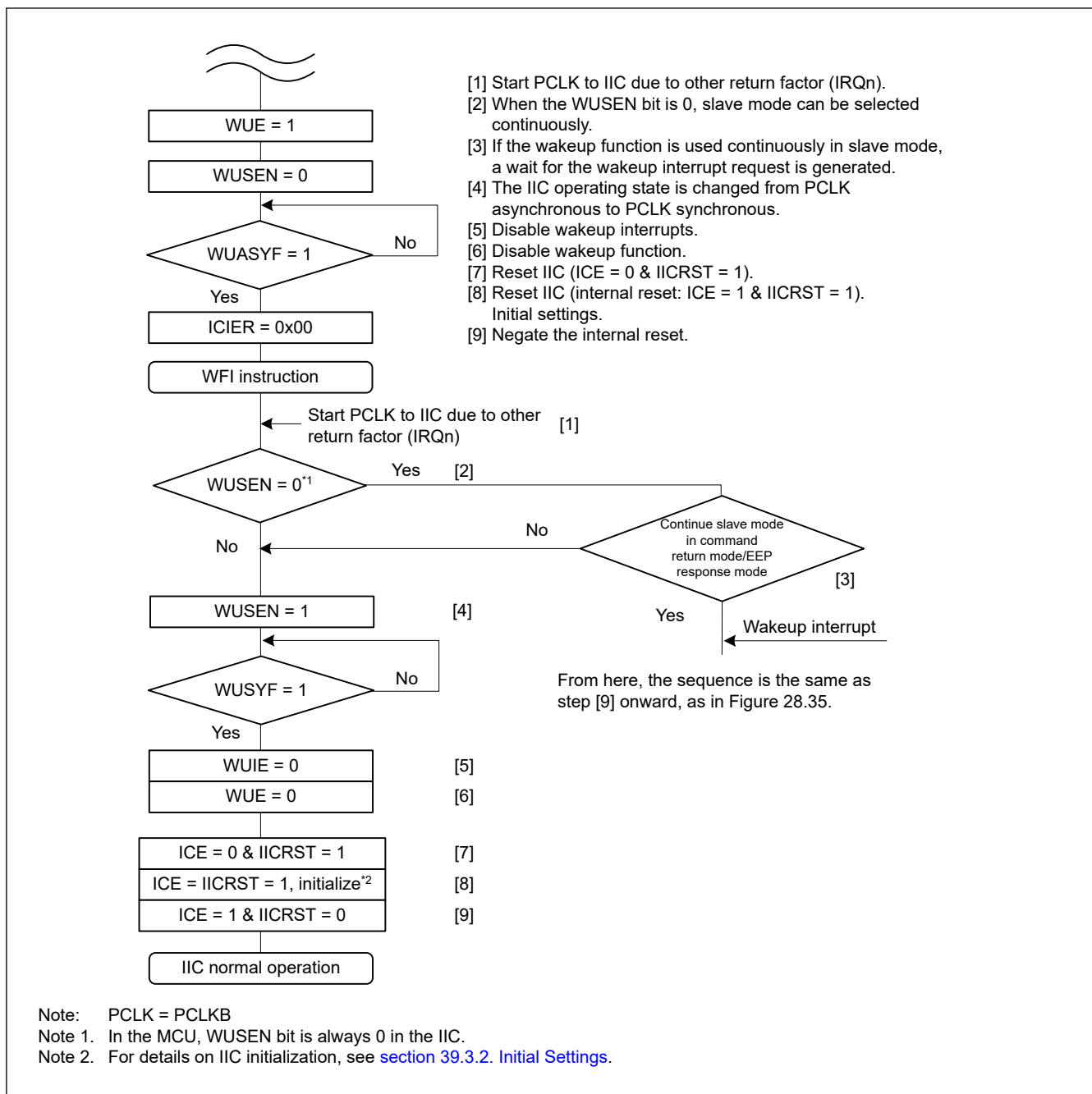


Figure 39.36 Example operation of command recovery and EEP response modes when wakeup is triggered by an interrupt other than IIC wakeup interrupt, for example, the IRQn

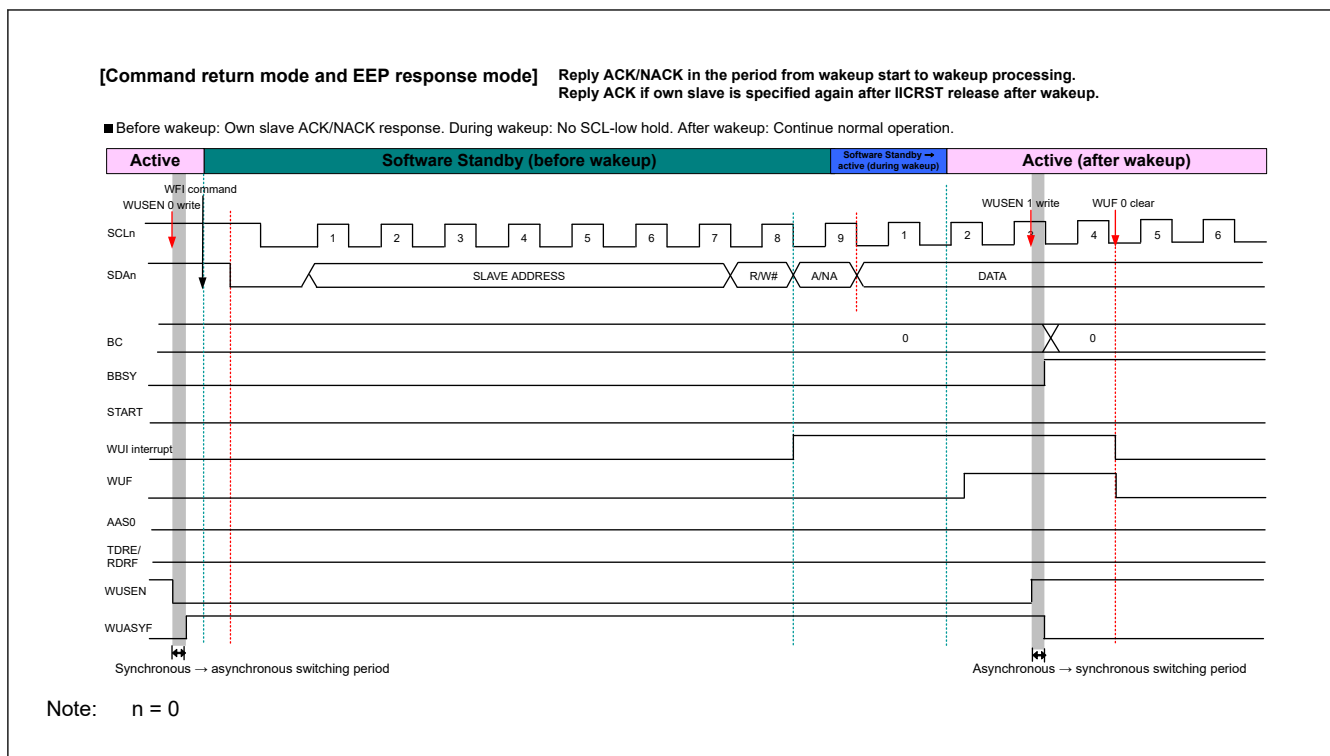


Figure 39.37 Timing of command recovery and EEP response modes

39.9 Automatic Low-Hold Function for SCL

39.9.1 Function to Prevent Wrong Transmission of Transmit Data

If the I²C Bus Shift Register (ICDRS) is empty and data has not been written to the I²C Bus Transmit Data Register (ICDRT) with the IIC in transmission mode (TRS bit = 1 in ICCR2), the SCLn line is automatically held low over the subsequent intervals. This low-hold period is extended until the transmit data is written, which prevents the unintended transmission of erroneous data.

Master transmit mode:

- Low-level interval after a start or restart condition is issued
- Low-level interval between the 9th clock cycle of one transfer and the 1st clock cycle of the next.

Slave transmit mode:

- Low-level interval between the 9th clock cycle of one transfer and the 1st clock cycle of the next.

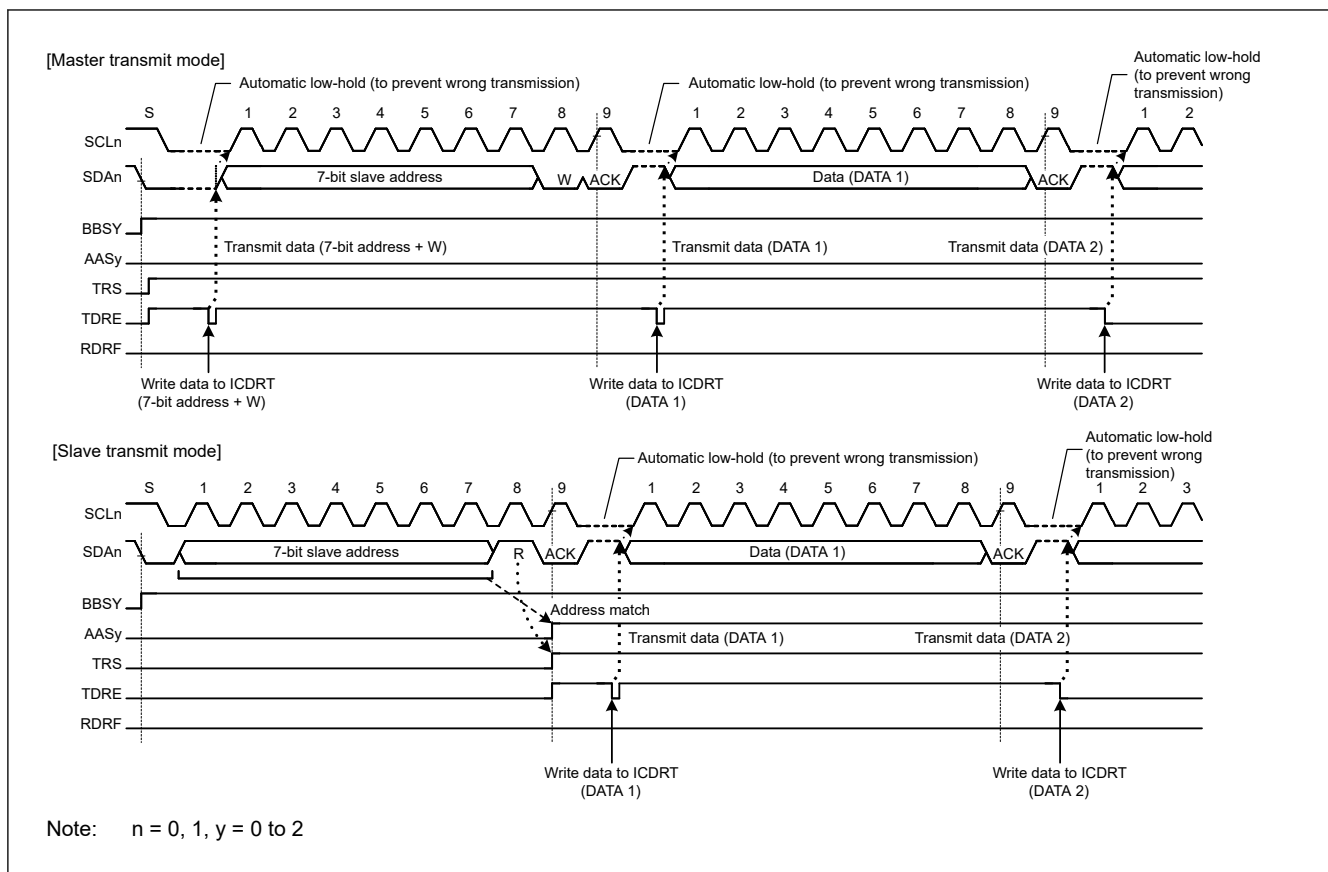
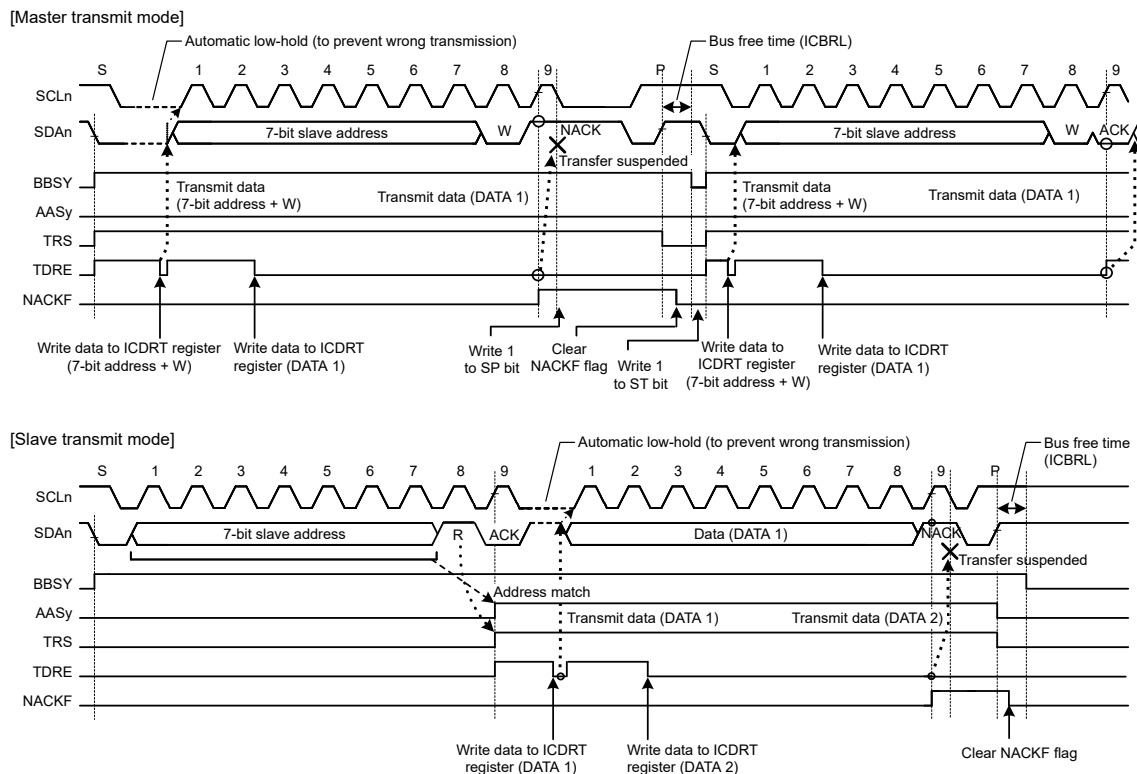


Figure 39.38 Automatic low-hold operation in transmit mode

39.9.2 NACK Reception Transfer Suspension Function

This function suspends transfer operation when NACK is received in transmit mode (TRS bit = 1 in ICCR2). This function is enabled when the NACKF bit in ICFER is set to 1. If the next transmit data is already written (TDRE flag = 0 in ICSR2) when NACK is received, the next data transmission on the falling edge of the 9th SCL clock cycle is automatically suspended. This prevents the SDAn line output level from being held low when the MSB of the next transmit data is 0.

If the transfer operation is suspended by this function (NACKF flag = 1 in ICSR2), transmit and receive operations are discontinued. To restore transmit or receive operation, after issuing the restart condition, you need to set the NACKF flag to 0 and try again, or set the NACKF flag to 0 after issuing the stop condition and then start again from issuing the start condition.



Note: n = 0, 1, y = 0 to 2

Figure 39.39 Suspension of data transfer when NACK is received, when NACK = 1

39.9.3 Function to Prevent Failure to Receive Data

If response processing is delayed when receive data (ICDRT) read is delayed for a period of one transfer frame or more with receive data full (RDRF flag = 1 in ICSR2) in receive mode (TRS = 0 in ICCR2), the IIC holds the SCLn line low automatically immediately before the next data is received to prevent a failure to receive data.

This function is enabled even if the read processing of the final receive data is delayed and, in the meantime, the IIC slave address is designated after a stop condition is issued. This function does not interfere with other communication because the IIC does not hold the SCLn line low when a mismatch with its own slave address occurs after a stop condition is issued.

Periods in which the SCLn line is held low can be selected with a combination of the WAIT and RDRFS bits in ICMR3.

(1) 1-byte receive operation and automatic low-hold function using the WAIT bit

When the WAIT bit in ICMR3 is set to 1, the IIC performs a 1-byte receive operation using the WAIT bit function. Additionally, when the ICMR3.RDRFS bit is 0, the IIC automatically sends the ACKBT bit value in ICMR3 for the acknowledge bit in the period from the falling edge of the 8th SCL clock cycle to the falling edge of the 9th SCL clock cycle, and automatically holds the SCLn line low on the falling edge of the 9th SCL clock cycle using the WAIT bit function. This low-hold is released by reading data from ICDRT, which enables byte-wise receive operation.

The WAIT bit function is enabled for receive frames after a match with the IIC slave address, including the general call address and host address, is obtained in master or slave receive mode.

(2) 1-byte receive operation (ACK/NACK transmission control) and automatic low-hold function using the RDRFS bit

When the RDRFS bit in ICMR3 is set to 1, the IIC performs a 1-byte receive operation using the RDRFS bit function. When the RDRFS bit is set to 1, the RDRF flag in ICSR2 is set to 1 (receive data full) on the rising edge of the eighth SCL clock cycle, and the SCLn line is automatically held low on the falling edge of the eighth SCL clock cycle. This low-hold

is released by writing a value to the ACKBT bit in ICMR3, but cannot be released by reading data from ICDRR, which enables receive operation through the ACK or NACK transmission control based on the data received in byte units.

The RDRFS bit function is enabled for receive frames after a match with the IIC slave address, including the general call address and host address, is obtained in master or slave receive mode.

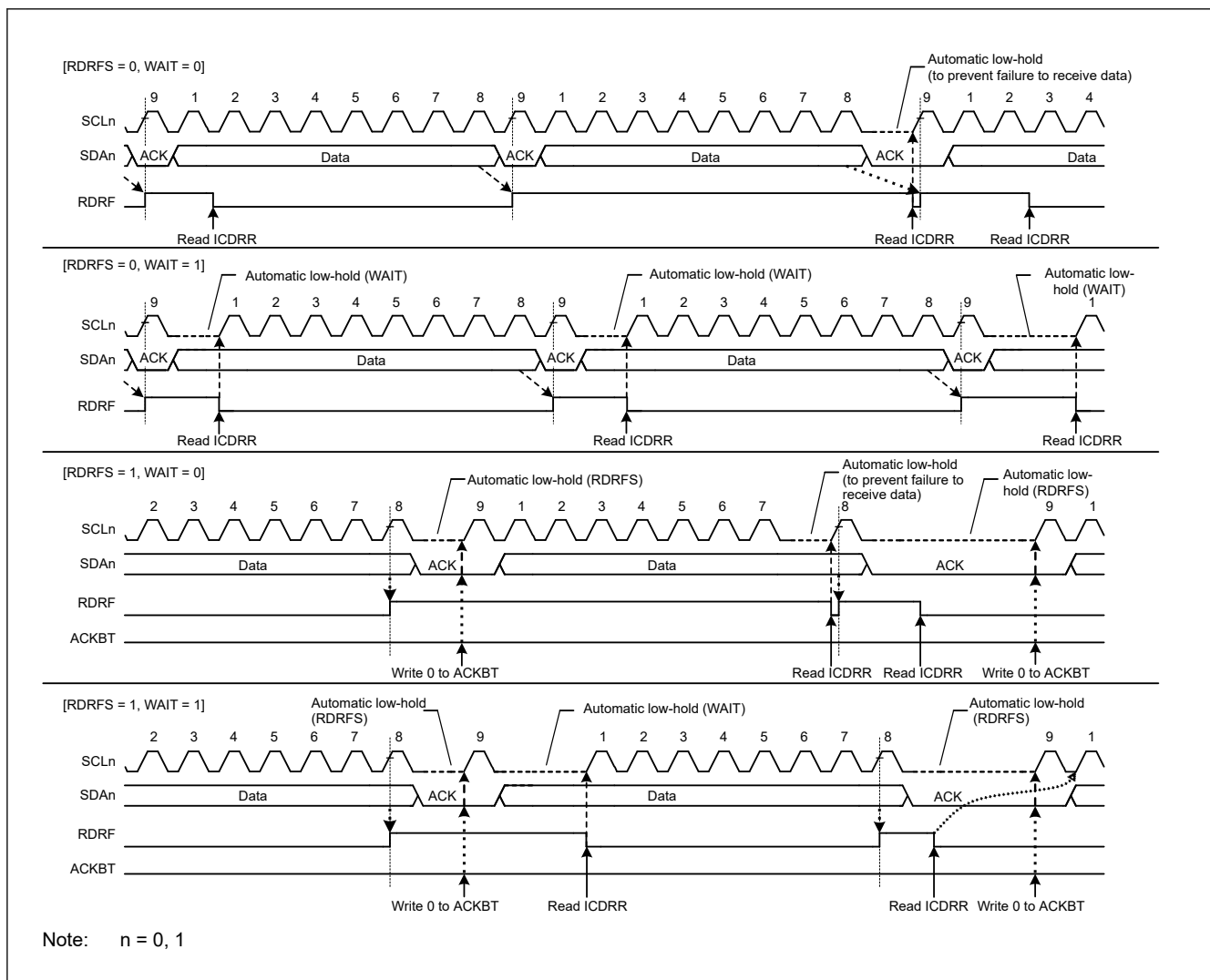


Figure 39.40 Automatic low-hold operation in receive mode using the RDRFS and WAIT bits

39.10 Arbitration-Lost Detection Functions

In addition to the normal arbitration-lost detection function defined by the I²C bus standard, the IIC provides functions to prevent double-issue of a start condition, detect arbitration-lost during transmission of NACK, and detect arbitration-lost in slave transmit mode.

39.10.1 Master Arbitration-Lost Detection (MALE Bit)

The IIC drives the SDAn line low to issue a start condition. However, if the SDAn line was already driven low by another master device issuing a start condition, the IIC regards its own start condition as an error and considers this a loss in arbitration. Priority is given to transfer by the other master device. Similarly, if a request to issue a start condition is made by setting the ST bit in ICCR2 to 1 while the bus is busy (BBSY flag = 1 in ICCR2), the IIC regards this as a double-issuing-of-start-condition error and considers itself to have lost in arbitration. This prevents a failure of transfer resulting from a start condition being issued while transfer is in progress.

When a start condition is issued successfully, if the transmit data including the address bits (internal SDA output level) and the level on the SDAn line do not match (high output as the internal SDA output, meaning the SDAn pin is in the high-impedance state and a low level is detected on the SDAn line), the IIC loses in arbitration.

After a loss in arbitration of mastership, the IIC immediately enters slave receive mode. If a slave address, including the general call address, matches its own address at this time, the IIC continues in slave operation.

A loss in arbitration of mastership is detected when the following conditions are met while the MALE bit in ICFER is 1 (master arbitration-lost detection enabled).

[Master arbitration-lost conditions]

- Mismatching of the internal level for output on SDA and the level on the SDA_n line after a start condition was issued by setting the ST bit in ICCR2 to 1 while the BBSY flag in ICCR2 is set to 0 (erroneous issuing of a start condition)
- Setting of the ST bit in ICCR2 to 1 (start condition double-issue error) while the BBSY flag is 1
- When the transmit data excluding acknowledge (internal SDA output level) does not match the level on the SDA_n line in master transmit mode (MST and TRS bits = 11b in ICCR2).

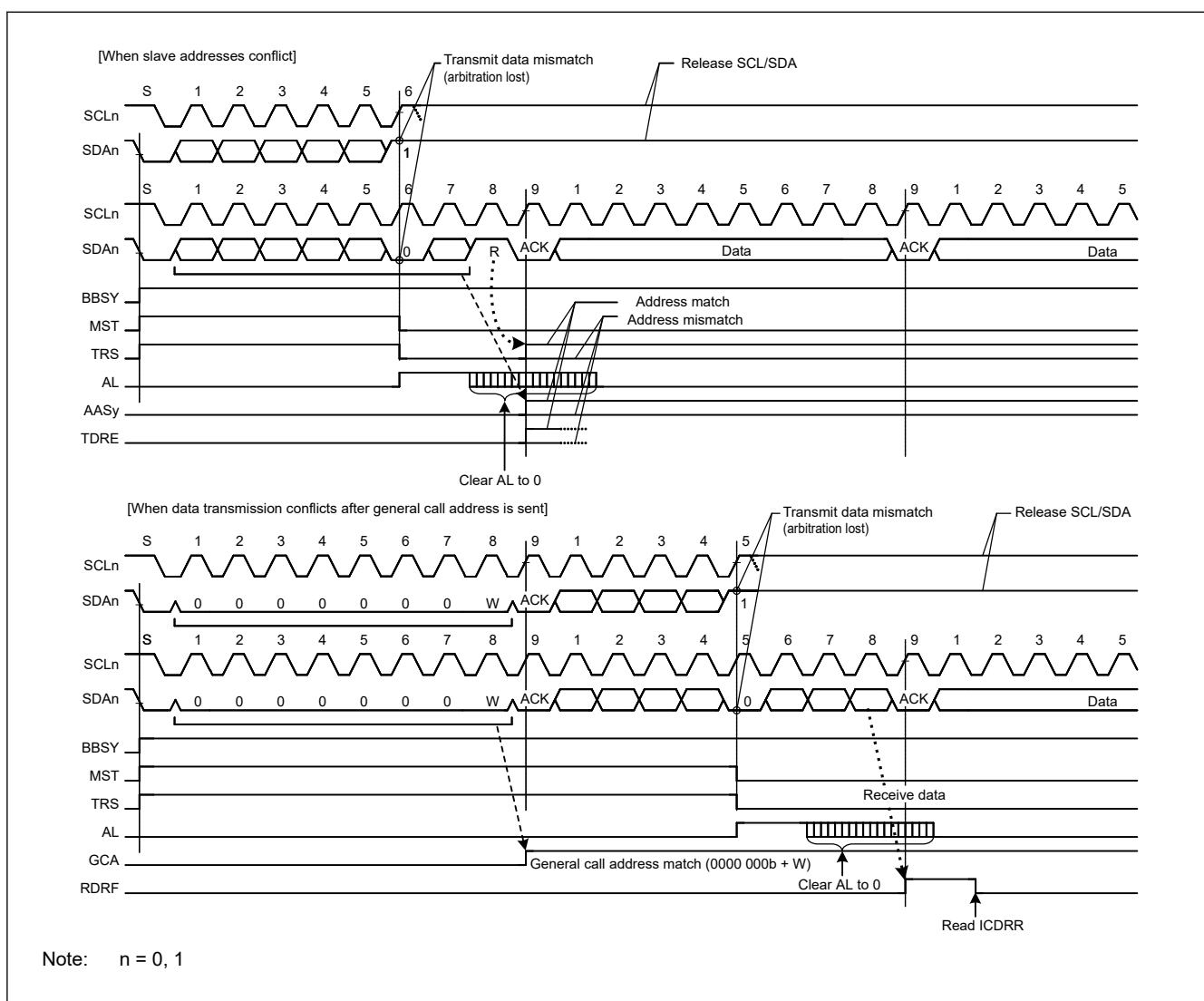


Figure 39.41 Examples of master arbitration-lost detection when MALE = 1

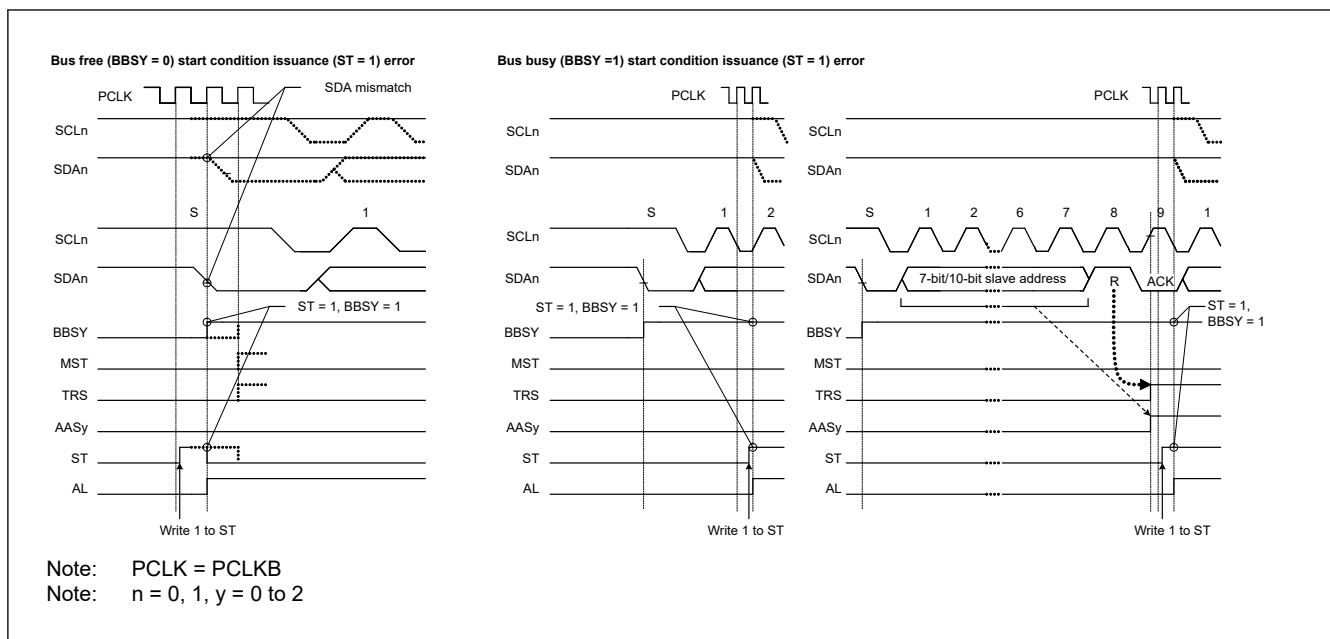


Figure 39.42 Arbitration-lost when start condition is issued when MALE = 1

39.10.2 Function to Detect Loss of Arbitration During NACK Transmission (NALE Bit)

This function causes arbitration to be lost if the internal SDA output level does not match the level on the SDA_n line (high output as the internal SDA output, meaning the SDA_n pin is in the high-impedance state) and the low level is detected on the SDA_n line during transmission of NACK in receive mode. Arbitration is lost because of a conflict between NACK and ACK transmissions when two or more master devices receive data from the same slave device simultaneously in a multi-master system. Such a conflict occurs when multiple master devices send or receive the same information through a single slave device. Figure 39.43 shows an example of arbitration-lost detection during transmission of NACK.

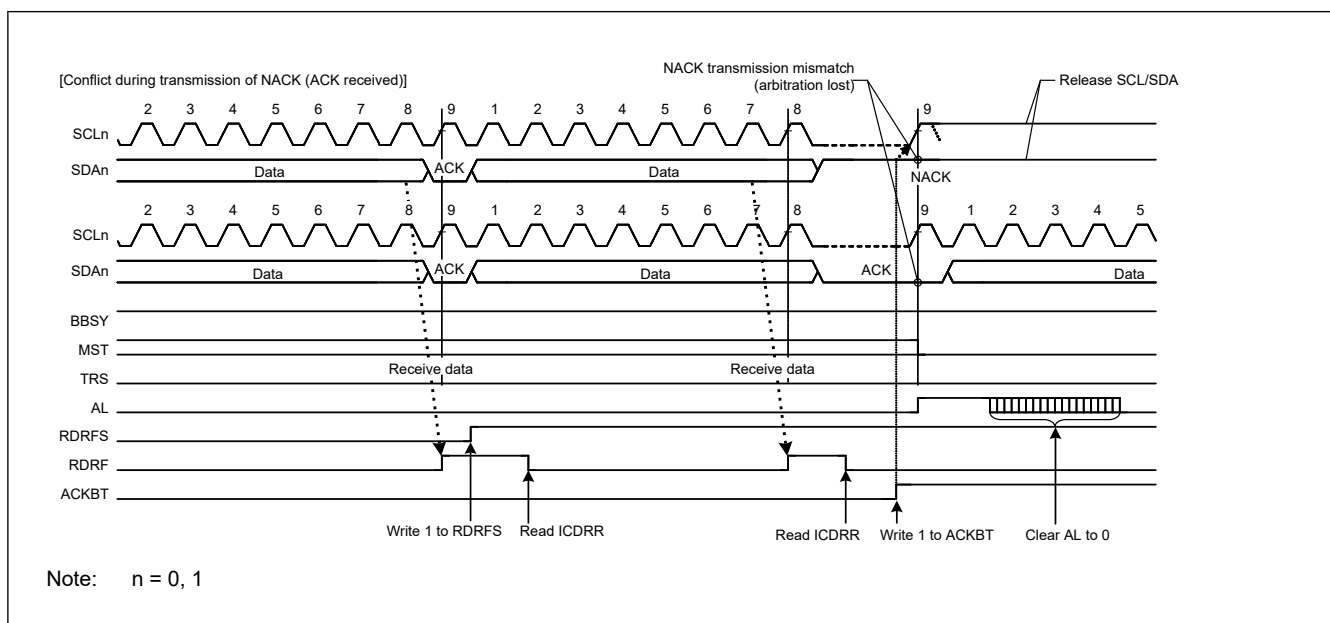


Figure 39.43 Example of arbitration-lost detection during transmission of NACK when NALE = 1

The following explains arbitration-lost detection using an example in which two master devices (master A and master B) and a single slave device are connected through the bus. In this example, master A receives 2 bytes of data from the slave device, and master B receives 4 bytes of data from the slave device.

If master A and master B access the slave device simultaneously, because the slave address is identical, arbitration is not lost in either master A or master B during access to the slave device. Therefore, both master A and master B recognize that they have obtained the bus mastership and operate as such. Master A sends NACK when it has received 2 final bytes of

data from the slave device. Meanwhile, master B sends ACK because it has not received the required 4 bytes of data. At this time, the NACK transmission from master A and the ACK transmission from master B conflict. In general, if a conflict like this occurs, master A cannot detect the ACK transmitted by master B and issues a stop condition. The stop condition issue conflicts with the SCL clock output of master B, which disrupts communication.

When the IIC receives ACK during transmission of NACK, it detects a defeat in conflict with other master devices and causes arbitration to be lost. If arbitration is lost during transmission of NACK, the IIC immediately cancels the slave match condition and enters slave receive mode. This prevents a stop condition from being issued, preventing a communication failure on the bus.

Similarly, in the ARP command processing of SMBus, the function to detect loss of arbitration during transmission of NACK is also available for eliminating the extra clock cycle processing, such as 0xFF transmission processing, which is required if the UDID (Unique Device Identifier) of the assigned address does not match in the Get UDID general processing after the Assign Address command.

The IIC detects arbitration-lost during transmission of NACK when the following condition is met with the NALE bit in ICFER set to 1 (arbitration-lost detection during NACK transmission enabled).

[Condition for arbitration-lost during NACK transmission]

- When the internal SDA output level does not match the SDA_n line (ACK is received) during transmission of NACK (ACKBT bit = 1 in ICMR3).

39.10.3 Slave Arbitration-Lost Detection (SALE Bit)

This function causes arbitration to be lost if the transmit data (internal SDA output level) and the level on the SDA_n line do not match (high output as the internal SDA output, meaning the SDA_n pin is in the high-impedance state), and the low level is detected on the SDA_n line in slave transmit mode. This arbitration-lost detection function is mainly used when transmitting a UDID (Unique Device Identifier) over an SMBus.

When the IIC loses slave arbitration, the IIC is immediately released from the slave-matched state and enters slave receive mode. This function can detect conflicts of data during transmission of UDIDs over an SMBus and eliminates subsequent redundant processing for the transmission of 0xFF.

The IIC detects slave arbitration-lost when the following condition is met with the SALE bit in ICFER set to 1 (slave arbitration-lost detection enabled).

[Condition for slave arbitration-lost]

- When transmit data excluding acknowledge (internal SDA output level) does not match the SDA_n line in slave transmit mode (MST and TRS bits = 01b in ICCR2).

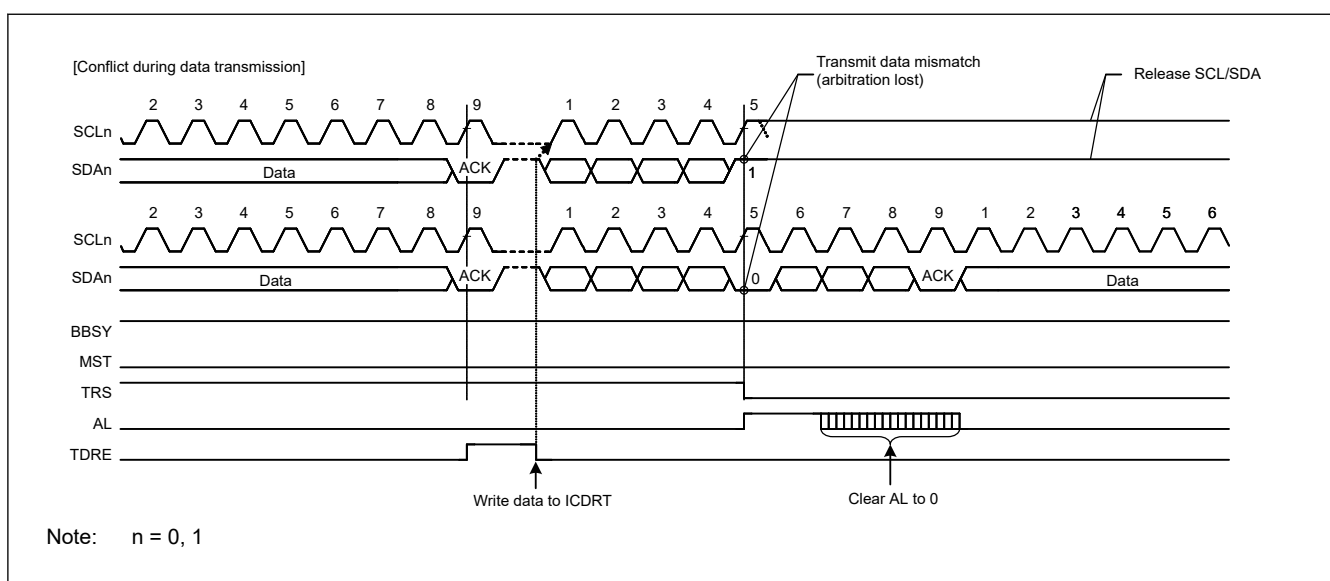


Figure 39.44 Example of slave arbitration-lost detection when SALE = 1

39.11 Start, Restart, and Stop Condition Issuing Function

39.11.1 Issuing a Start Condition

The IIC issues a start condition when the ST bit in ICCR2 is set to 1. When the ST bit is set to 1, a start condition request is made, and the IIC issues a start condition when the BBSY flag in ICCR2 is 0 (bus free state). When a start condition is issued normally, the IIC automatically shifts to the master transmit mode.

To issue a start condition:

1. Drive the SDA_n line low (high level to low level).
2. Ensure that the time set in ICBRH and the start condition hold time elapse.
3. Drive the SCL_n line low (high level to low level).
4. Detect low level on the SCL_n line and ensure the low-level period of the SCL_n line set in ICBRL elapses.

39.11.2 Issuing a Restart Condition

The IIC issues a restart condition when the RS bit in ICCR2 is set to 1. When the RS bit is set to 1, a restart condition request is made, and the IIC issues a restart condition when the BBSY flag in ICCR2 is 1 (bus busy state) and the MST bit in ICCR2 is 1 (master mode).

To issue a restart condition:

1. Release the SDA_n line.
2. Ensure the low-level period of the SCL_n line set in ICBRL elapses.
3. Release the SCL_n line (low level to high level).
4. Detect a high level on the SCL_n line and ensure the time set in ICBRL and the restart condition setup time elapse.
5. Drive the SDA_n line low (high level to low level).
6. Ensure the time set in ICBRH and the restart condition hold time elapse.
7. Drive the SCL_n line low (high level to low level).
8. Detect a low level on the SCL_n line and ensure the low-level period of the SCL_n line set in ICBRL elapses.

Note: When issuing restart condition requests, write the slave address to ICDRT after confirming that ICCR2.RS = 0. Data written while ICCR2.RS = 1 is not forwarded because of the retransmission condition before the occurrence.

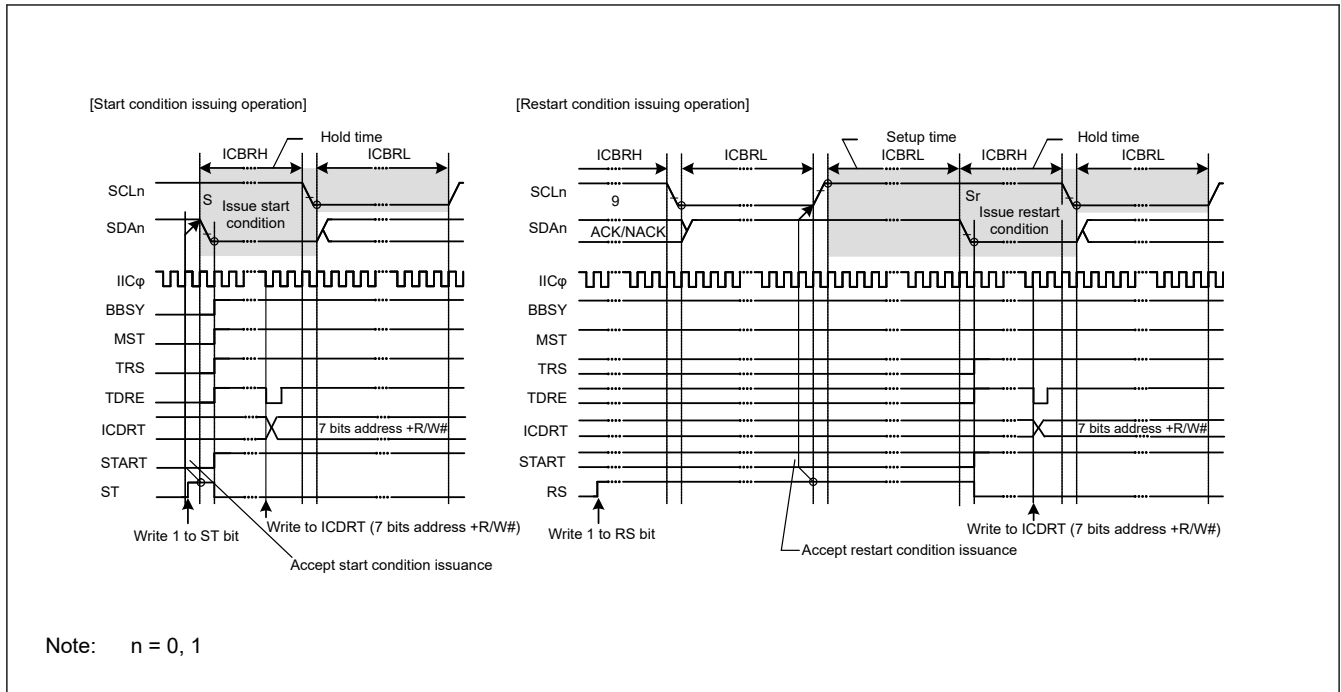


Figure 39.45 Start and restart condition issue timing using the ST and RS bits

Figure 39.46 shows the operation timing when a restart condition is issued after the master transmission.

[To issue a restart condition after the master transmission:]

1. Initialize the IIC using the procedure in [section 39.3.2. Initial Settings](#).
2. Read the BBSY flag in ICCR2 to check that the bus is open, then set the ST bit in ICCR2 to 1 (start condition issuance request). On receiving the request, the IIC issues a start condition. At the same time, the BBSY and the START flags in ICSR2 are automatically set to 1 and the ST bit is automatically set to 0. If the start condition is detected and the internal levels for the SDA output state and the levels on the SDA n line match while the ST bit is 1, the IIC recognizes that a start condition is successfully issued as requested by the ST bit has been successfully completed. The MST and TRS bits in ICCR2 are automatically set to 1, placing the IIC in master transmit mode. The TDRE flag in ICSR2 is also automatically set to 1 when the TRS bit is set to 1.
3. Check that the TDRE flag in ICSR2 is 1, and then write the value for transmission (the slave address and the R/W# bit) to ICDRT. After the data for transmission is written to ICDRT, the TDRE flag is automatically set to 0, data is transferred from ICDRT to ICDRS, and the TDRE flag is again set to 1. After the byte containing the slave address and R/W# bit has been transmitted, the value of the TRS bit is automatically updated to select master transmit or master receive mode according to the value of the transmitted R/W# bit. If the value of the R/W# bit is 0, the IIC continues in master transmit mode. If the ICSR2.NACKF flag is 1 at this time, indicating that no slave device recognized the address or that there was an error in communications, write 1 to ICCR2.SP bit to issue a stop condition. To transmit data with an address in the 10-bit format, start by writing 1111 0b, the 2 upper bits of the slave address, and W to ICDRT as the first address transmission. Then, as the second address transmission, write the 8 lower bits of the slave address to ICDRT.
4. After confirming that the TDRE flag in ICSR2 is 1, write data for transmission to the ICDRT register. The IIC automatically holds the SCLn line low until data for transmission is ready, a restart condition is issued or a stop condition is issued.
5. After all bytes of data for transmission are written to the ICDRT register, wait until the value of the TEND flag in ICSR2 returns to 1. Then after checking that the START flag in ICSR2 is 1, set the START flag in ICSR2 to 0.
6. Set the RS bit in ICCR2 to 1 (restart condition issue request). On receiving the request, the IIC issues a restart condition.
7. After checking that the START flag in ICSR2 is 1, write the value for transmission (the slave address and the R/W# bit) to ICDRT.

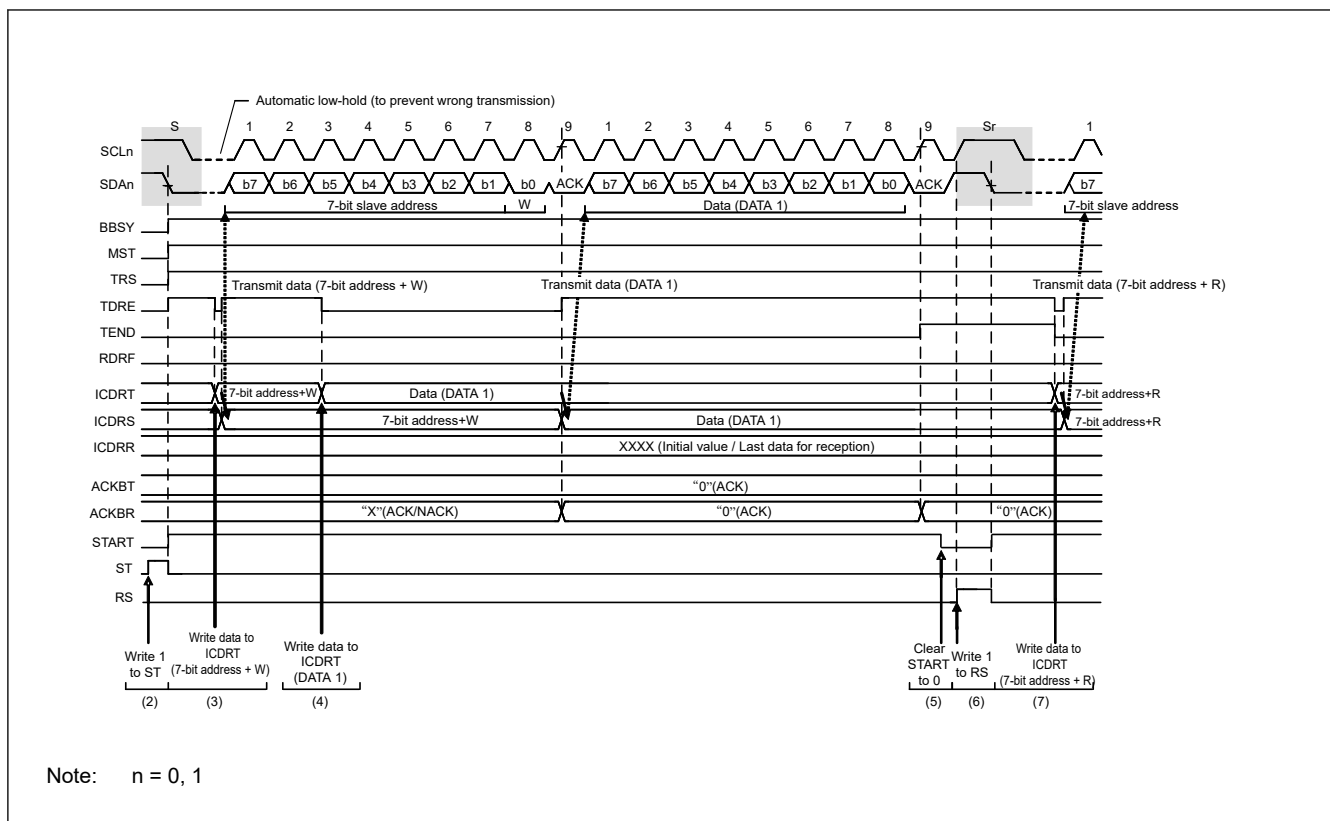


Figure 39.46 Restart condition issue timing after master transmission.

39.11.3 Issuing a Stop Condition

The IIC issues a stop condition when the SP bit in ICCR2 is set to 1. When the SP bit is set to 1, a stop condition request is made, and the IIC issues a stop condition when the BBSY flag in ICCR2 is 1 (bus busy state) and the MST bit in ICCR2 is 1 (master mode).

To issue a stop condition:

1. Drive the SDAn line low (high level to low level).
2. Ensure the low-level period of the SCLn line set in ICBRL elapses.
3. Release the SCLn line (low level to high level).
4. Detect a high level on the SCLn line and ensure the time set in ICBRH and the stop condition setup time elapse.
5. Release the SDAn line (low level to high level).
6. Ensure the time set in ICBRL and the bus free time elapse.
7. Clear the BBSY flag to 0 to release the bus mastership.

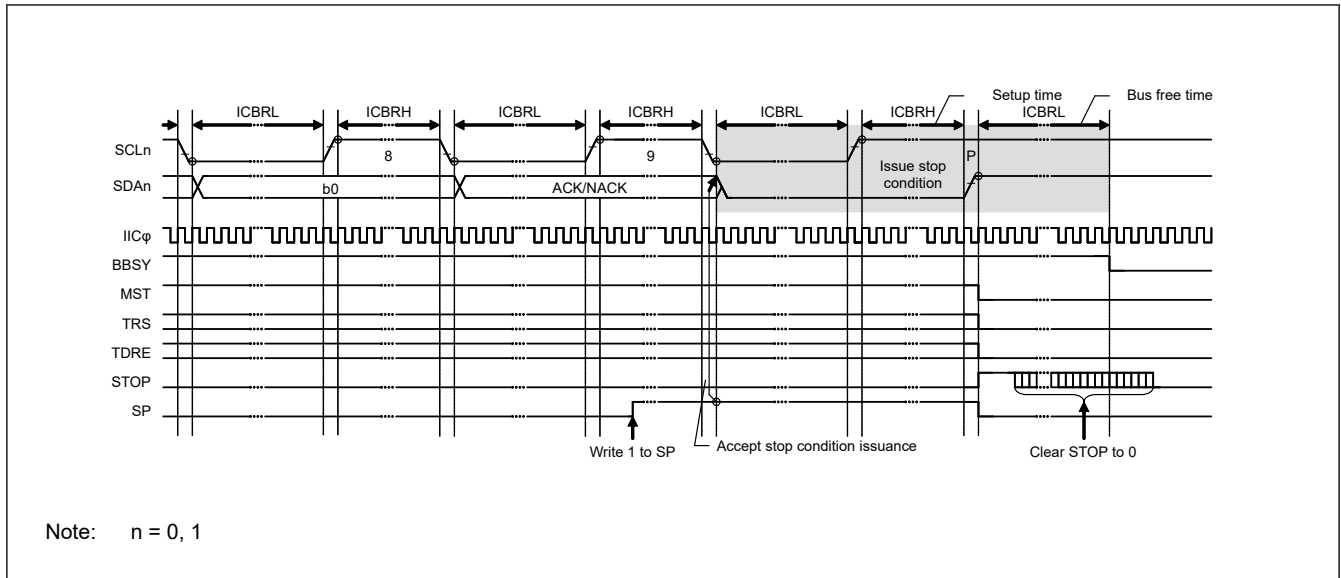


Figure 39.47 Stop condition issue timing using the SP bit

39.12 Bus Hanging

If the clock signals from the master and slave devices are out of synchronization because of noise or other factors, the I²C bus might hang with a fixed level on the SCLn line or SDA n line.

To manage bus hanging, the IIC has a timeout function to detect hanging by monitoring the SCLn line, and a function for outputting an extra SCL clock cycle to release the bus from:

- A timeout function to detect hanging by monitoring the SCLn line
- The IIC reset function
- An internal reset function.

By checking the SCLO, SDAO, SCLI, and SDAI bits in ICCR1, it is possible to see whether the IIC or its communicating partner is placing the low level on the SCLn or SDA n line.

39.12.1 Timeout Function

The timeout function can detect when the SCLn line is stuck longer than the predetermined time. The IIC can detect an abnormal bus state by monitoring that the SCLn line is stuck low or high for a predetermined time.

The timeout function monitors the SCLn line state and counts the low- or high-level period using the internal counter. The timeout function resets the internal counter each time the SCLn line changes (rises or falls), but continues to count unless the SCLn line changes. If the internal counter overflows because no SCLn line changes, the IIC can detect the timeout and report the bus hung state.

This timeout function is enabled when the ICFER.TMOE bit is 1. It detects a hung state when the SCLn line is stuck low or high during the following conditions:

- The bus is busy (ICCR2.BBSY flag is 1) in master mode (ICCR2.MST bit is 1)
- The IIC slave address is detected (ICSR1 register is not 0x00) and the bus is busy (ICCR2.BBSY flag is 1) in slave mode (ICCR2.MST bit is 0)
- The bus is open (ICCR2.BBSY flag is 0) while a start condition is requested (ICCR2.ST bit is 1).

The internal counter of the timeout function uses the internal reference clock (IICφ) set in the CKS[2:0] bits in ICMR1 as a count source. It functions as a 16-bit counter when long mode is selected (TMOS bit = 0 in ICMR2) or a 14-bit counter when short mode is selected (TMOS bit = 1).

The SCLn line level (low, high, or both levels) during which this counter is activated can be selected in the TMOH and TMOL bits in ICMR2. If both TMOL and TMOH bits are set to 0, the internal counter is disabled.

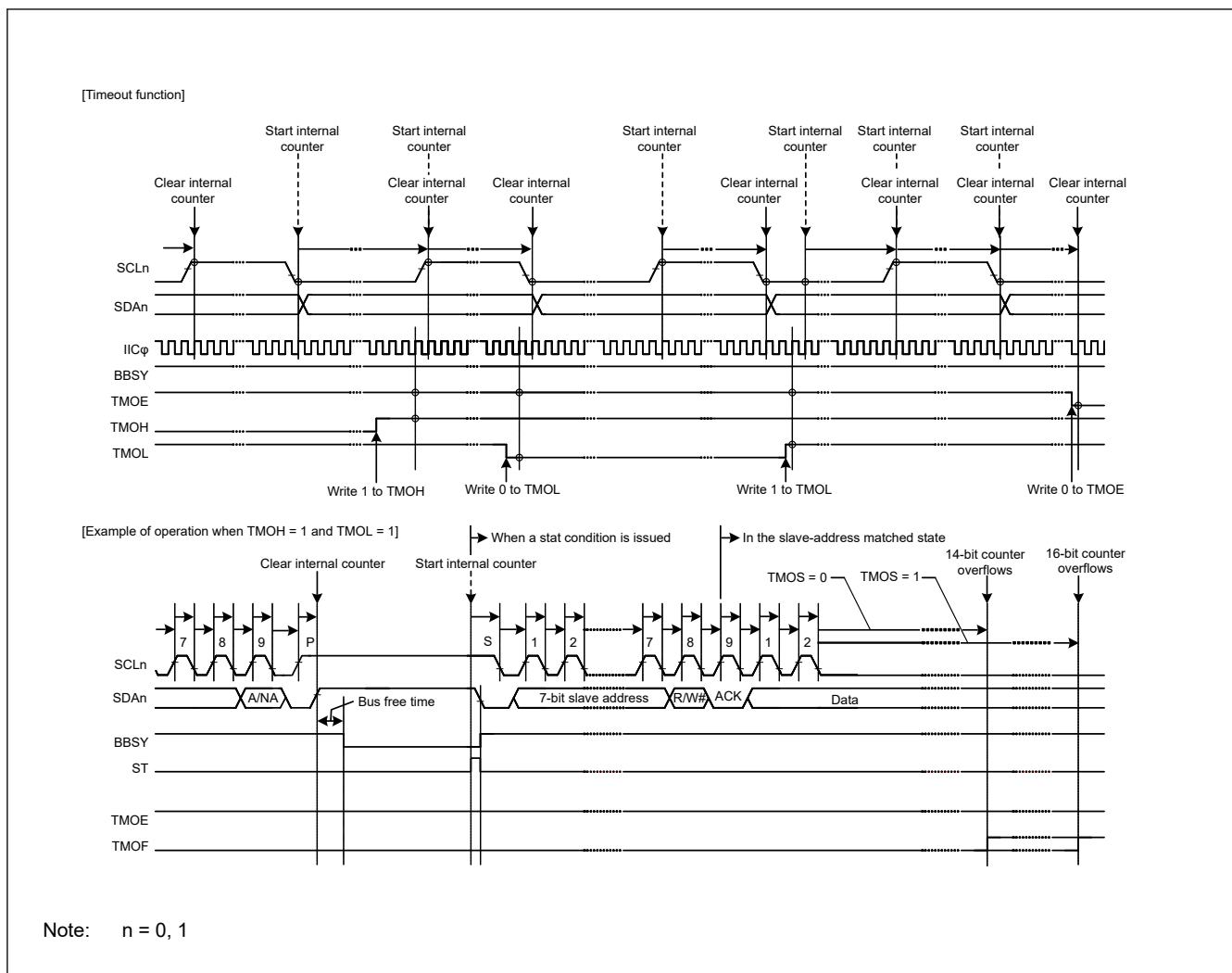


Figure 39.48 Timeout function using the TMOE, TMOS, TMOH, and TMOL bits

39.12.2 Extra SCL Clock Cycle Output Function

In master mode, this function outputs extra SCL clock cycles to release the SDAn line of the slave device from being held low because the master is out of synchronization with the slave device. This function is mainly used in master mode to release the SDAn line of the slave device from being fixed low by including extra cycles of SCL output from the IIC. It uses single cycles of the SCL clock for a bus error where the IIC cannot issue a stop condition because the slave device is holding the SDAn line at the low level. Do not use this function in normal situations. Using it when communications are proceeding correctly leads to malfunctions.

When the CLO bit in ICCR1 is set to 1 in master mode, a single cycle of the SCL clock at the frequency specified in the CKS[2:0] bits in ICMR1, and in the ICBRH and ICBLR registers, is output as an extra clock cycle. After output of this single cycle of the SCL clock, the CLO bit is automatically set to 0. At this time, if ICCR2.BBSY = 1, the SCL pin goes low, and when ICCR2.BBSY = 0, the SCL pin goes high. After confirming that the CLO bit is 0 by software, write 1 to the CLO bit to output the additional clock continuously.

When the IIC module is in master mode and the slave device is holding the SDAn line low because synchronization with the slave device is lost because of effects like noise, the output of a stop condition is not possible. This function can be used to output extra cycles of SCL one by one to make the slave device release the SDAn line from being held low, and so recover the bus from an unusable state. Release of the SDAn line by the slave device can be monitored by reading the SDAI bit in ICCR1. After confirming the release of the SDAn line by the slave device, complete communications by reissuing the stop condition.

[Output conditions for using the CLO bit in ICCR1]

- When the bus is open (BBSY flag in ICCR2 = 0) or in master mode (MST bit = 1 and BBSY flag = 1 in ICCR2)

- When the communication device does not hold the SCLn line low.

Figure 39.49 shows the operation timing of the extra SCL clock cycle output function (CLO bit).

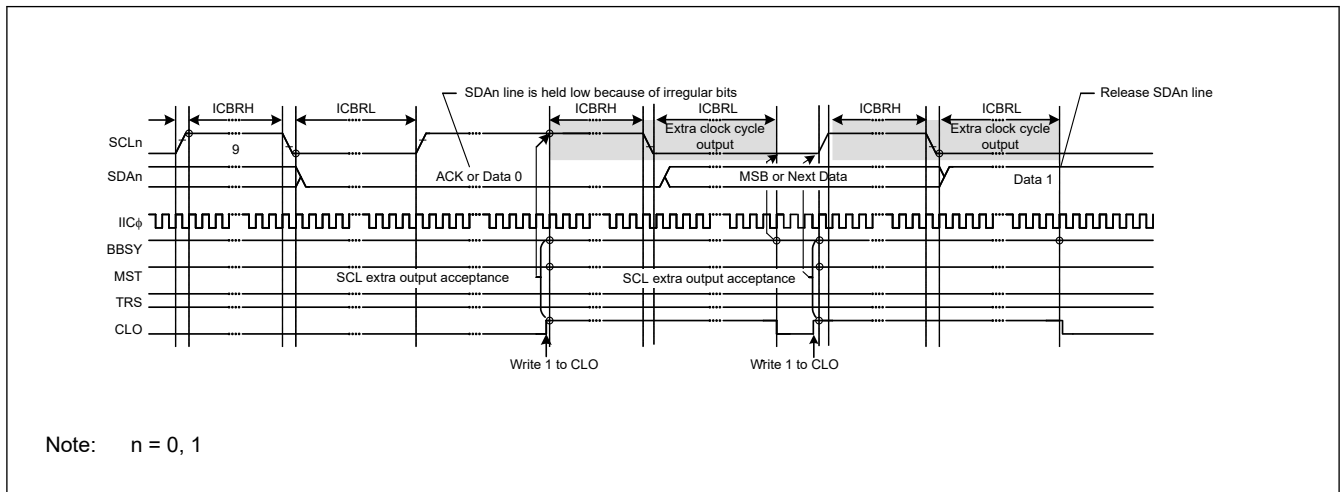


Figure 39.49 Extra SCL clock cycle output function using the CLO bit

39.12.3 IIC Reset and Internal Reset

The IIC module incorporates a function for resetting itself. It uses two types of resets:

- An IIC reset, which initializes all registers, including the BBSY flag in ICCR2.
- An internal reset, which releases the IIC from the slave-address matched state and initializes the internal counter while saving other settings.

After issuing a reset, always set the IICRST bit in ICCR1 to 0. Both types of resets are valid for release from bus-hung states, because both restore the output state of the SCLn and SDAn pins to the high-impedance state.

Issuing a reset during slave operation might lead to a loss of synchronization between the master device clock and the slave device clock, so avoid this when possible. In addition, monitoring of the bus state, such as for the presence of a start condition, is not possible during an IIC reset (ICE and IICRST bits = 01b in ICCR1).

For a detailed description of the IIC and internal resets, see [section 39.15. State of Registers When Issuing Each Condition](#).

39.13 SMBus Operation

The IIC supports data communication conforming to the SMBus Specification (version 2.0). To perform SMBus communication, set the SMBS bit in ICMR3 to 1. To use the transfer rate within a range of 10 to 100 kbps of the SMBus standard, set the CKS[2:0] bits in ICMR1, the ICBRH, and ICBRL registers. In addition, specify the values in the DLCS bit in ICMR2 and the SDDL[2:0] bits in ICMR2 to meet the data hold time specification of 300 ns or more. When the IIC is used only as a slave device, the transfer rate setting is not required, but ICBRL must be set to a value longer than the data setup time (250 ns).

For the SMBus device default address (1100 001b), use one of the slave address registers L0 to L2 (SARL0, SARL1, and SARL2), and set the associated FS bit (7- or 10-bit address format select) in SARUy (y = 0 to 2) to 0 (7-bit address format).

When transmitting the UDID (Unique Device Identifier), set the SALE bit in ICFER to 1 to enable the slave arbitration-lost detection function.

39.13.1 SMBus Timeout Measurement

(1) Measuring slave device timeout

The following period (timeout interval: $T_{\text{LOW:SEXT}}$) must be measured for slave devices in SMBus communication:

- From start condition to stop condition.

To measure timeout for slave devices, measure the period from start condition detection to stop condition detection with the GPT using the IIC start condition detection interrupt (STIn) and stop condition detection interrupt (SPIn). The measured timeout period must be within the total clock low-level period [slave device] $T_{LOW:SEXT}$: 25 ms (maximum) of the SMBus standard.

If the time measured with the GPT exceeds the clock low-level detection timeout $T_{TIMEOUT}$: 25 ms (minimum) of the SMBus standard, the slave device must release the bus by writing 1 to the IICRST bit in ICCR1 to issue an internal reset of the IIC. When an internal reset is issued, the IIC stops driving the bus for the SCLn and SDAn pins, making them output high-impedance, which releases the bus.

(2) Measuring master device timeout

The following periods (timeout interval: $T_{LOW:MEXT}$) must be measured for master devices in SMBus communication:

- From start condition to acknowledge bit
- Between acknowledge bits
- From acknowledge bit to stop condition.

To measure timeout for master devices, measure these periods with the GPT using the IIC start condition detection interrupt (STIn), stop condition detection interrupt (SPIn), transmit end interrupt (IICn_TEI), or receive data full interrupt (IICn_RXI). The measured timeout period must be within the total clock low-level extended period (master device) $T_{LOW:MEXT}$: 10 ms (maximum) of the SMBus standard, and the total of all $T_{LOW:MEXT}$ values from start condition to stop condition must be within $T_{LOW:SEXT}$: 25 ms (maximum).

For the ACK receive timing (rising edge of the 9th SCL clock cycle), monitor the TEND flag in ICSR2 in master transmit mode (master transmitter) and the RDRF flag in ICSR2 in master receive mode (master receiver). Perform byte-wise transmit operations in master transmit mode, and hold the RDRFS bit in ICMR3 at 0 until the byte immediately before reception of the final byte in master receive mode. While the RDRFS bit is 0, the RDRF flag is set to 1 on the rising edge of the 9th SCL clock cycle.

If the period measured with the GPT exceeds the total clock low-level extended period (master device) $T_{LOW:MEXT}$: 10 ms (maximum) of the SMBus standard or the total of measured periods exceeds the clock low-level detection timeout $T_{TIMEOUT}$: 25 ms (minimum) of the SMBus standard, the master device must stop the transaction by issuing a stop condition. In master transmit mode, immediately stop the transmit operation (stop writing data to ICDRT).

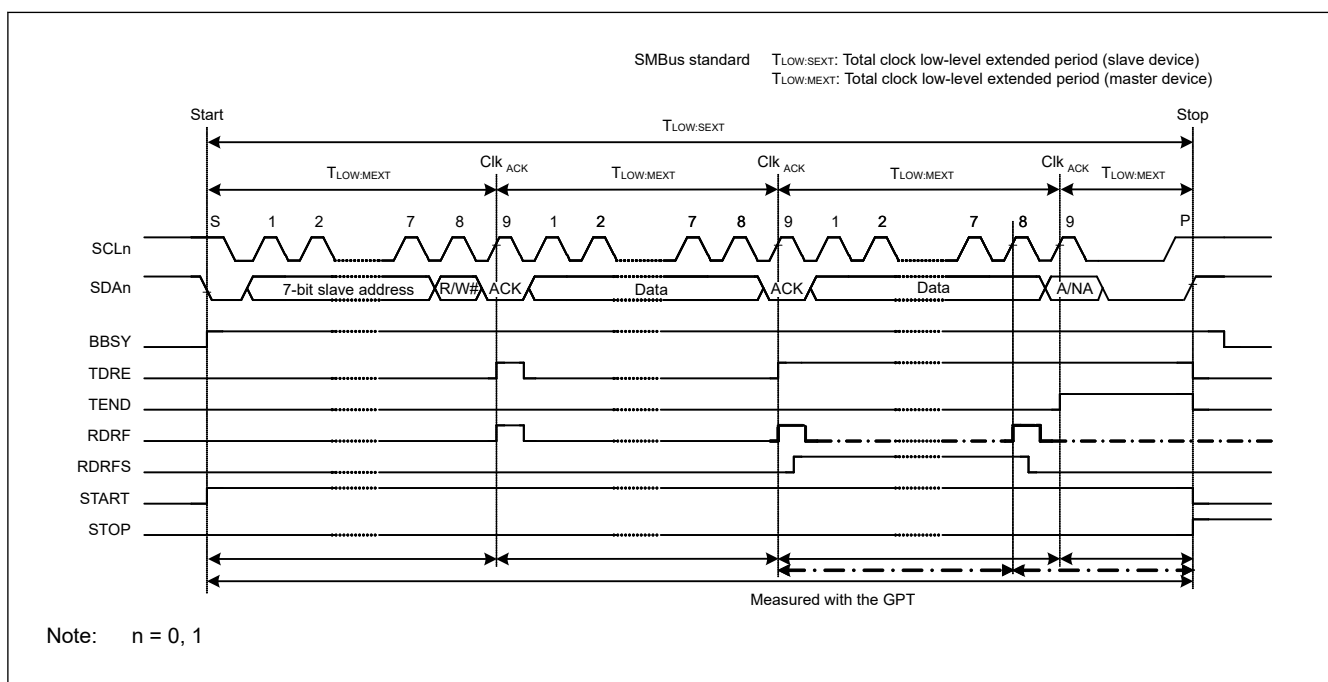


Figure 39.50 SMBus timeout measurement

39.13.2 Packet Error Code (PEC)

The MCU provides a CRC calculator that enables transmission of a Packet Error Code (PEC) or allows checking of the received data in SMBus data communication for the IIC. For the CRC-generating polynomials of the CRC calculator, see [section 47, Cyclic Redundancy Check \(CRC\)](#).

In master transmit mode, the PEC data can be generated by writing all transmit data to the CRC Data Input Register (CRCDIR) in the CRC calculator.

In master receive mode, the PEC data can be checked by writing all receive data to CRCDIR in the CRC calculator and comparing the obtained value in the CRC Data Output Register (CRCDOR) with the received PEC data.

To send ACK or NACK based on the match or mismatch result when the final byte is received as a result of the PEC code check, set the RDRFS bit in ICMR3 to 1 before the rising edge of the 8th SCL clock cycle during reception of the final byte, and hold the SCLn line low on the falling edge of the 8th clock cycle.

39.13.3 SMBus Host Notification Protocol (Notify ARP Master Command)

In communicating over an SMBus, a slave device can temporarily act as a master device to notify the SMBus host (or ARP master) of its own slave address, or to request its own slave address from the SMBus host.

For a product using the MCU to operate as an SMBus host or ARP master, the host address (0001 000b) sent from the slave device must be detected as a slave address, and so the IIC provides a function for detecting the host address. To detect the host address as a slave address, set the SMBS bit in ICMR3 and the HOAE bit in ICSEI to 1. Operation after the host address is detected is the same as normal slave operation.

39.14 Interrupt Sources

The IIC issues five types of interrupt requests:

- Transfer error or event generation (arbitration-lost, NACK detection, timeout detection, start condition detection, and stop condition detection)
- Receive data full
- Transmit data empty
- Transmit end
- Address match during wakeup function

[Table 39.10](#) lists details about the interrupt requests. The receive data full and transmit data empty interrupts can activate data transfer by the DTC or DMAC.

Table 39.10 Interrupt sources

Symbol	Interrupt source	Interrupt flag	DTC or DMAC activation	Interrupt condition
IICn_EEI* ⁵	Transfer error or event occurrence	AL	Not possible	AL = 1, ALIE = 1
		NACKF		NACKF = 1, NAKIE = 1
		TMOF		TMOF = 1, TMOIE = 1
		START		START = 1, STIE = 1
		STOP		STOP = 1, SPIE = 1
IICn_RXI* ² * ⁵	Receive data full	RDRF	Possible	RDRF = 1, RIE = 1
IICn_TXI* ¹ * ⁵	Transmit data empty	RDRF	Possible	TDRE = 1, TIE = 1
IICn_TEI* ³ * ⁵	Transmit end	TEND	Not possible	TEND = 1, TEIE = 1
IIC0_WUI* ⁴	Slave address match during wakeup function	WUF	Not possible	Slave address match Slave receive complete RWAK operation ASY0 = 1 WUIE = 1

Note: There is a delay between the execution of a write instruction for a peripheral module by the CPU and the actual writing to the module. When an interrupt flag is cleared or masked, read the relevant flag again to check whether clearing or masking is complete, then return from interrupt handling. Not doing so creates the possibility of repeated processing of the same interrupt.

- Note 1. Because IICn_TXI is edge-detected, it does not require clearing. Additionally, the TDRE flag in ICSR2 (condition for IICn_TXI) is automatically set to 0 when transmit data is written to the ICDRT register or a stop condition is detected (STOP flag = 1 in ICSR2).
- Note 2. Because IICn_RXI is edge-detected, it does not require clearing. Additionally, the RDRF flag in ICSR2 (condition for IICn_RXI) is automatically set to 0 when data is read from ICDRR.
- Note 3. When using the IICn_TEI interrupt, clear the TEND flag in ICSR2 in the IICn_TEI interrupt handling. The TEND flag in ICSR2 automatically is set to 0 when transmit data is written to the ICDRT register or a stop condition is detected (STOP flag = 1 in ICSR2).
- Note 4. Only channel 0 has a wakeup function, so IIC0_WUI is for channel 0 only.
- Note 5. Channel number (n = 0 to 2).

Clear or mask each flag during interrupt handling.

39.14.1 Buffer Operation for IICn_TXI and IICn_RXI Interrupts

If the conditions for generating an IICn_TXI or IICn_RXI interrupt are satisfied while the associated IR flag is 1, the interrupt request is not output for the ICU but is saved internally. One request per source can be saved internally.

An interrupt request that is saved in the ICU is output when the ICU.IELSRn.IR flag becomes 0. Internally saved interrupt requests are automatically cleared under normal conditions. They can also be cleared by writing 0 to the interrupt enable bit within the associated peripheral module.

39.15 State of Registers When Issuing Each Condition

The IIC has two dedicated resets, IIC reset and Internal reset. Table 39.11 lists the registers states when issuing each condition.

Table 39.11 Register states when issuing each condition (1 of 2)

Registers		Reset	IIC reset (ICE = 0, IICRST = 1)	Internal reset (ICE = 1, IICRST = 1)	Start or restart condition detection	Stop condition detection
ICCR1	ICE, IICRST	In reset	Saved	Saved	Saved	Saved
	SCLO, SDAO		In reset	In reset		
	Others			Saved		
ICCR2	BBSY	In reset	In reset	Saved	Set	In reset
	ST, RS			In reset	In reset	Saved
	SP				Set or saved	In reset
	TRS					
	MST					
ICMR1	BC[2:0]	In reset	In reset	In reset	In reset	Saved
	Others			Saved	Saved	
ICMR2		In reset	In reset	Saved	Saved	Saved
ICMR3	ACKBT	In reset	In reset	Saved	Saved	In reset
	Others					Saved
ICFER		In reset	In reset	Saved	Saved	Saved
ICSER		In reset	In reset	Saved	Saved	Saved
ICIER		In reset	In reset	Saved	Saved	Saved
ICSR1		In reset	In reset	In reset	Saved	In reset
ICSR2	TEND	In reset	In reset	In reset	Saved	In reset
	TDRE				Set or saved	
	START				Set	
	STOP				Saved	Set
	Others				Saved	Saved
ICWUR		In reset	In reset	Saved	Saved	Saved

Table 39.11 Register states when issuing each condition (2 of 2)

Registers		Reset	IIC reset (ICE = 0, IICRST = 1)	Internal reset (ICE = 1, IICRST = 1)	Start or restart condition detection	Stop condition detection
SARL0, SARL1, SARL2 SARU0, SARU1, SARU2		In reset	In reset	Saved	Saved	Saved
ICBRH, ICBRL		In reset	In reset	Saved	Saved	Saved
ICDRT		In reset	In reset	Saved	Saved	Saved
ICDRR		In reset	In reset	Saved	Saved	Saved
ICDRS		In reset	In reset	In reset	Saved	Saved
Timeout function		In reset	In reset	In reset	Operating	Operating
Bus free time measurement		In reset	In reset	Operating	Operating	Operating
ICWUR2	WUSEN	In reset	In reset	Saved	Saved	Saved
	Others					Saved or set or reset

39.16 Event Link Output

The IIC0 module handles the event output for the Event Link Controller (ELC) for the following sources:

(1) Transfer error event

When a transfer error event occurs, the associated event signal can be output to another module by the ELC.

(2) Receive data full

When a receive data register becomes full, the associated event signal can be output to another module by the ELC.

(3) Transmit data empty

When a transmit data register becomes empty, the associated event signal can be output to another module by the ELC.

(4) Transmit end

On completion of the transfer, the associated event signal can be output to another module by the ELC.

39.16.1 Interrupt Handling and Event Linking

Each of the IIC interrupt types (see [Table 39.10](#)) has an enable bit to control enabling and disabling of the associated interrupt signal. An interrupt request signal is output to the CPU when an interrupt source condition is satisfied while the associated enable bit is set.

The associated event link output signals are sent to other modules as event signals by the ELC when the interrupt source conditions are satisfied, regardless of the interrupt enable bit settings. For details on interrupt sources, see [Table 39.10](#).

39.17 Usage Notes

39.17.1 Settings for the Module-Stop Function

The Module Stop Control Register B (MSTPCRB) can enable or disable IIC operation. The IIC is initially stopped after reset. Releasing the module-stop state enables access to the registers. For details, see [section 11, Low Power Mode](#).

39.17.2 Notes on Starting Transfer

If the IR flag associated with the IIC interrupt is 1 when transfer is started (ICCR1.ICE bit = 1), follow the procedure in this section to clear the interrupts before enabling operations. Starting transfer with the IR flag set to 1 while the ICCR1.ICE bit is 1 leads to an interrupt request being internally saved after transfer starts, and this can lead to unexpected behavior of the IR flag.

To clear interrupts before starting transfer operation:

1. Confirm that the ICCR1.ICE bit is 0.

2. Set the relevant interrupt enable bits, such as ICIER.TIE to 0.
3. Read the relevant interrupt enable bits, such as ICIER.TIE, and confirm that the value is 0.
4. Set the IR flag to 0.

39.17.3 Software Standby Mode

When using a wakeup function, the following setting is not required:

- When the IIC is in communication, set ICCR1.ICE to 0 to stop communication before transitioning to Software Standby mode.
- After returning from Software Standby mode, set ICCR1.ICE to 1 to enable communication.